BMED6208P: Medical Image Computing

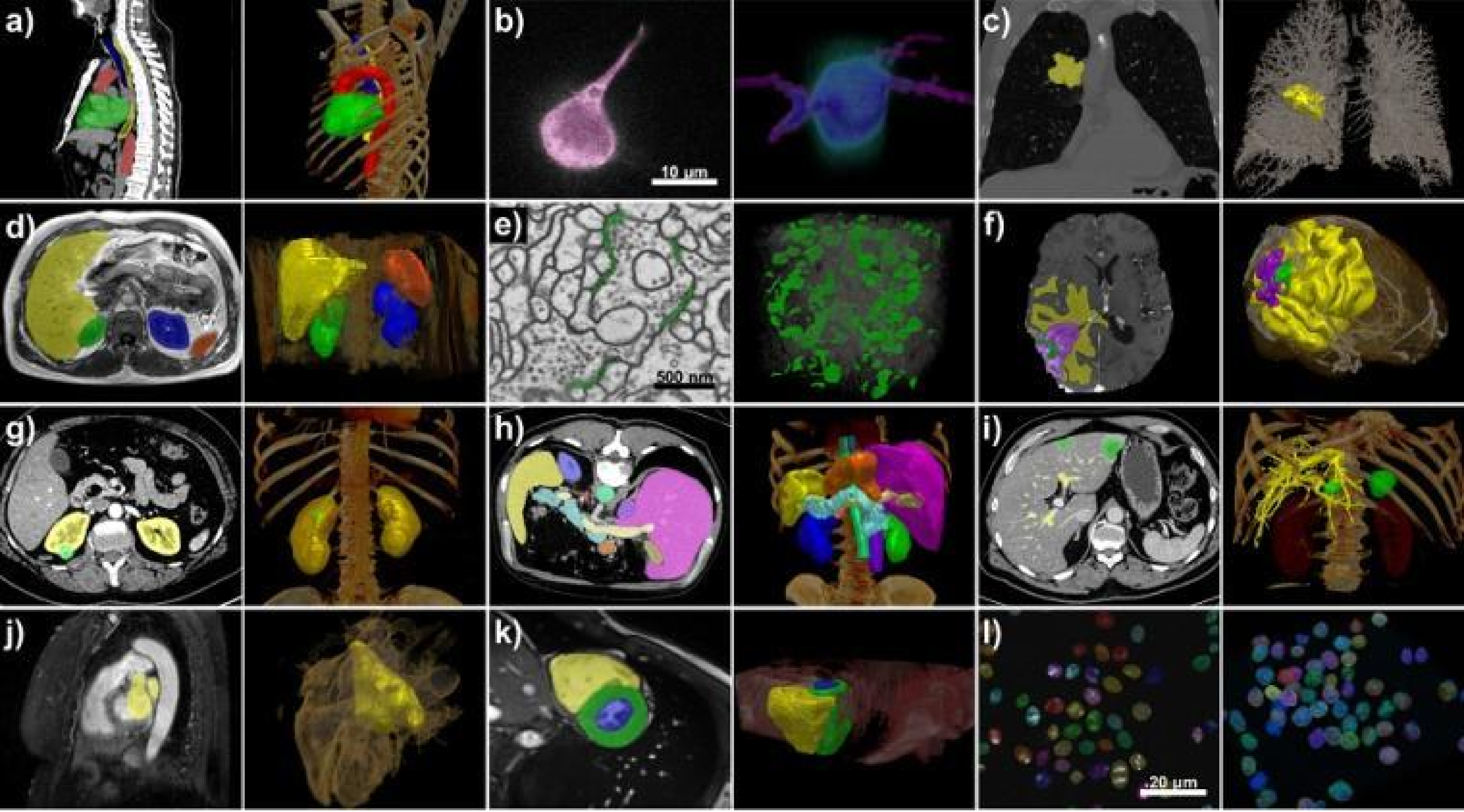
Medical Image Segmentation

Instructor: Professor S. Kevin Zhou

Talk: Fenghe Tang & Mingyue Zhao

What is medical image segmentation?

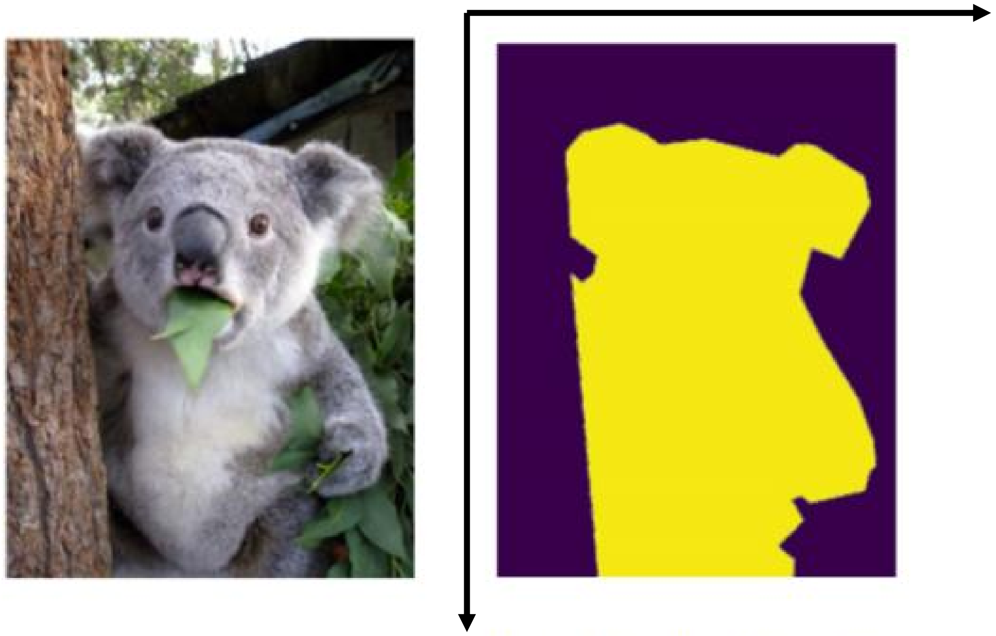
Extracting an object or objects from a medical image.


a)
b)
10 µm
c)
d)
e)
500 nm
f)
g)
h)
i)
j)
k)
l)
20 µm

How to represent an object?

Key assumptions of an object

Pixel
Representation

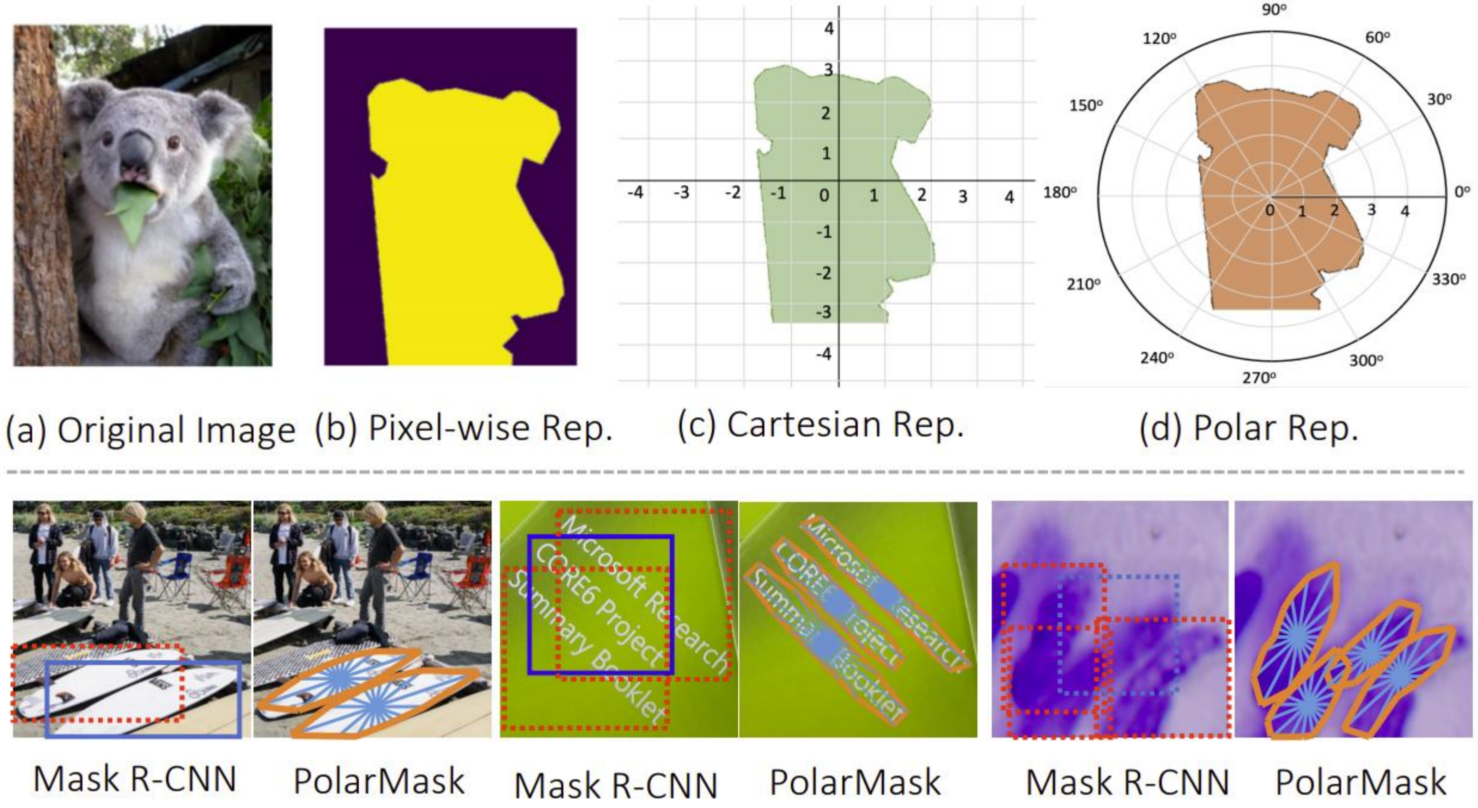


(a) Original Image (b) Pixel-wise Rep.

Key assumptions of an object

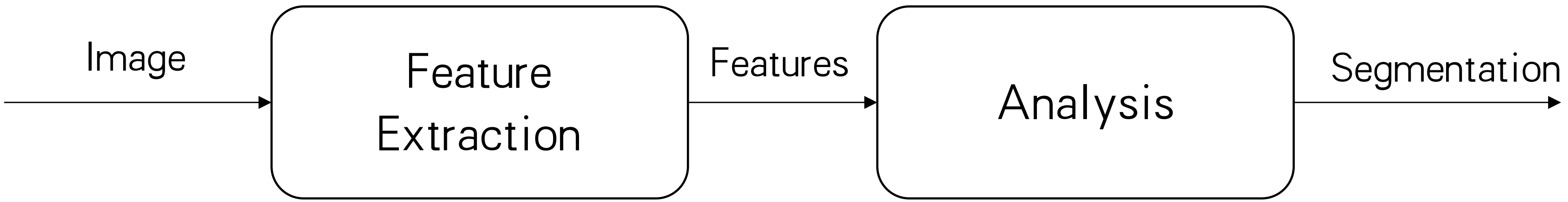
- Pixels belonging to an object **share the same semantic meaning** -> similar intensity, appearance, feature, etc.
- Pixels belonging to an object typically stay together
- The boundary of an object is rather visible
- The boundary of an object is typically smooth
- The object lives in a certain functional or to-be-defined space

Polar
Representation



Xie et al. PolarMask++: Enhanced Polar Representation for Single-Shot Instance Segmentation and Beyond. TPAMI 2021

A common pipeline



- Feature crafting
- Feature selection
- Feature learning

- Supervised classification
- Unsupervised clustering
- Iterative optimization (active)
- Graph theory

Image segmentation methods

- **Thresholding and Ostu's Method**
- Watershed Transform
- **Region Growing**
- **Clustering**
- **Mean Shift**

- Active Contour
- Variational or **level set** methods
- Active Shape Model & Active Appearance Model (AAM)

- Normalized Cut
- Graph Cut
- **Random walker**

- Multi–atlas

- Neural network

Thresholding

Thresholding and Ostu's method

Thresholding

- m(x,y)= T[I(x,y)>t], where t is a threshold.
- Key： how to determine the threshold t?
- Variant： m(x,y)= T[F(I(x,y))>t]

Otsu method

- based on a minimum variation

$$\sigma^2(t) = \omega_{bg}(t)\sigma_{bg}^2(t) + \omega_{fg}(t)\sigma_{fg}^2(t)$$

$$\omega_{bg}(t) = \frac{P_{BG}(t)}{P_{all}}$$

$$\omega_{fg}(t) = \frac{P_{FG}(t)}{P_{all}}$$

$$\sigma^2(t) = \frac{\sum(x_i-\overline{x})^2}{N-1}$$

Region Growing

Region growing

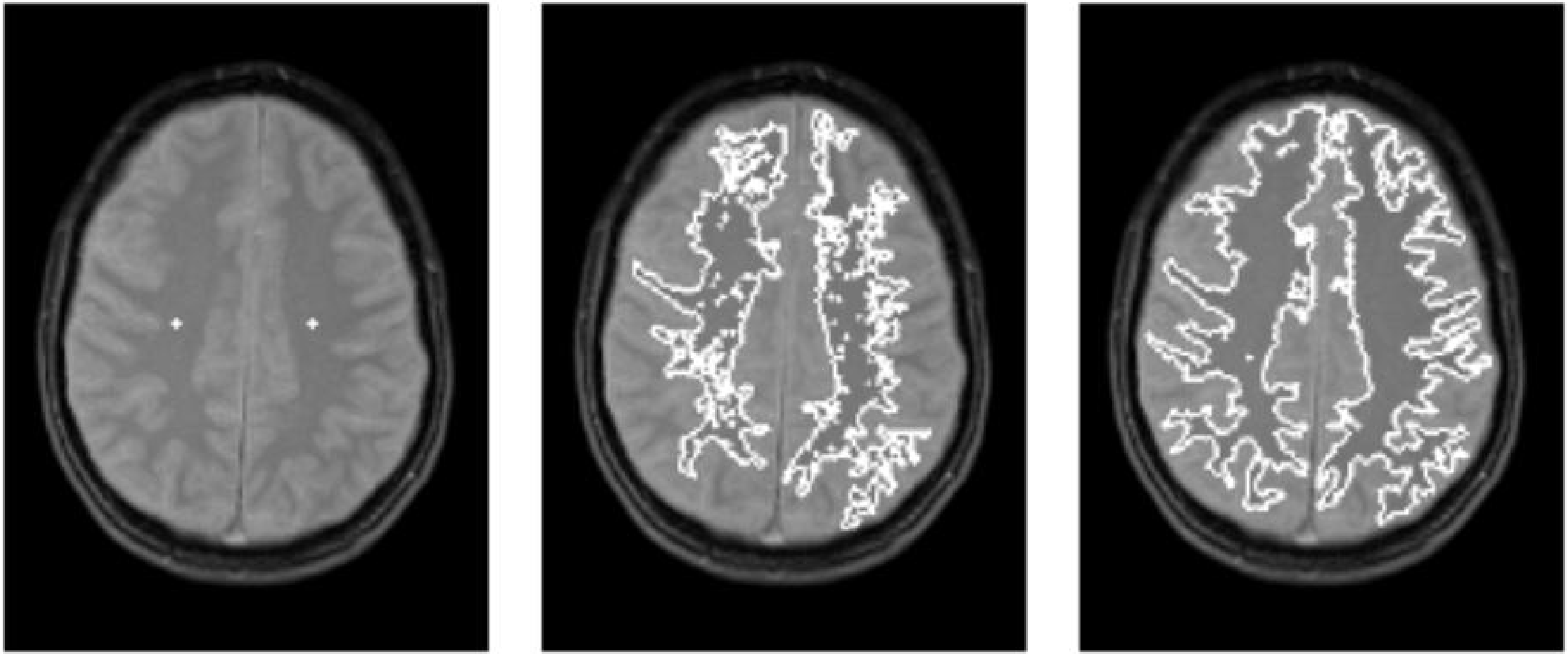
- The region is iteratively grown by comparing all unallocated neighboring pixels to the region.

<u>Procedure</u>

- Start from a single seed
- For an **unallocated neighboring** pixel **p**, compute its similarity **s[p]** with the current region
- Add the pixel with the largest similarity (bigger than a stopping threshold) into the region.

<u>Keys</u>

- Defining neighbor
- Defining similarity
- Defining stopping threshold

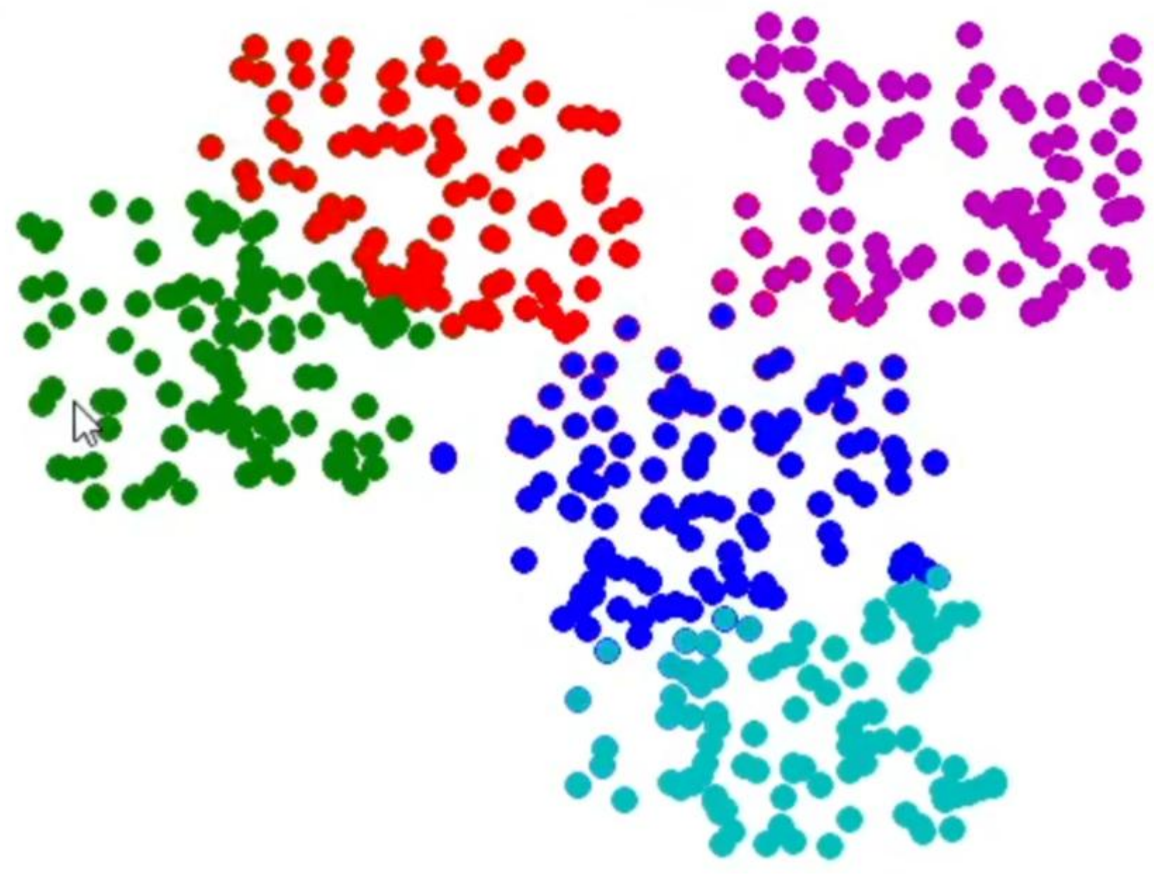


Clustering

K-means

Given # of clusters K

1. Initialize K cluster centers
2. Assign each data point to its nearest center
3. Update the cluster center using the newly assigned data
4. Repeat 2&3 till convergence



Pros and cons of K-means

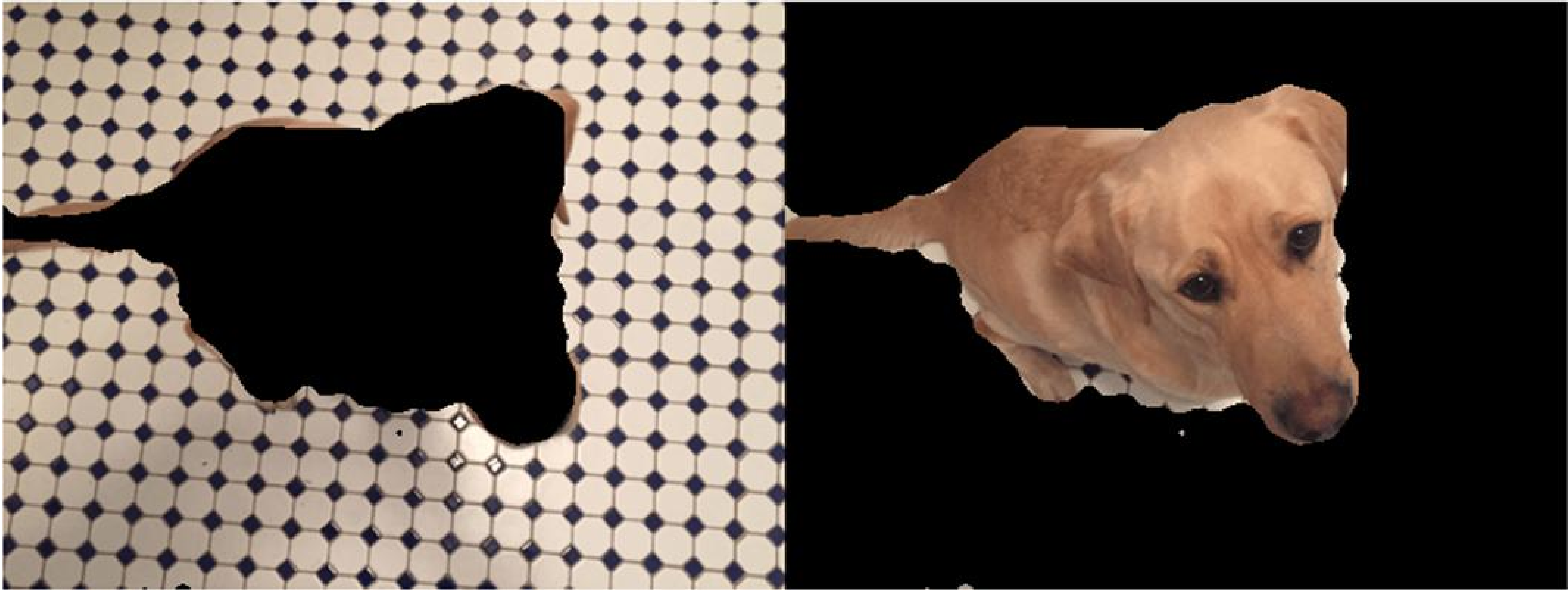
Pros

- Normally good performance
- Good scalability to big data
- Low computation complexity

Cons

- Need to know K
- Sensible to initial centers
- Sensible to outliers
- Less appropriate to diverse, imbalanced, or non-convex population

Gabor filter, then K-means clustering

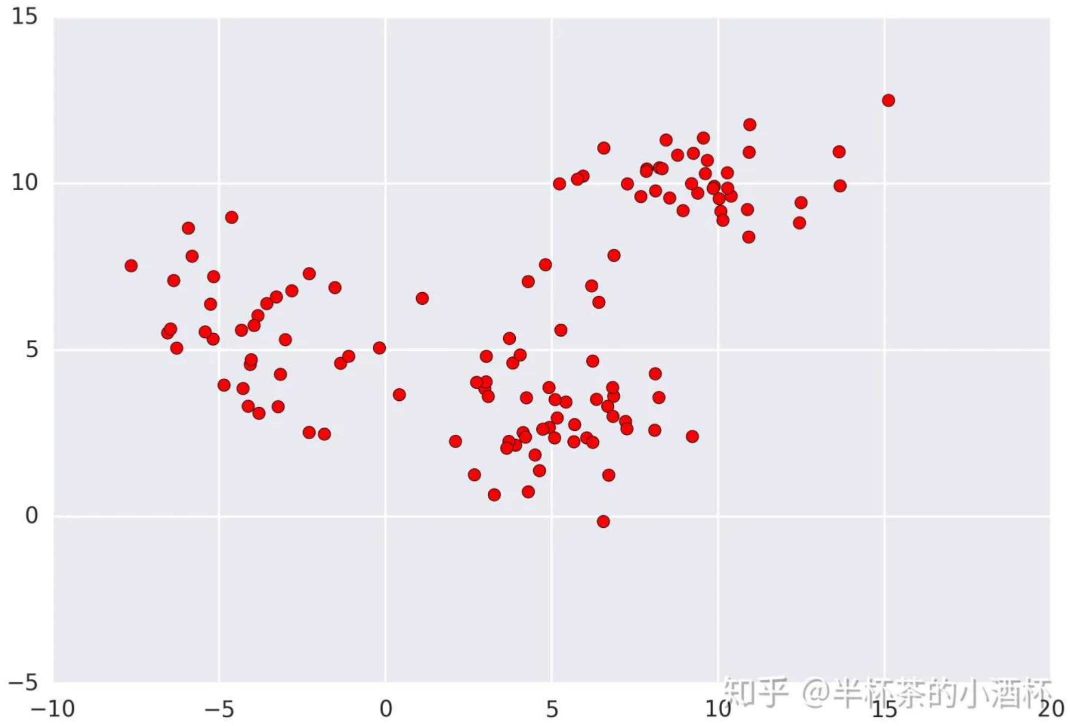


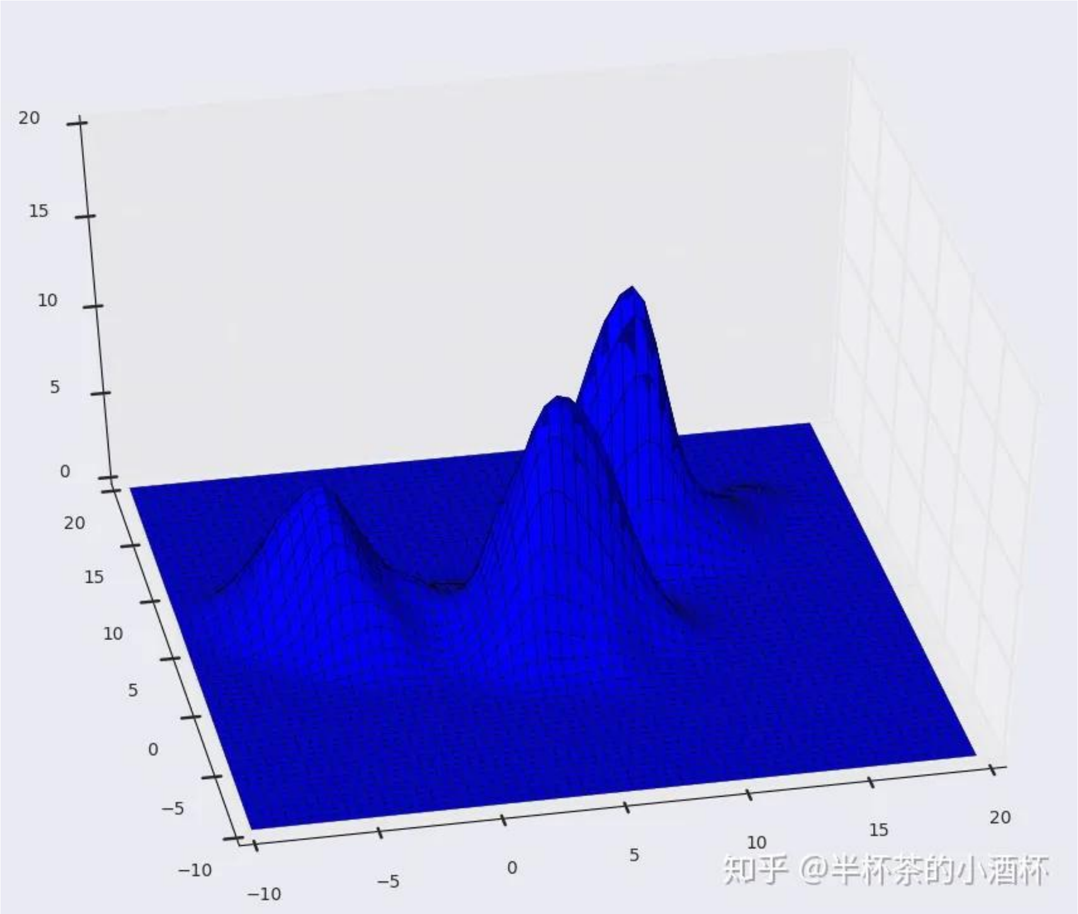
Jain, Anil K., and Farshid Farrokhnia. "Unsupervised Texture Segmentation Using Gabor Filters." Pattern Recognition 24, no. 12 (January 1991): 1167–86.
https://ww2.mathworks.cn/help/images/texture-segmentation-using-gabor-filters.html

Mean Shift

Comaniciu and Meer. Mean shift: a robust approach toward feature space analysis. IEEE TPAMI 2002.
https://zhuanlan.zhihu.com/p/81629406

Kernel density estimation





$$\hat{f}_K = \frac{1}{nh^d} \sum_i^n K(\frac{x - x_i}{h}) \quad K(x) = c_k k(||x||^2)$$

Mean shift algorithm

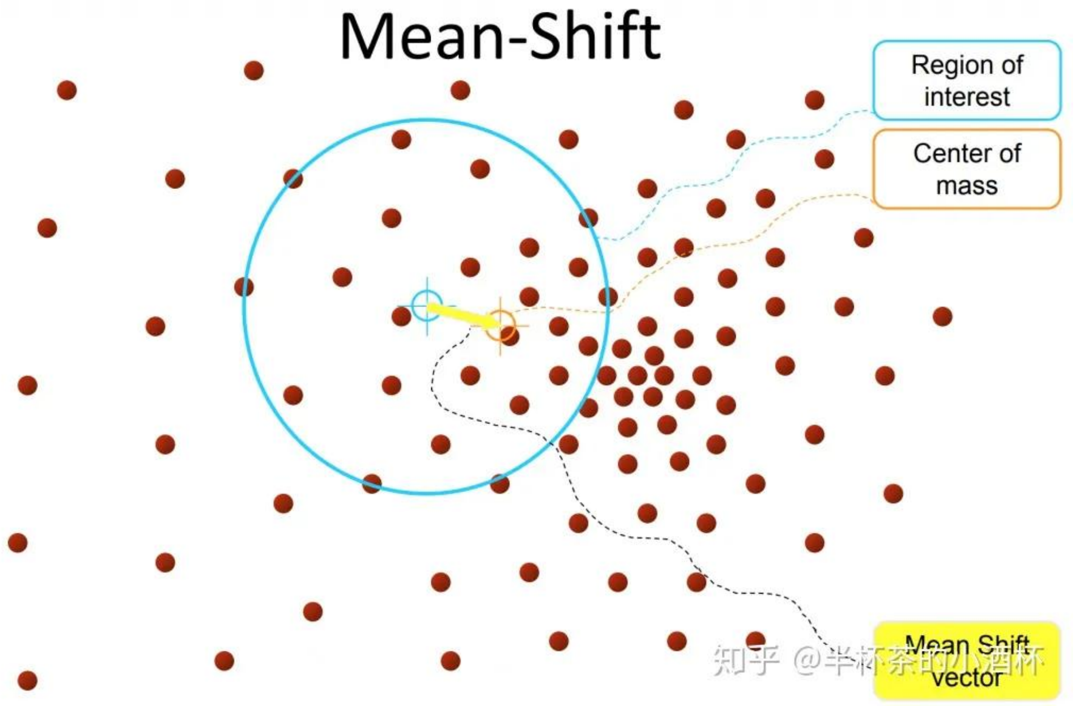
1. For a given point x_i, compute the mean shift

$$m(x_i) = \frac{\sum_{i=1}^{n} x_i g(||\frac{x-x_i}{h}||)^2}{\sum_{i=1}^{n} g(||\frac{x-x_i}{h}||)^2} - x \qquad g(x) = -k'(x)$$

2. Translate the density estimation window.

$$x_i^{t+1} = x_i^t + m(x_i^t)$$

3. Repeat 1&2 until convergence



Mumford—Shah Variational Method & Level Set

Mumford, D., Shah, J.: Optimal approximation by piecewise smooth functions and associated variational problems. Comm. Pure Appl. Math. 42 (1989) 577—685.
Osher, S., Sethian, J. A.: Fronts Propagating with Curvature—Dependent Speed: Algorithms Based on Hamilton—Jacobi Formulation. Journal of Computational Physics 79 (1988) 12—49.
Chan and Vese. Active contours without edges. IEEE Trans. Image Processing. 2001.

Mumford–Shah region partition

$$E(u,C) = \frac{1}{2}\int\limits_{\Omega}(u-f)^2 + \frac{\alpha}{2}\int\limits_{\Omega\backslash C}|\nabla u|^2 + \lambda\Gamma(C)$$

Variational approach
Piece–wise approximation

$$\mathcal{L}_{MS}(\chi;u) = \mathcal{E}(\chi,u) + \lambda\sum_{n=1}^{N}\int_{\Omega}|\nabla\chi_n(\mathbf{r})|d\mathbf{r}$$

$$\mathcal{E}(\chi,u) = \sum_{n=1}^{N}\int_{\Omega}|u(\mathbf{r}) - c_n|^2\chi_n(\mathbf{r})d\mathbf{r}$$

$$\sum_{n=1}^{N}\chi_n(\mathbf{r}) = 1, \quad \forall\mathbf{r}\in\Omega$$

$$c_n = \frac{\int_{\Omega}u(\mathbf{r})\chi_n(\mathbf{r})d\mathbf{r}}{\int_{\Omega}\chi_n(\mathbf{r})d\mathbf{r}}$$

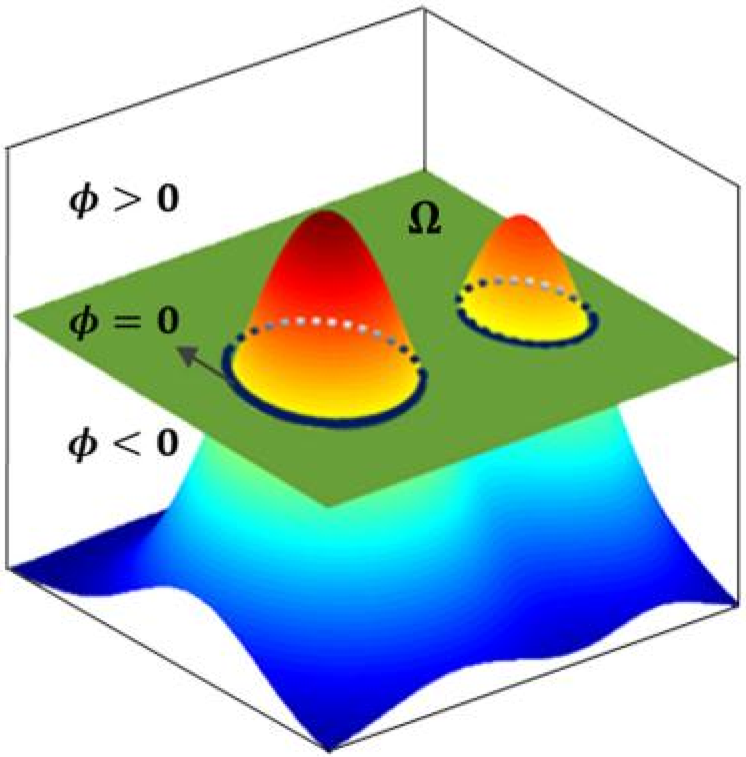
Multiple regions

Level set

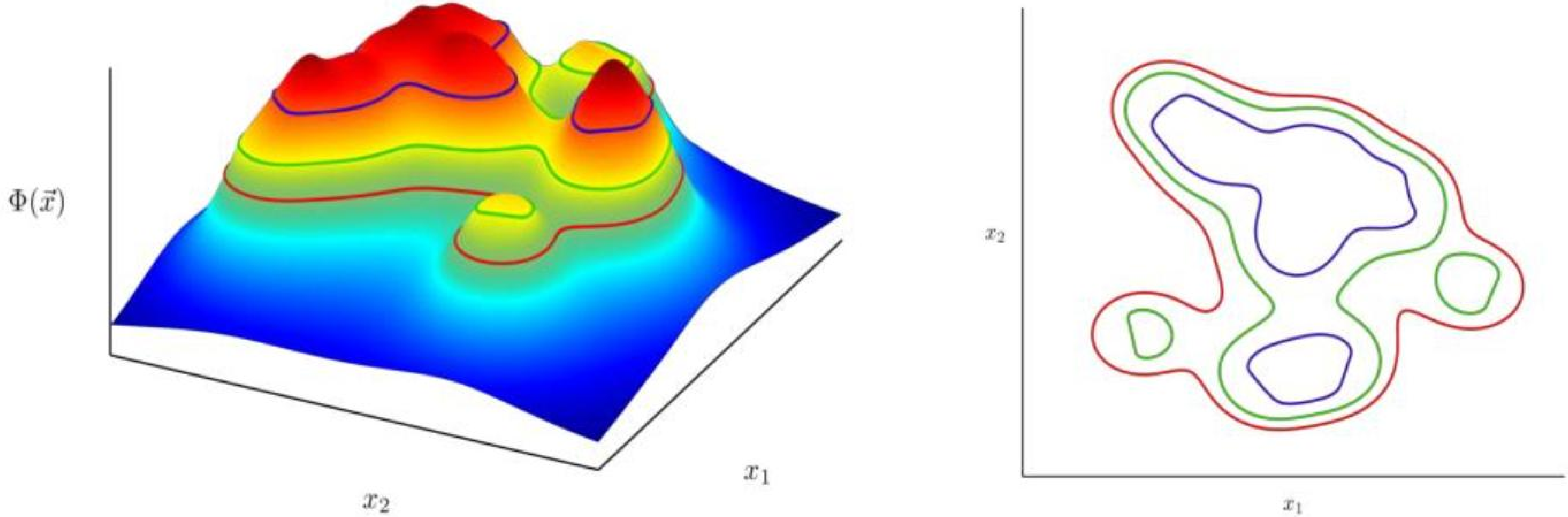
$$\chi_n = H(\Phi) = \begin{bmatrix} H(\phi_1) & \cdots & H(\phi_p) \end{bmatrix}$$

Heaviside function $H(\phi)=\pi[\phi \geqslant 0]$

$$\delta(x) = \frac{d}{dx} H(x)$$



$$H_{2,\varepsilon}(x) = \frac{1}{2}\left(1 + \frac{2}{\pi}\arctan\left(\frac{x}{\varepsilon}\right)\right), \ \delta_{2,\varepsilon}(x) = H'_{2,\varepsilon}(x) = \frac{1}{\pi} \cdot \frac{\varepsilon}{\varepsilon^2 + x^2}.$$



Able to represent any topology

Optimization or evolution

$$F(\phi, c_1, c_2) = \mu \cdot \text{length}\{\phi = 0\} + \nu \cdot \text{area}\{\phi \geq 0\}$$
$$+ \lambda_1 \int_{\phi \geq 0} |u_0 - c_1|^2 dxdy + \lambda_2 \int_{\phi < 0} |u_0 - c_2|^2 dxdy.$$

$$\text{length}\{\phi = 0\} = \int_\Omega |\nabla H(\phi)| = \int_\Omega \delta(\phi)|\nabla \phi|, \quad \int_{\phi \geq 0} |u_0 - c_1|^2 dxdy = \int_\Omega |u_0 - c_1|^2 H(\phi) dxdy$$
$$\text{area}\{\phi \geq 0\} = \int_\Omega H(\phi) dxdy, \quad \int_{\phi < 0} |u_0 - c_2|^2 dxdy = \int_\Omega |u_0 - c_2|^2 (1 - H(\phi)) dxdy$$

$$F(\phi, c_1, c_2) = \mu \int_\Omega \delta(\phi)|\nabla \phi| + \nu \int_\Omega H(\phi) dxdy$$
$$+ \lambda_1 \int_\Omega |u_0 - c_1|^2 H(\phi) dxdy + \lambda_2 \int_\Omega |u_0 - c_2|^2 (1 - H(\phi)) dxdy$$

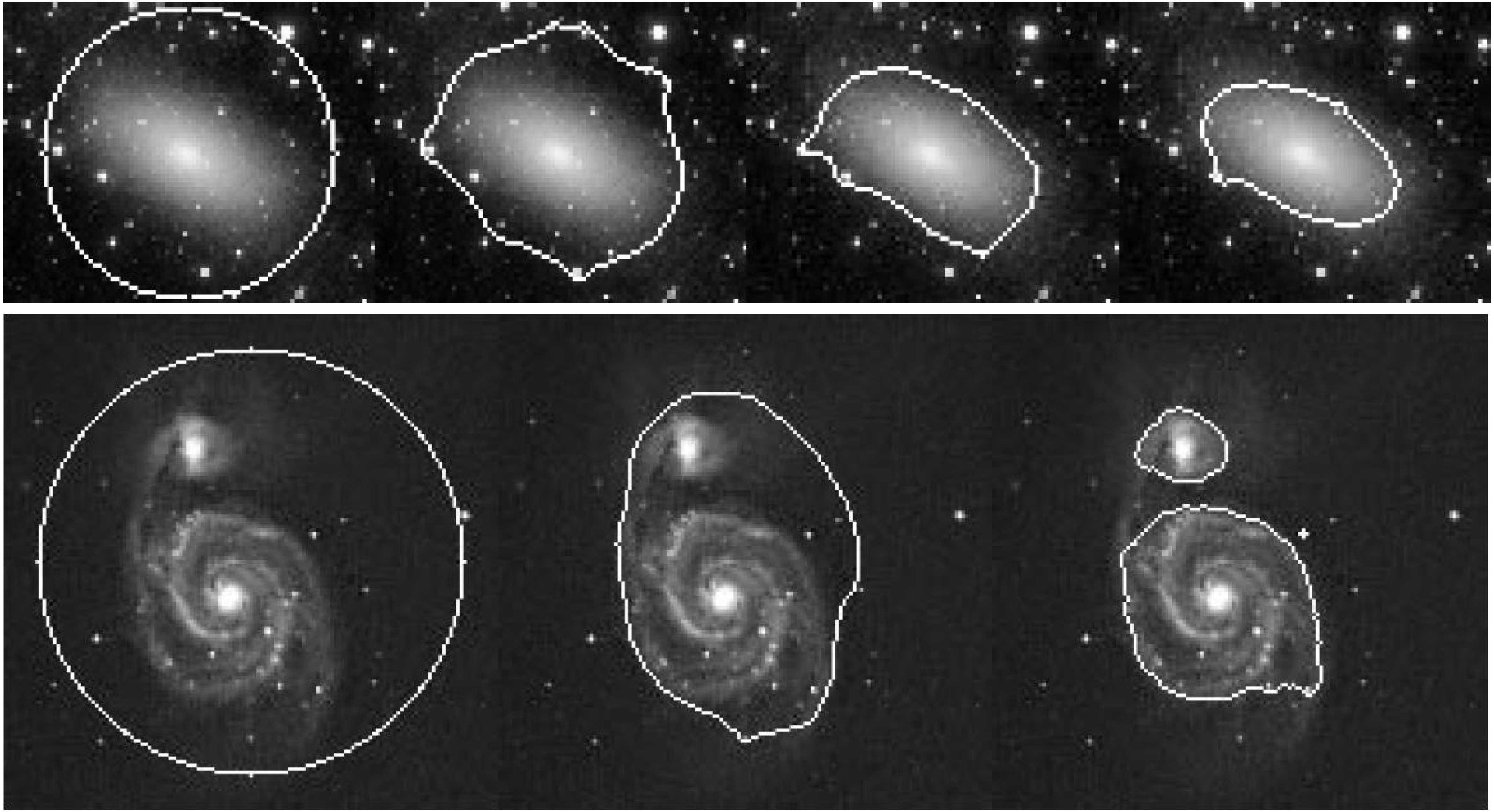
Optimization or evolution

$$c_1(\phi) = \frac{\int_\Omega u_0 H(\phi) dxdy}{\int_\Omega H(\phi(x,y)) dxdy} \qquad \text{(the average of } u_0 \text{ in } \{\phi \geq 0\}\text{)}$$

$$c_2(\phi) = \frac{\int_\Omega u_0 (1 - H(\phi)) dxdy}{\int_\Omega (1 - H(\phi(x,y))) dxdy} \quad \text{(the average of } u_0 \text{ in } \{\phi < 0\}\text{)}$$

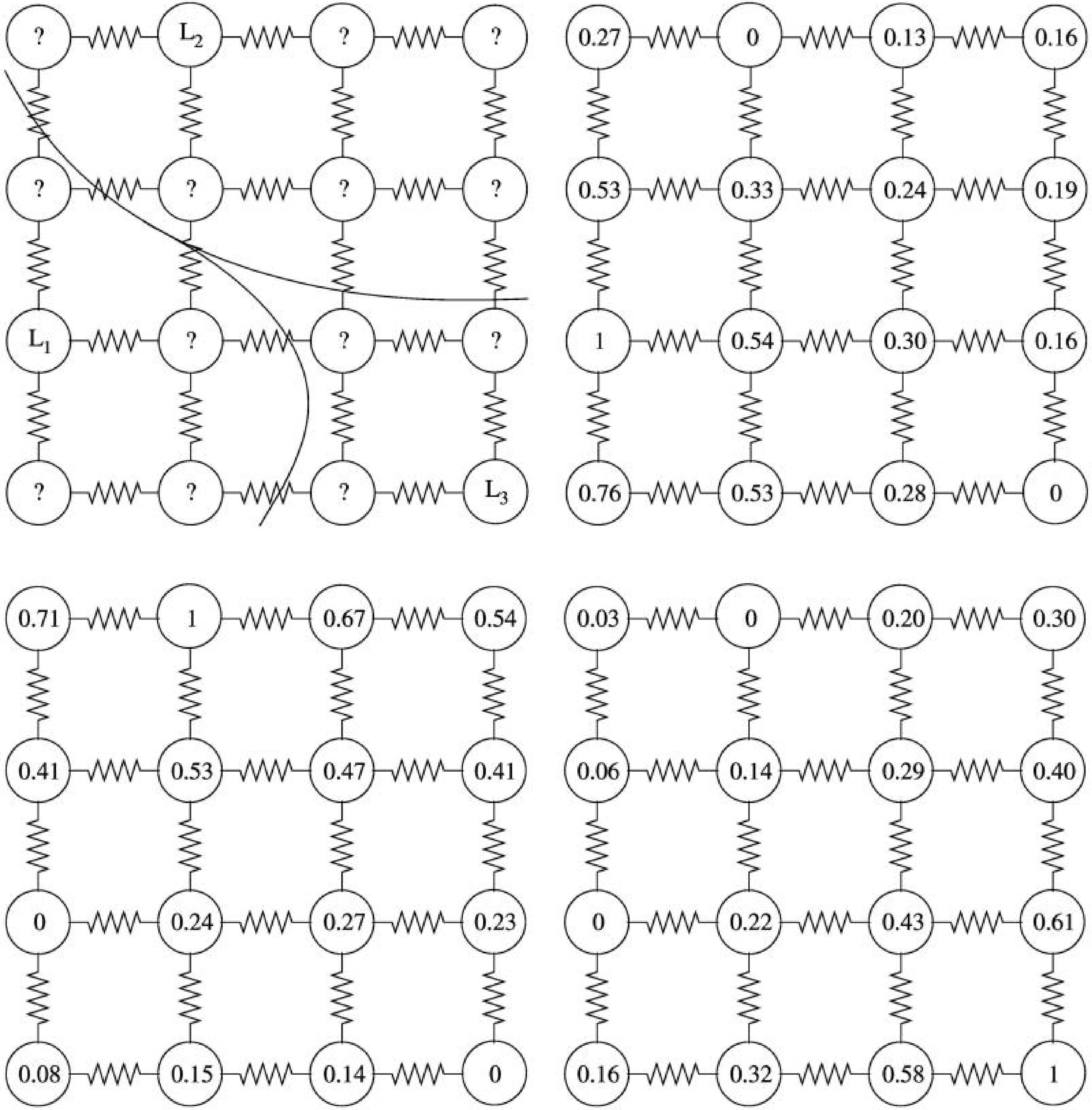
$$\frac{\partial \phi}{\partial t} = \delta(\phi) \left[\mu \mathrm{div} \left(\frac{\nabla \phi}{|\nabla \phi|} \right) - \nu - \lambda_1 (u_0 - c_1)^2 + \lambda_2 (u_0 - c_2)^2 \right] \text{ in } \Omega$$

$$\frac{\delta(\phi)}{|\nabla \phi|} \frac{\partial \phi}{\partial n} = 0 \text{ on } \partial \Omega.$$



Random Walk

L Grady. Random Walks for Image Segmentation. IEEE TPAMI 2006.



? L2 ? ?
? ? ? ?
L1 ? ? ?
? ? ? L3
0.27 0 0.13 0.16
0.53 0.33 0.24 0.19
1 0.54 0.30 0.16
0.76 0.53 0.28 0
0.71 1 0.67 0.54
0.41 0.53 0.47 0.41
0 0.24 0.27 0.23
0.08 0.15 0.14 0
0.03 0 0.20 0.30
0.06 0.14 0.29 0.40
0 0.22 0.43 0.61
0.16 0.32 0.58 1

Algorithm

1. Compute the Laplacian matrix L;
2. Solve the following for potential function
3. Pick the label with the maximum potential

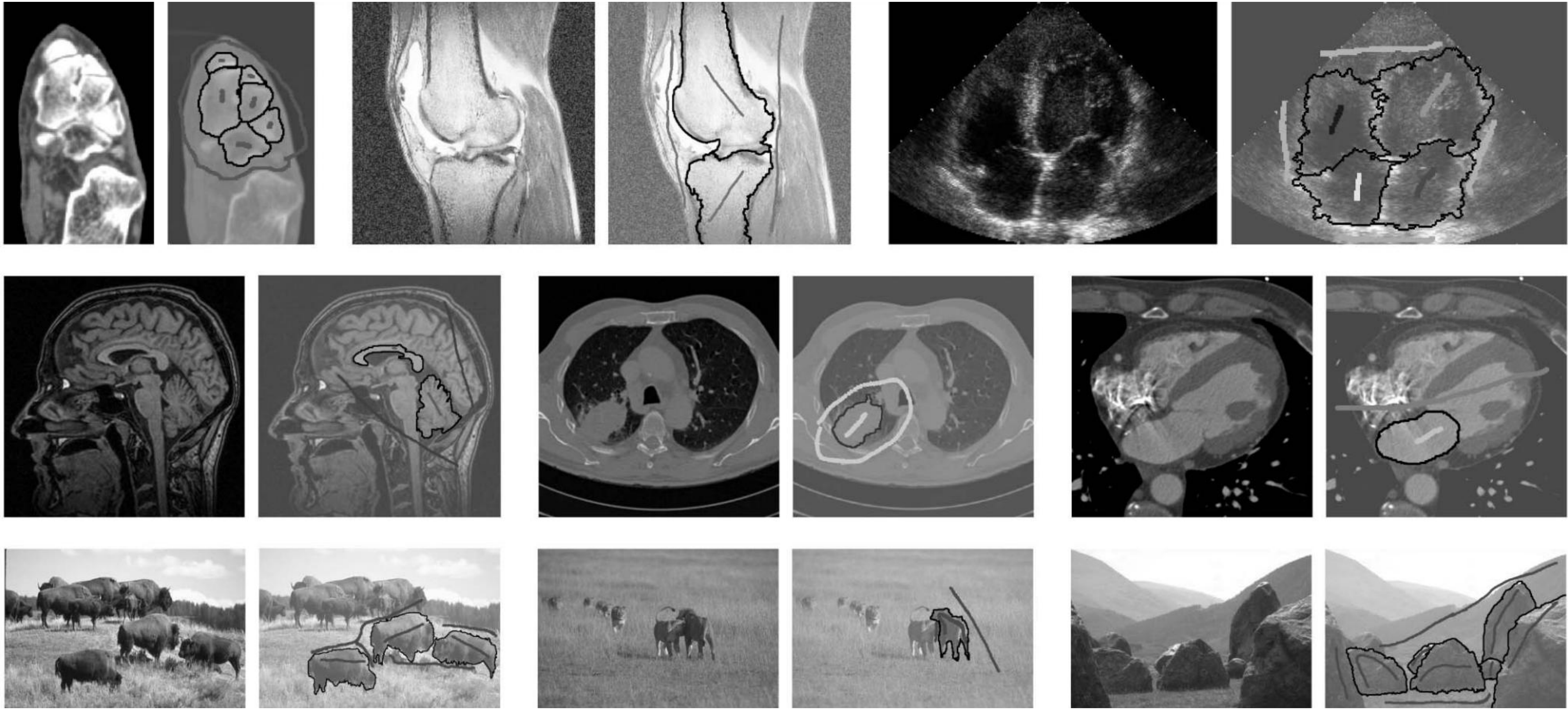
$$D[x] = \frac{1}{2}(Ax)^T C(Ax) = \frac{1}{2}x^T L x = \frac{1}{2}\sum_{e_{ij}\in E} w_{ij}(x_i - x_j)^2,$$

$$D[x_U] = \frac{1}{2}\left[x_M^T x_U^T\right]\begin{bmatrix} L_M & B \\ B^T & L_U \end{bmatrix}\begin{bmatrix} x_M \\ x_U \end{bmatrix}$$

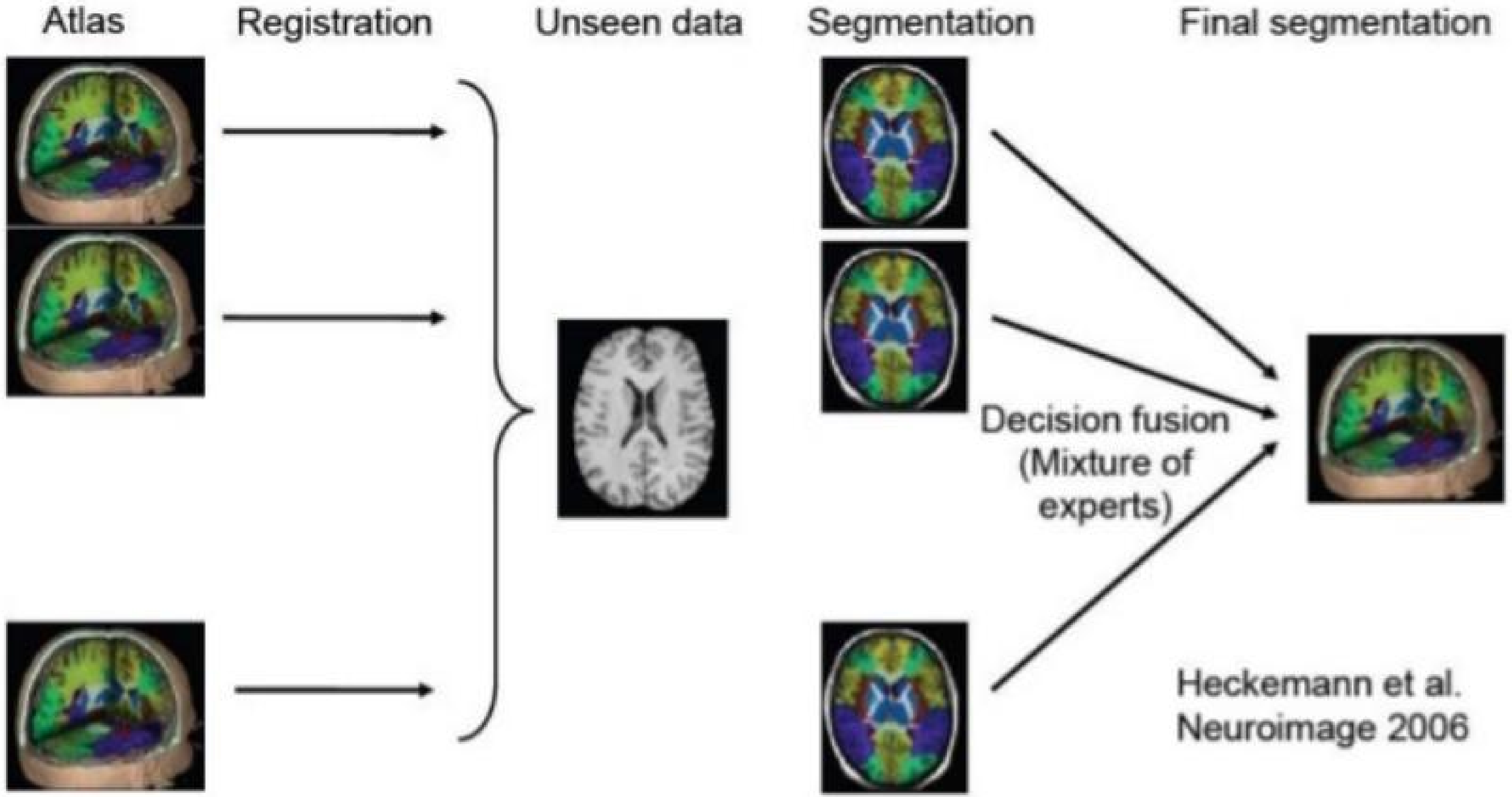
$$= \frac{1}{2}\left(x_M^T L_M x_M + 2x_U^T B^T x_M + x_U^T L_U x_U\right)$$

$$L_U x_U = -B^T x_M$$

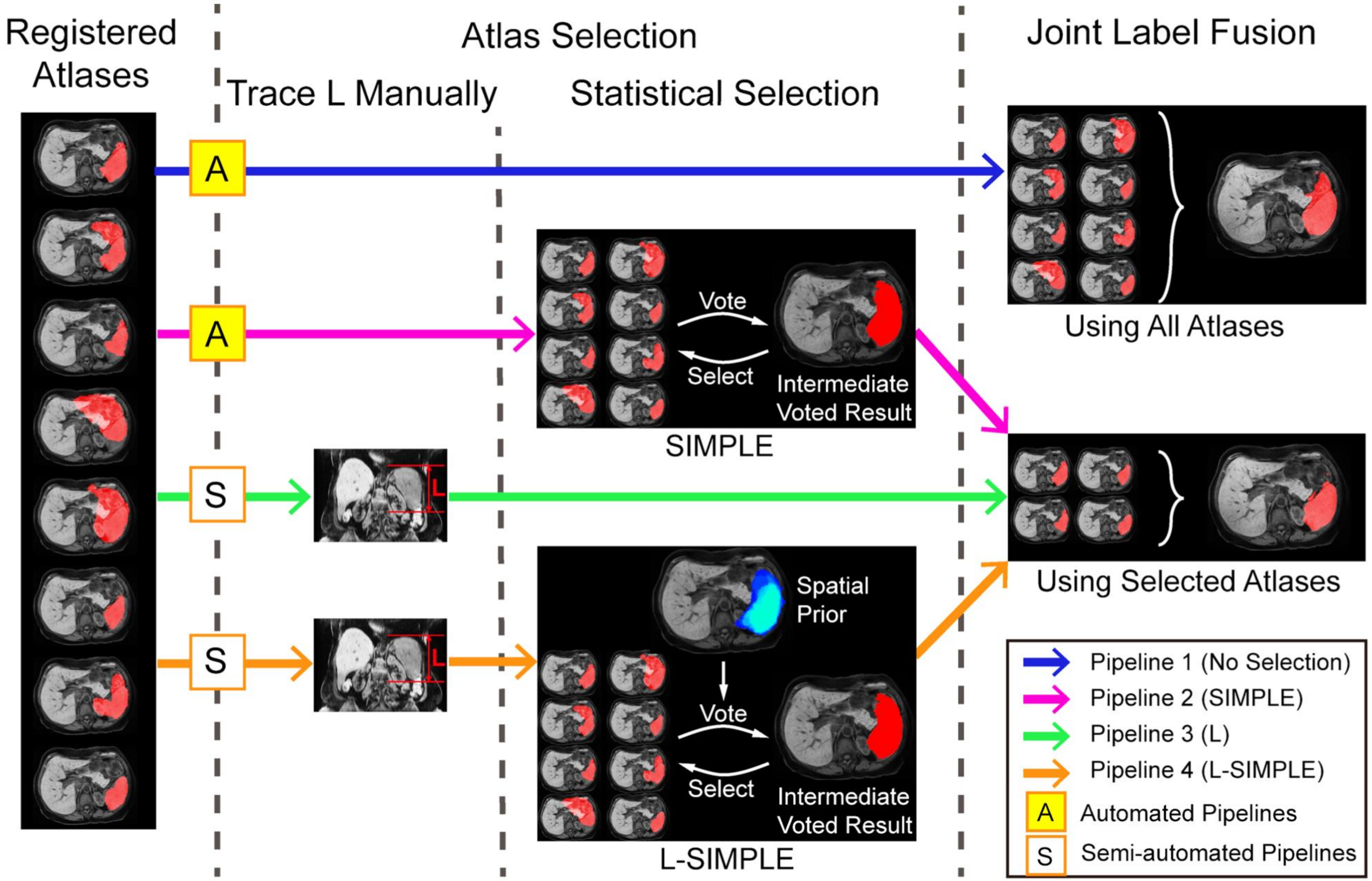
$$L_U x^s = -B^T m^s.$$ m^s is the indicator function for the label s



Multi-Atlas Segmentation (MAS)



https://blog.qure.ai/notes/brain-anatomy-segmentation



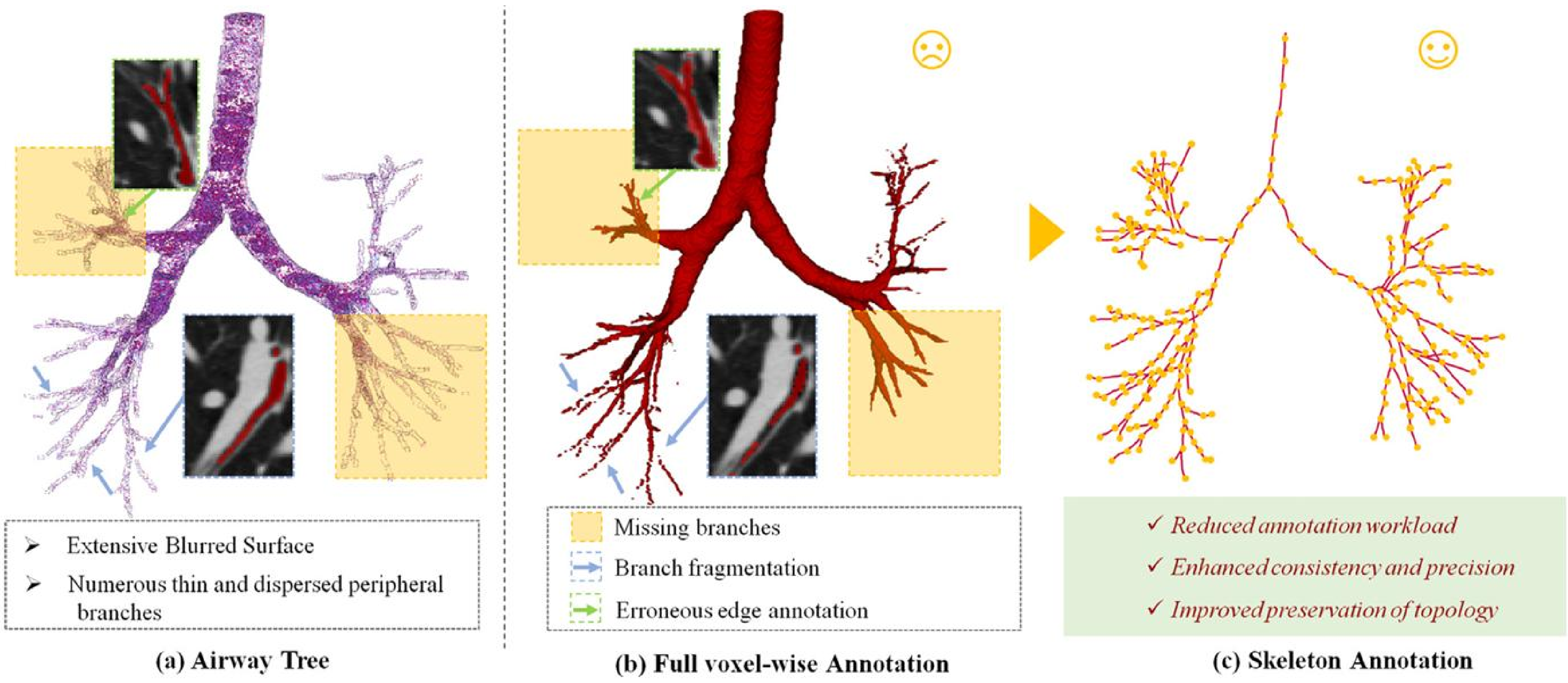
https://my.vanderbilt.edu/masi/category/multi-atlas-segmentation/

?

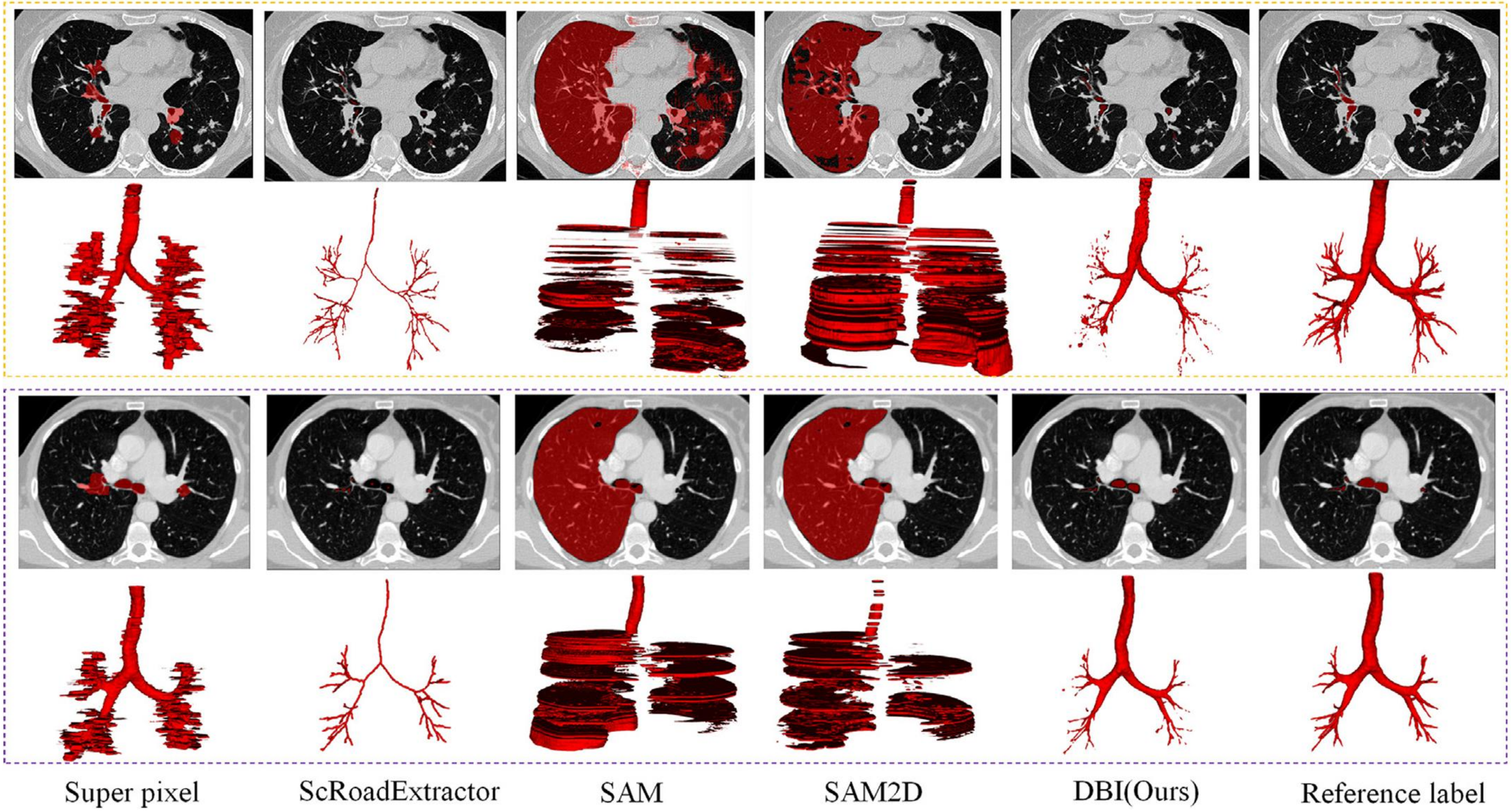
Are traditional segmentation methods still useful in the era of deep learning?

Segmentation with limited supervision/without supervision

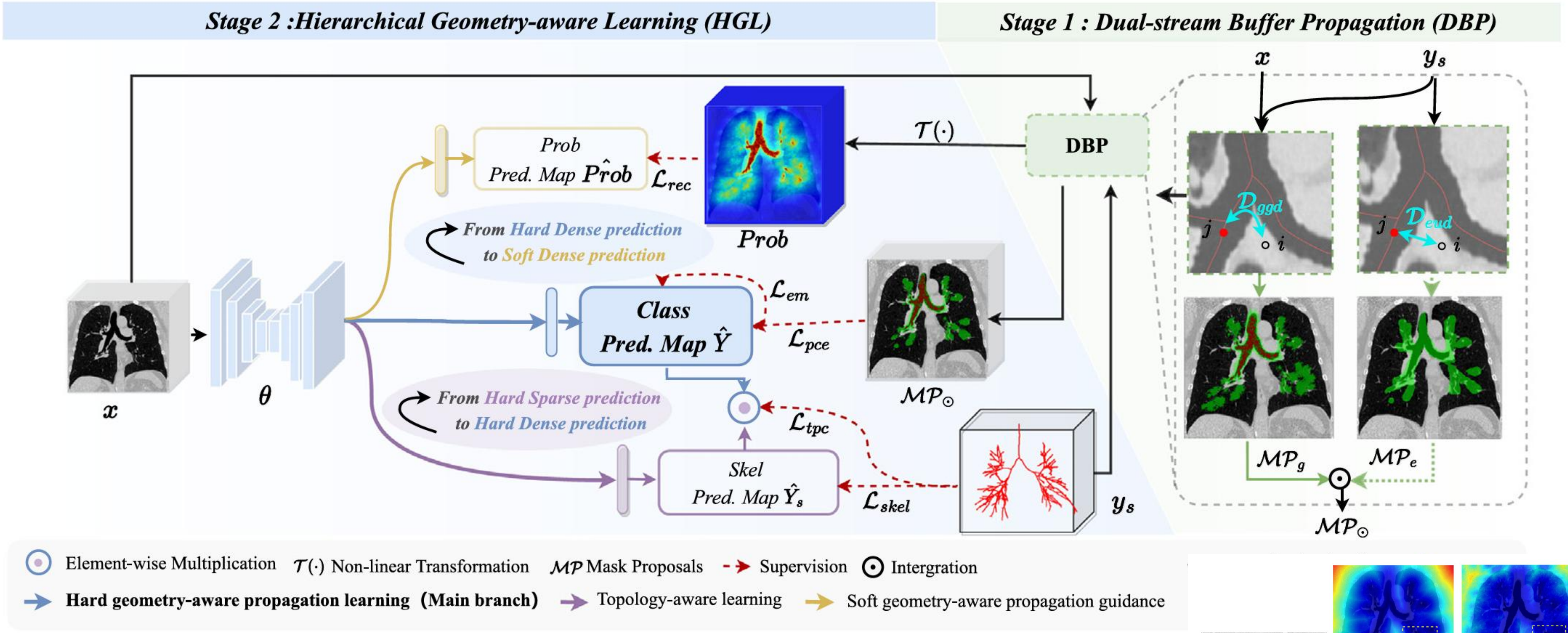
Sparse supervised learning

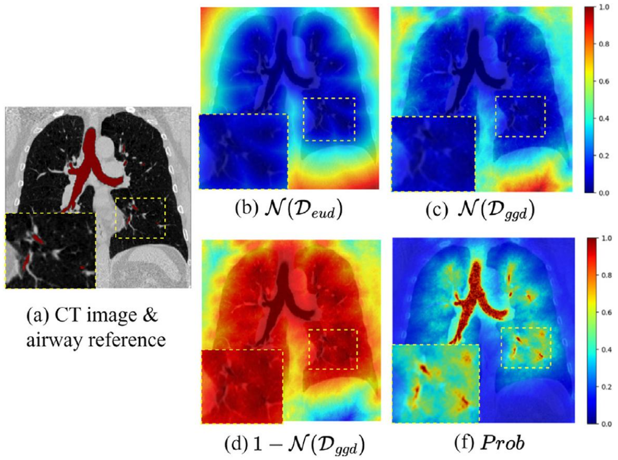


Zhao M, Li H et al. Skeleton2Mask: Skeleton-supervised airway segmentation, Medical Image Analysis,2025.



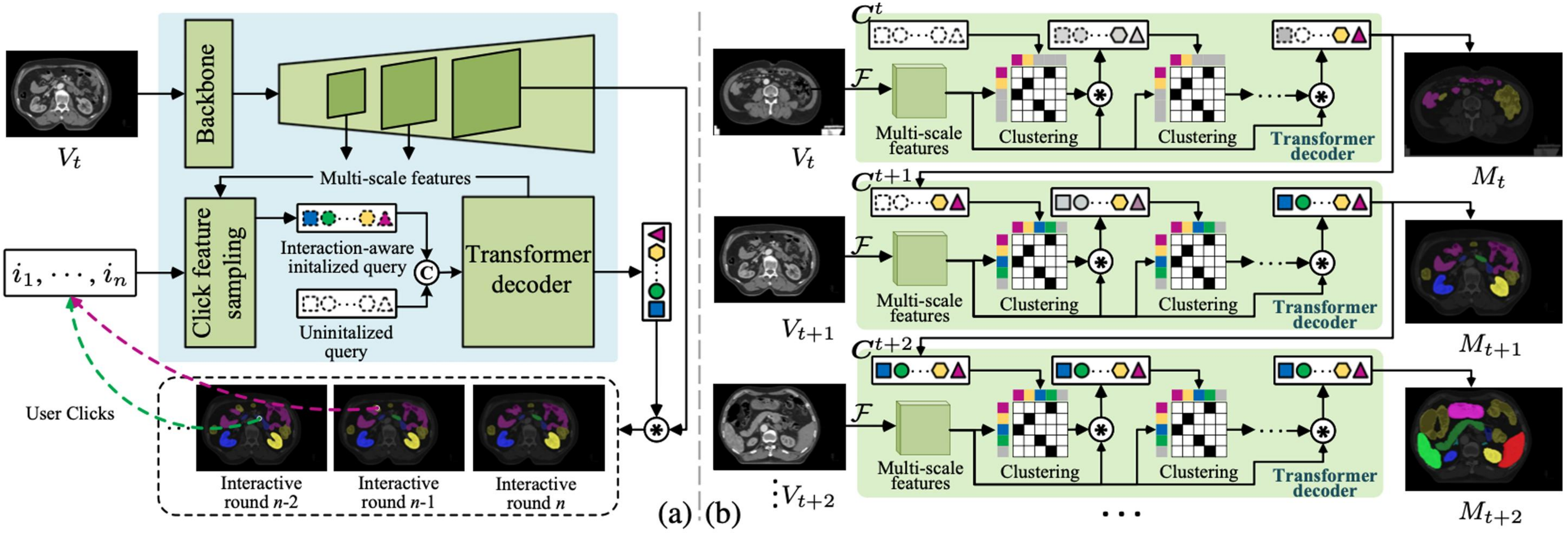
Zhao M, Li H et al. Skeleton2Mask: Skeleton-supervised airway segmentation, Medical Image Analysis,2025.





Zhao M, Li H et al. Skeleton2Mask: Skeleton-supervised airway segmentation, Medical Image Analysis,2025.

Interactive segmentation

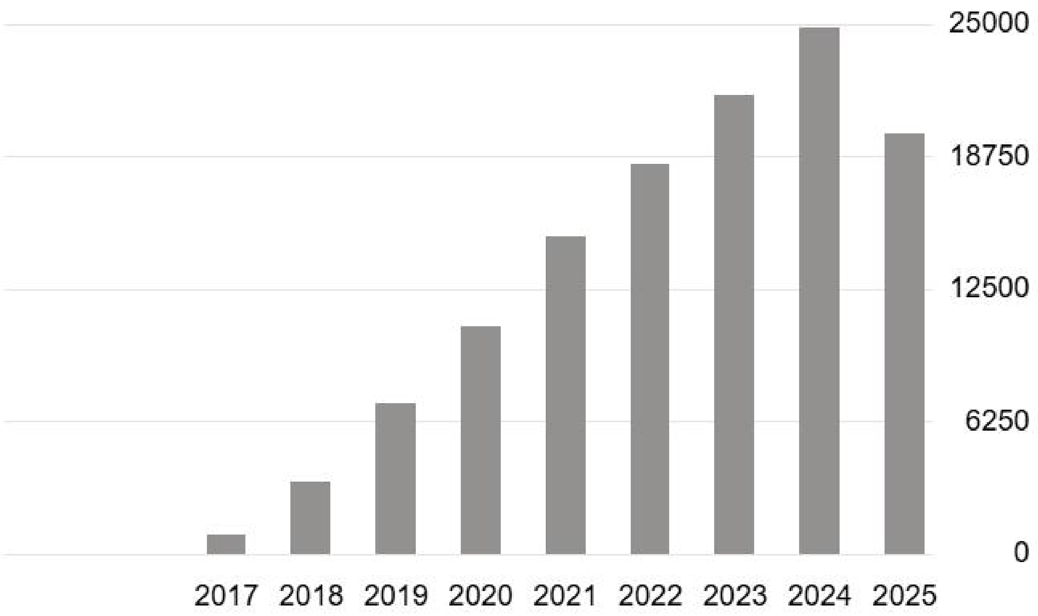


Ding Y, Li L et al. Clustering Propagation for Universal Medical Image Segmentation,CVPR,2025.

Supervised Method: U-Net

Using supervised machine learning

- $m(x,y) = \mathrm{T}[F(I(x,y)) > t]$
- Learn $F(.)$



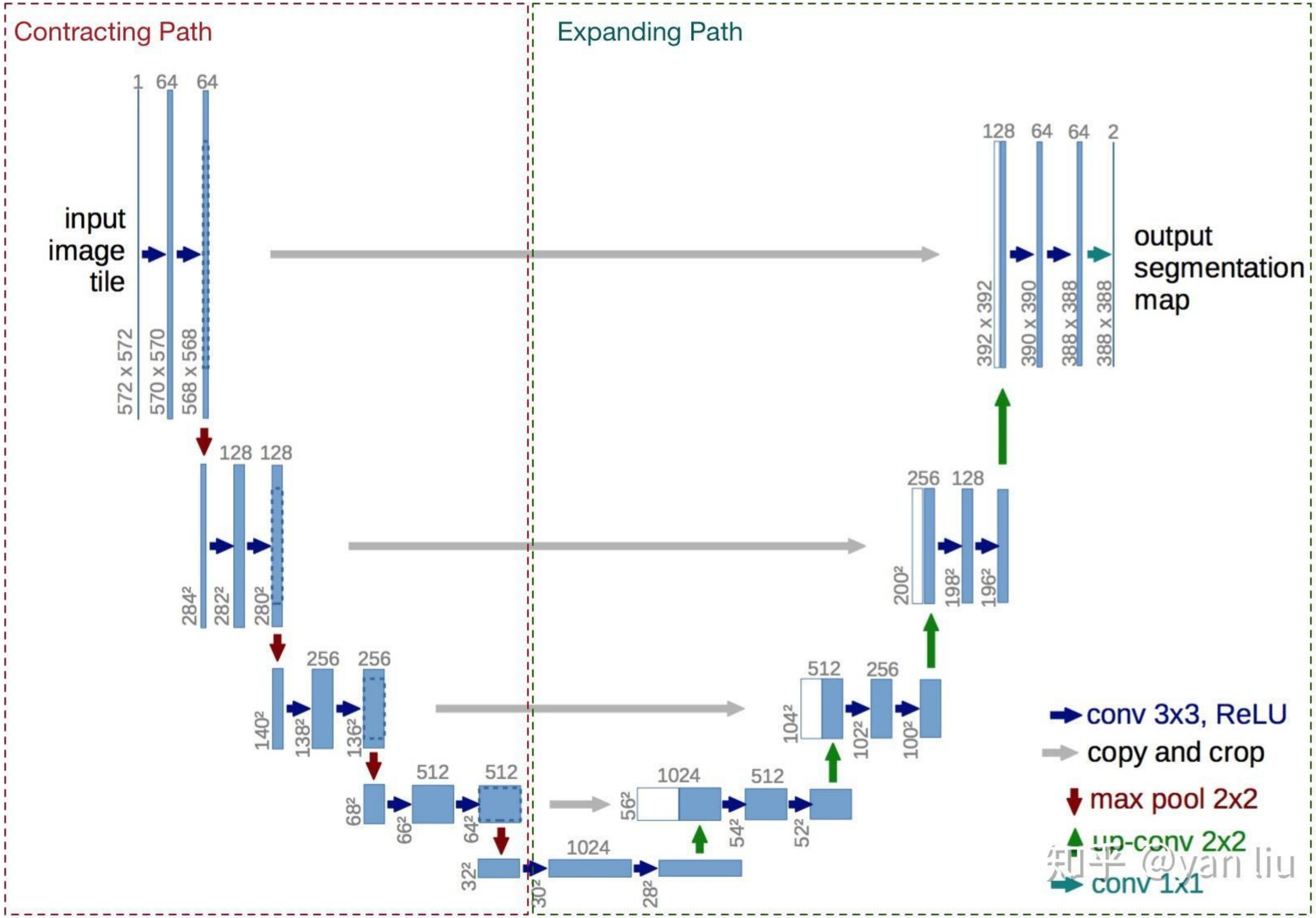
U-net: Convolutional networks for biomedical image segmentation

International Conference On Medical Image Computing And Computer-assisted Intervention

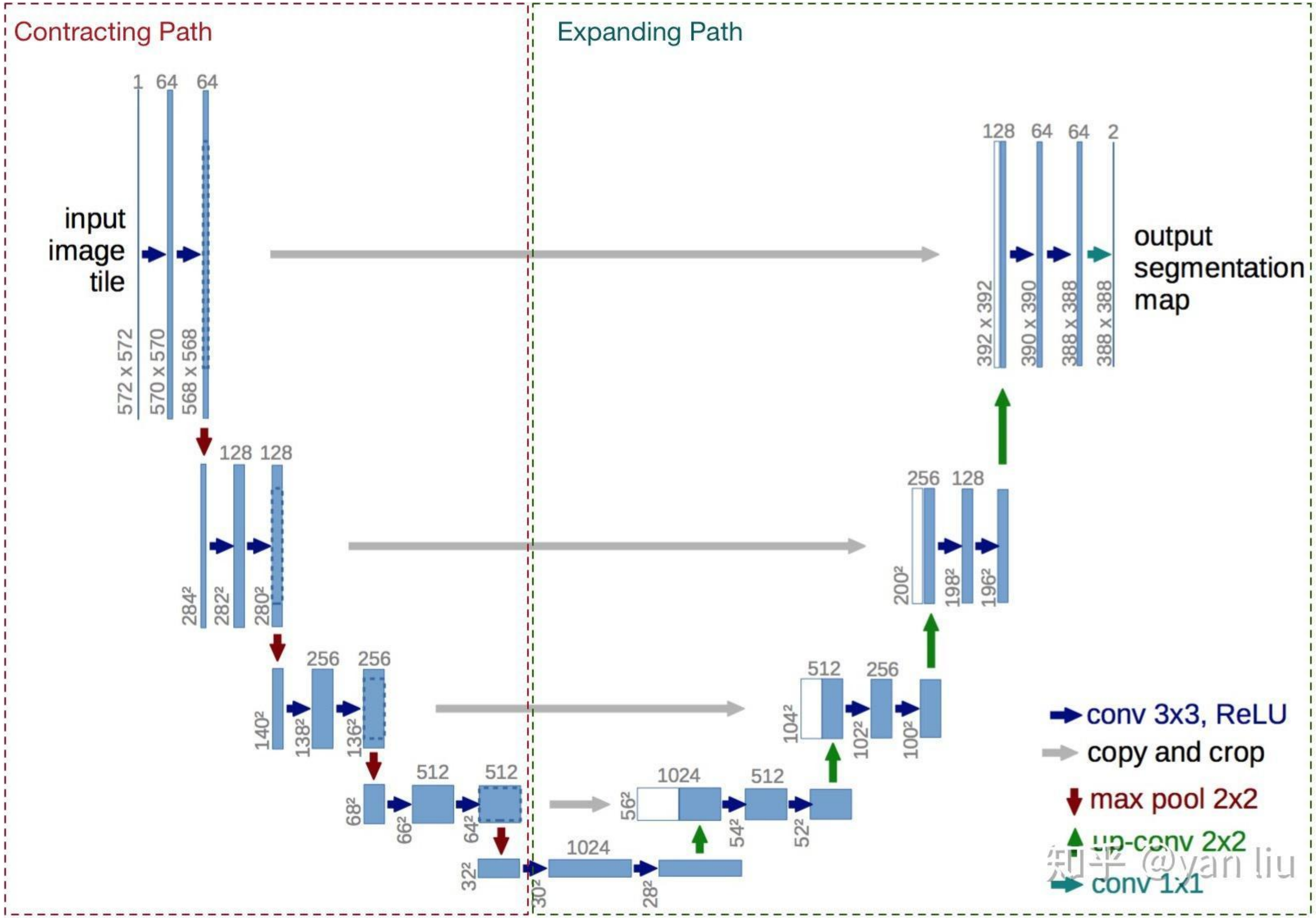
O Ronneberger, P Fischer, T Brox - International Conference on Medical …, 2015 - Springer

… We demonstrate the application of the u-net to three different segmentation tasks. The first … The u-net (averaged over 7 rotated versions of the input data) achieves without any further …

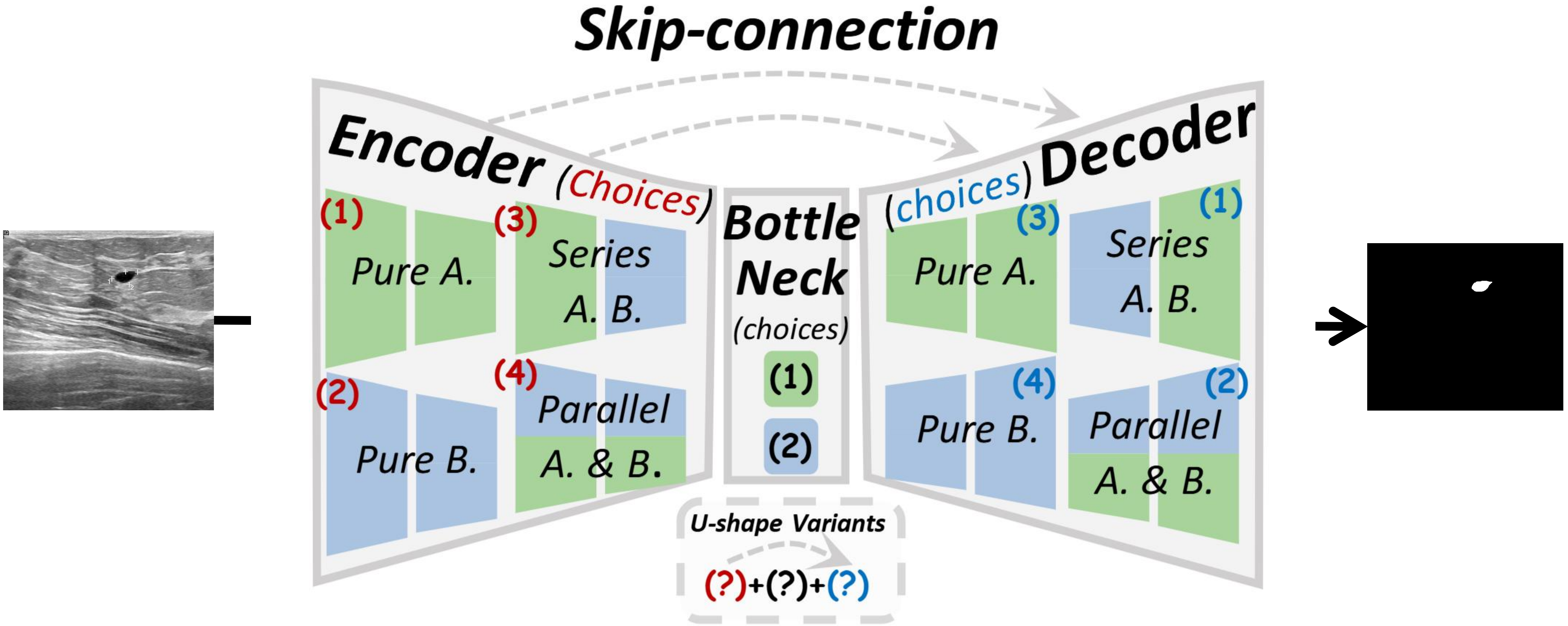
☆ 保存 99 引用 被引用次数：122975 相关文章 所有 28 个版本 easyScholar文献收藏



U-Net: Convolutional Networks for Biomedical Image Segmentation

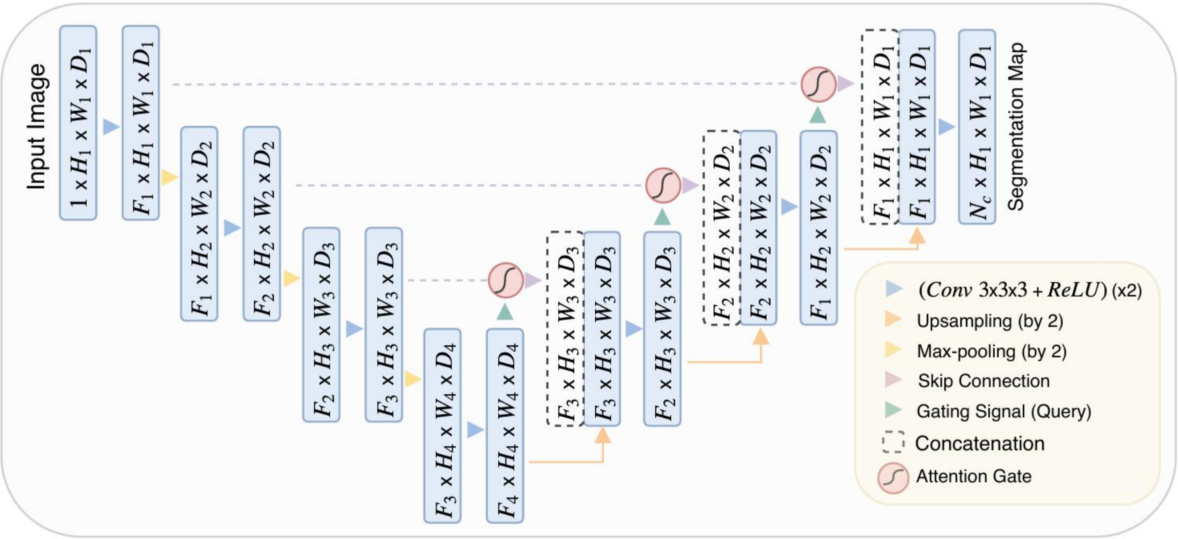


U-Net: Convolutional Networks for Biomedical Image Segmentation

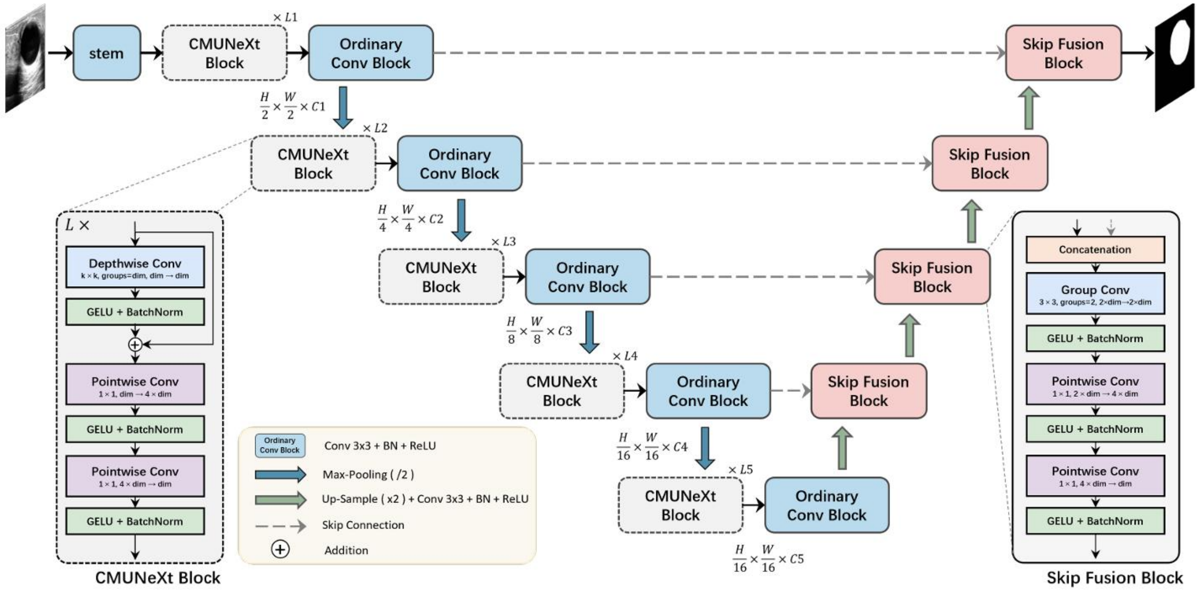


U-Net: Encoder + Decoder + Skip-Connection

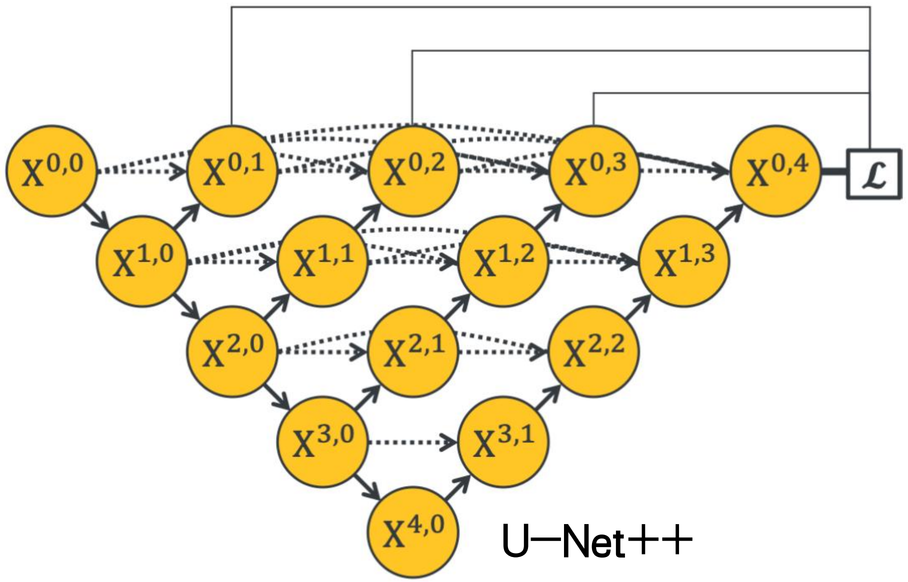
U-Net Variants



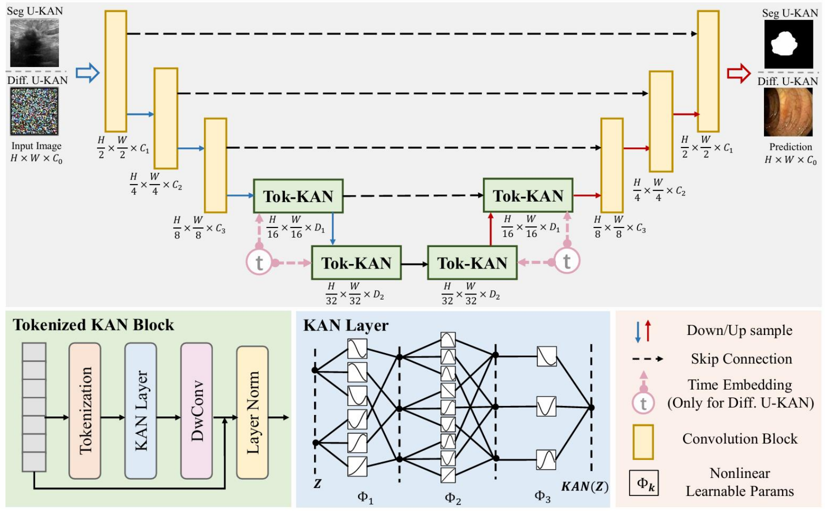
Attention U-Net



CMUNeXt

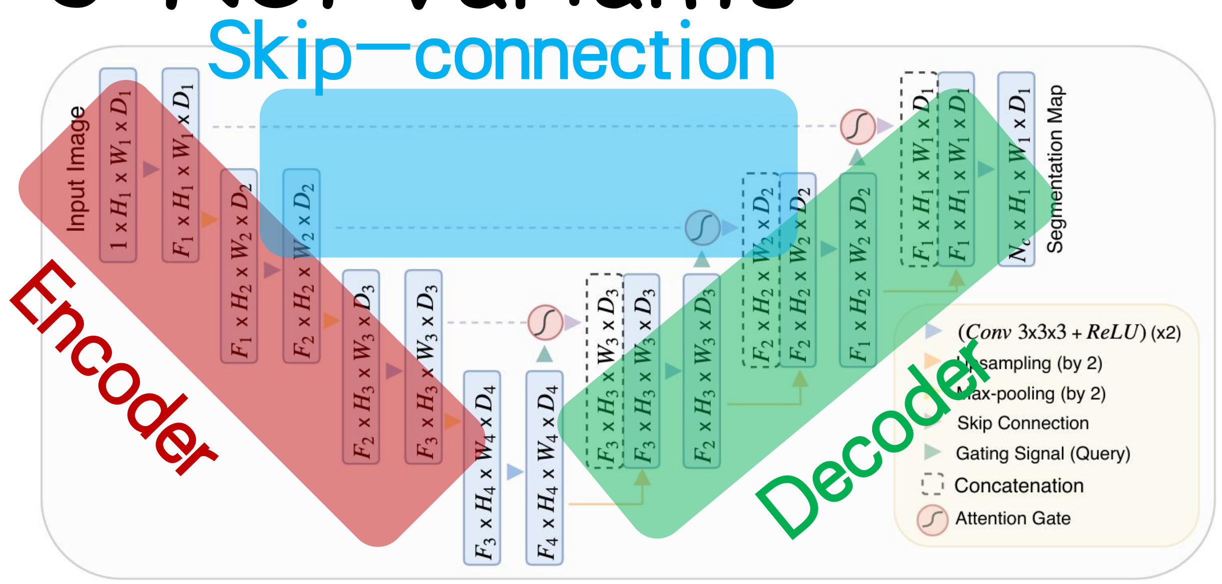


U-Net++

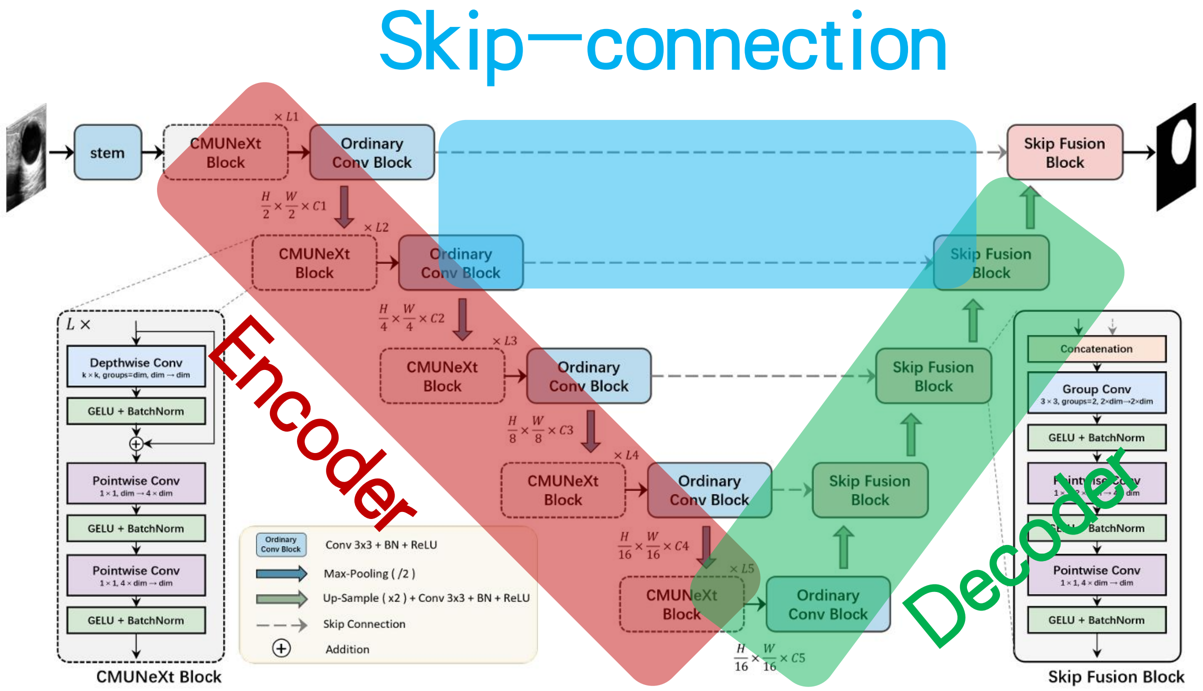


U-KAN

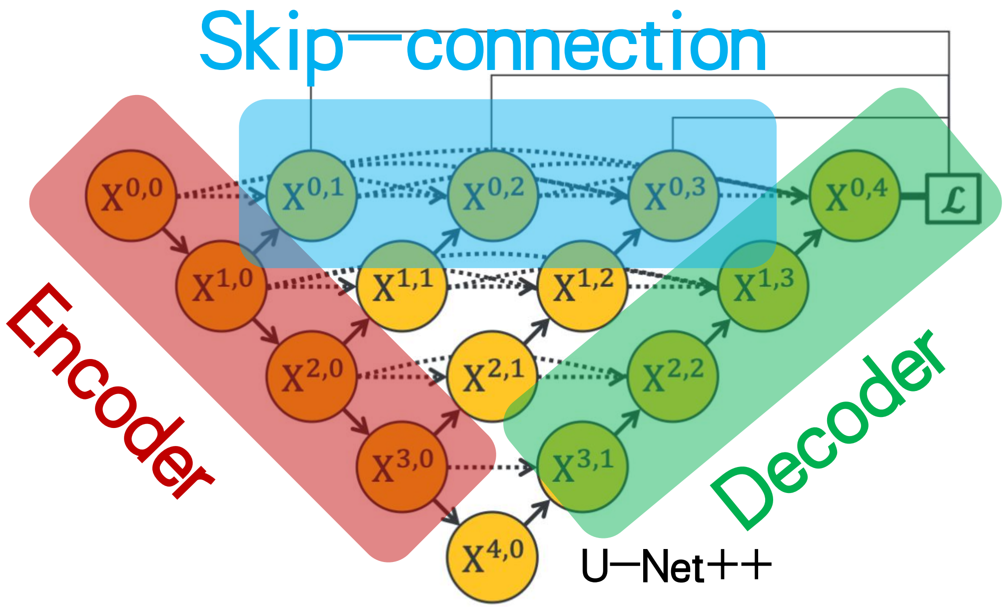
U-Net Variants



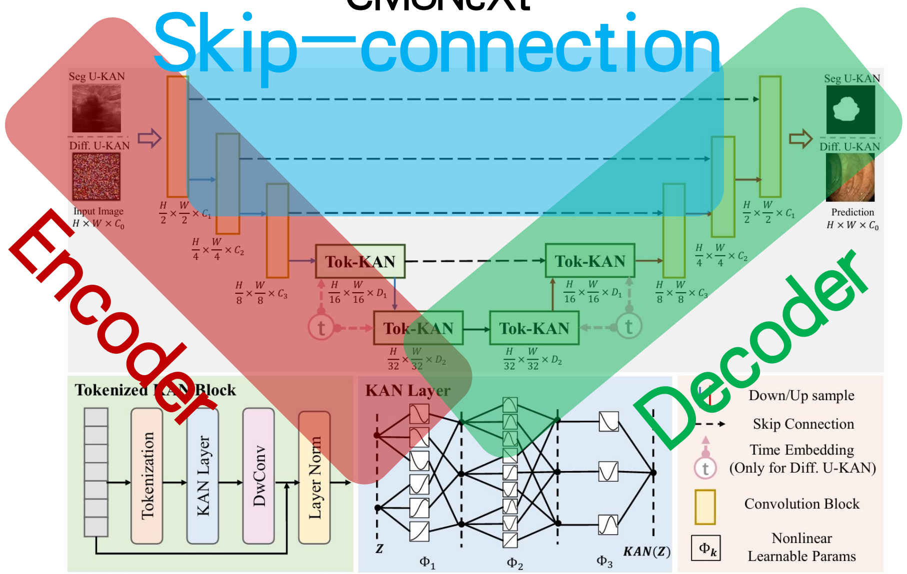
Attention U—Net



CMUNeXt

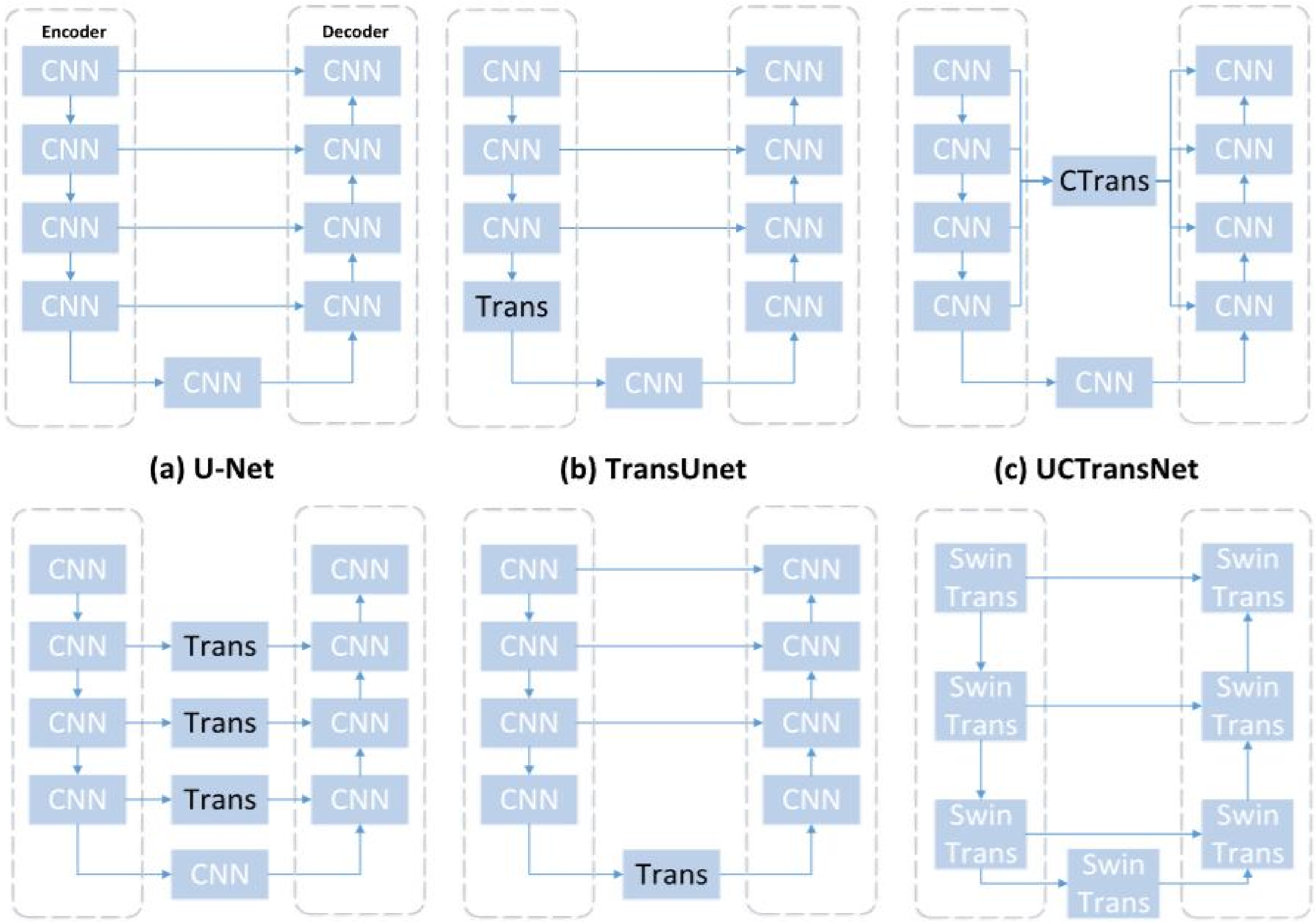


U—Net++



U—KAN

Hybrid U-Net Variants



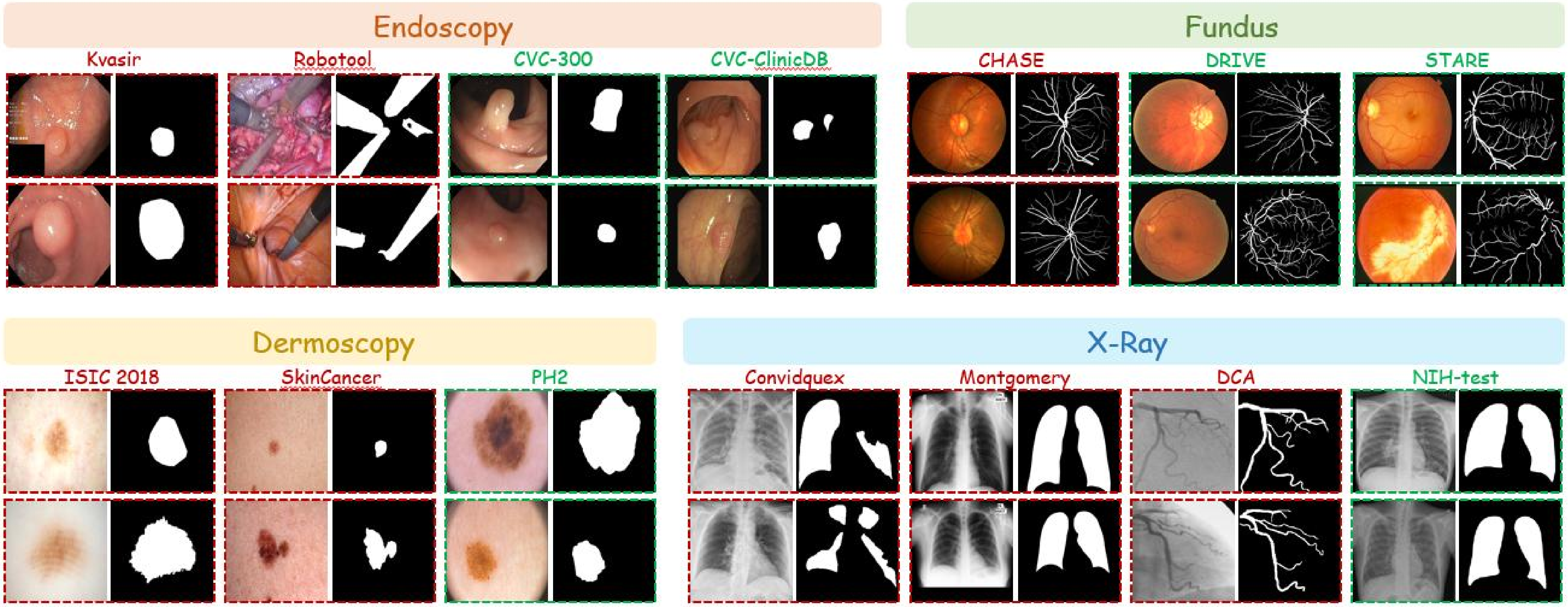
(a) U-Net (b) TransUnet (c) UCTransNet

(d) HT-Net (e) TransBTS (f) Swin-Unet

Xiao H, Li L, Liu Q, et al. Transformers in medical image segmentation: A review[J]. Biomedical Signal Processing and Control, 2023, 84: 104791.

Why skip-connections important ?

1. Fine-grained spatial information recovery → blur boundary in medical image
2. Enhanced multi-scale feature fusion → Large variation in target scales



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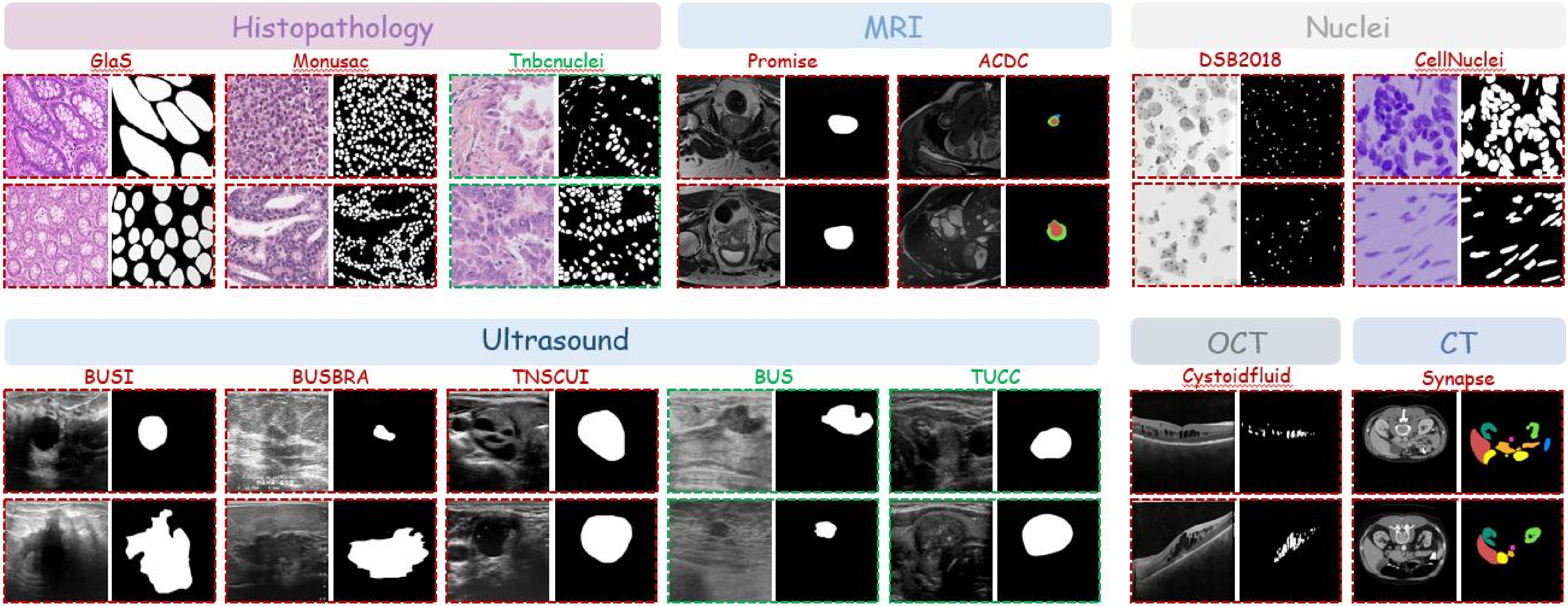
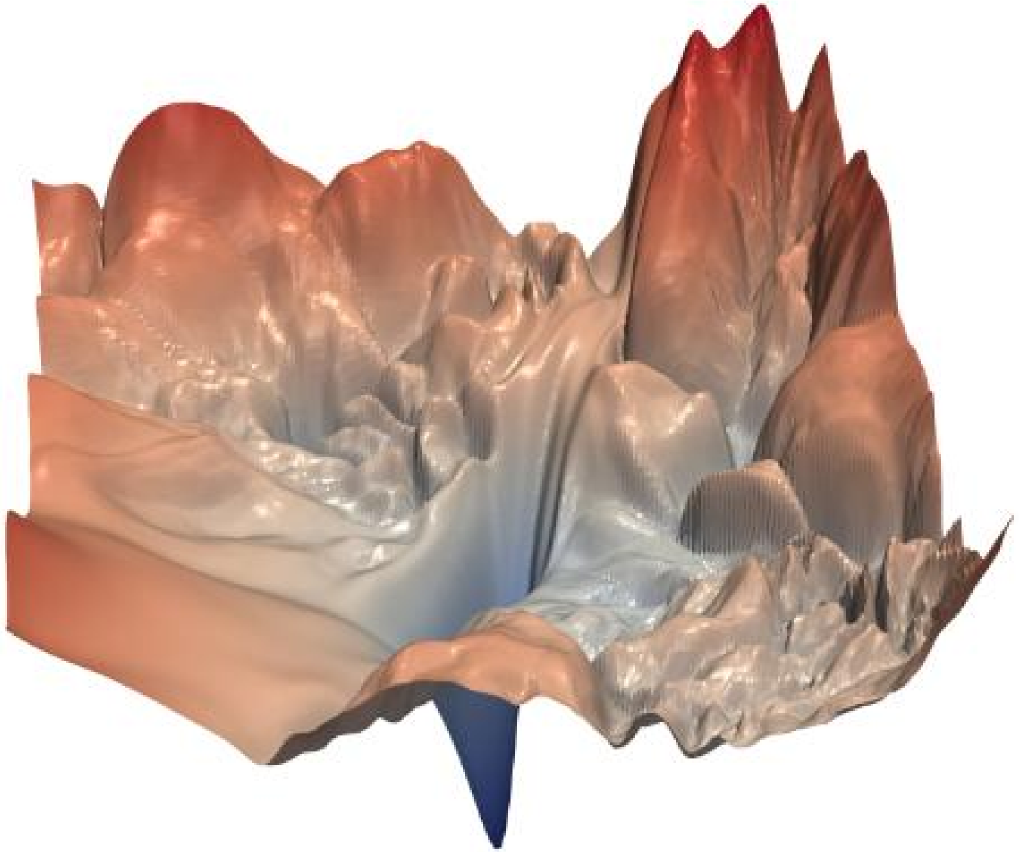
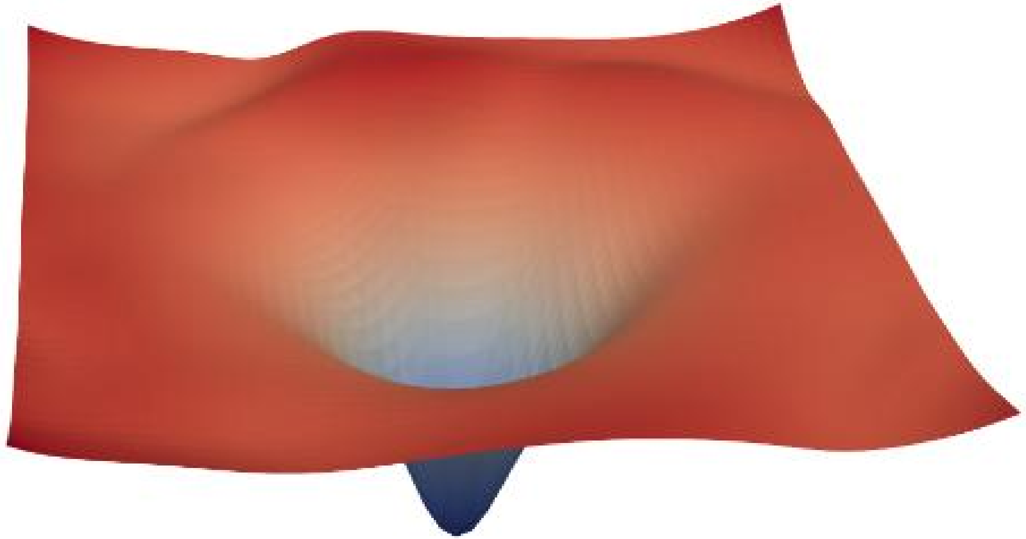


Why skip-connections important ?

3. Mitigating vanishing gradients/training difficulties → limited annotate images



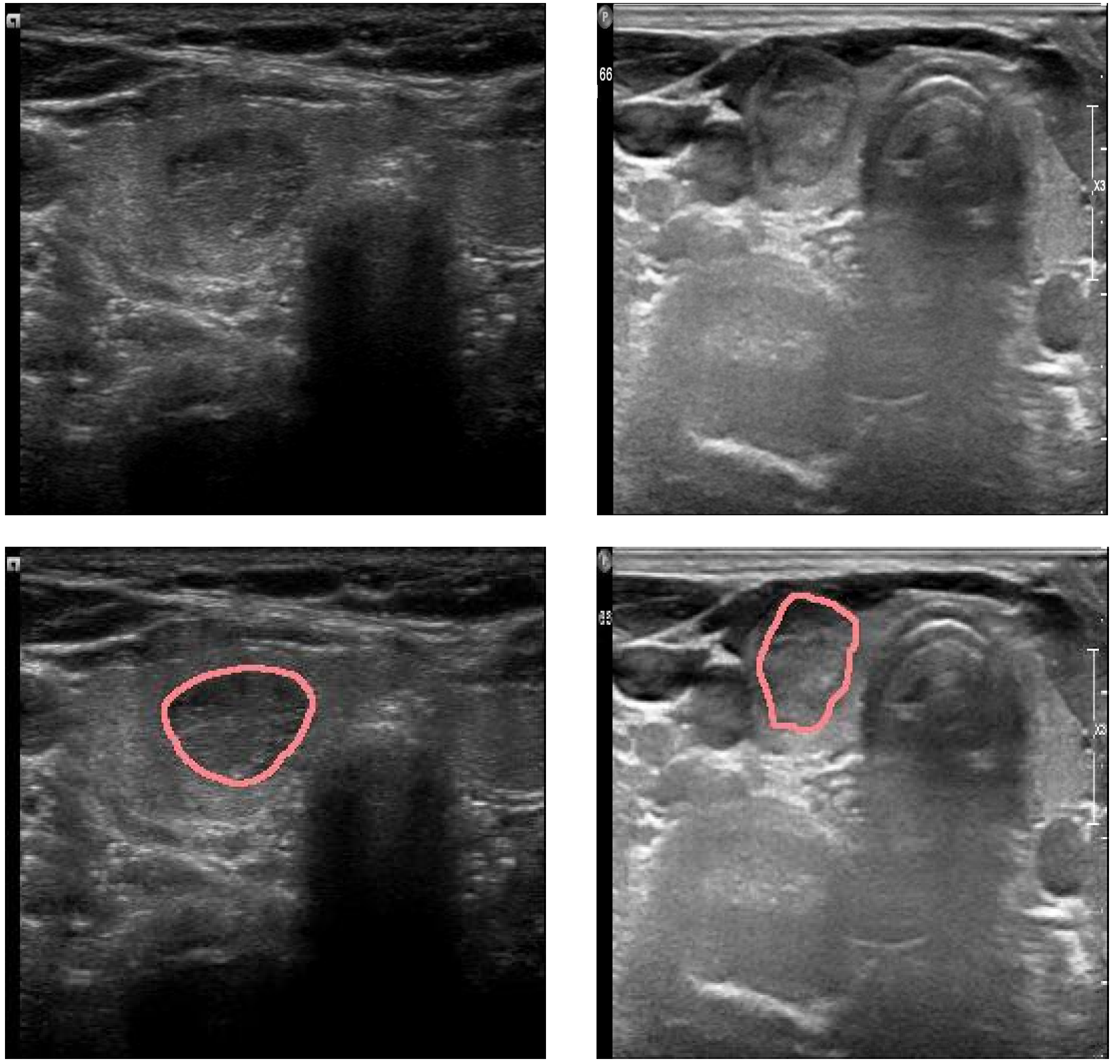
(a) without skip connections

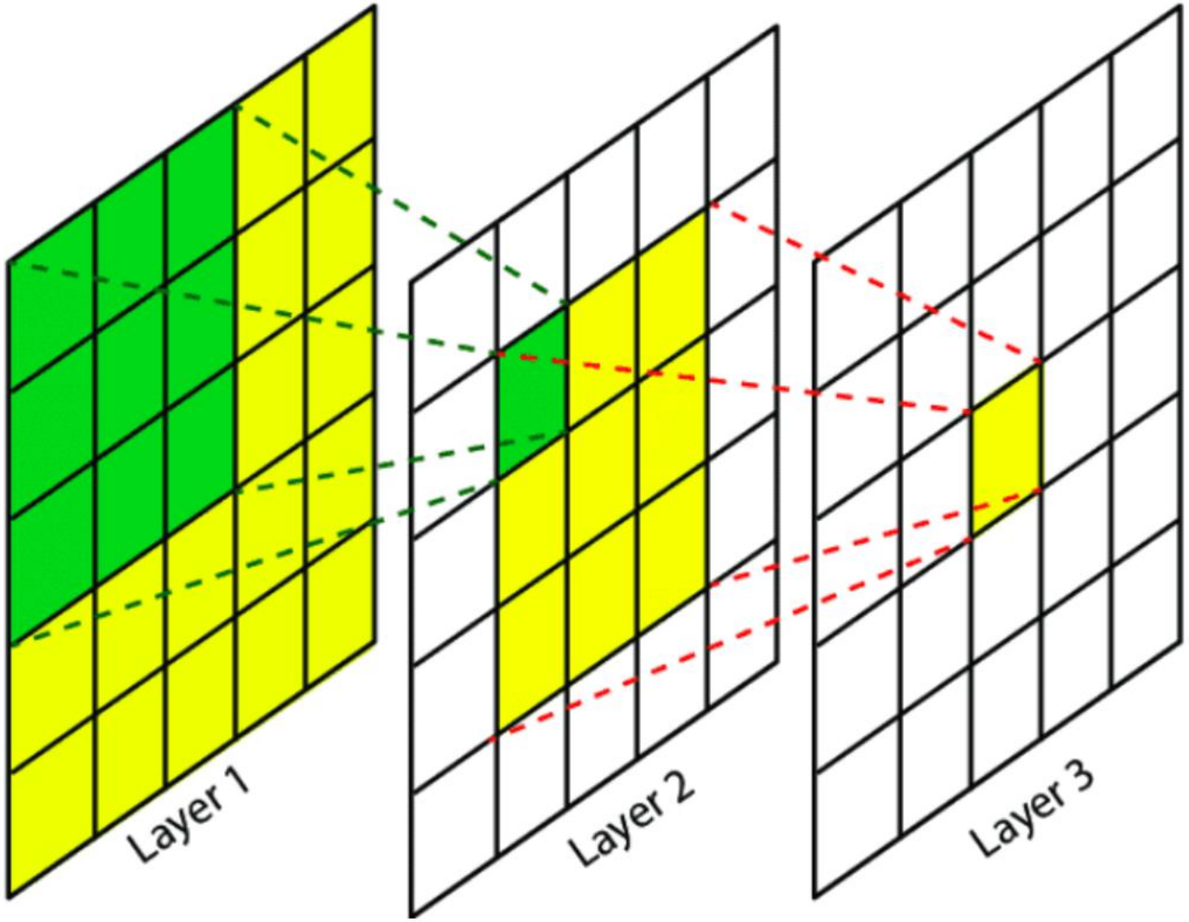


(b) with skip connections

Li H, Xu Z, Taylor G, et al. Visualizing the loss landscape of neural nets[J]. Advances in neural information processing systems, 2018, 31.

Why receptive field important ?

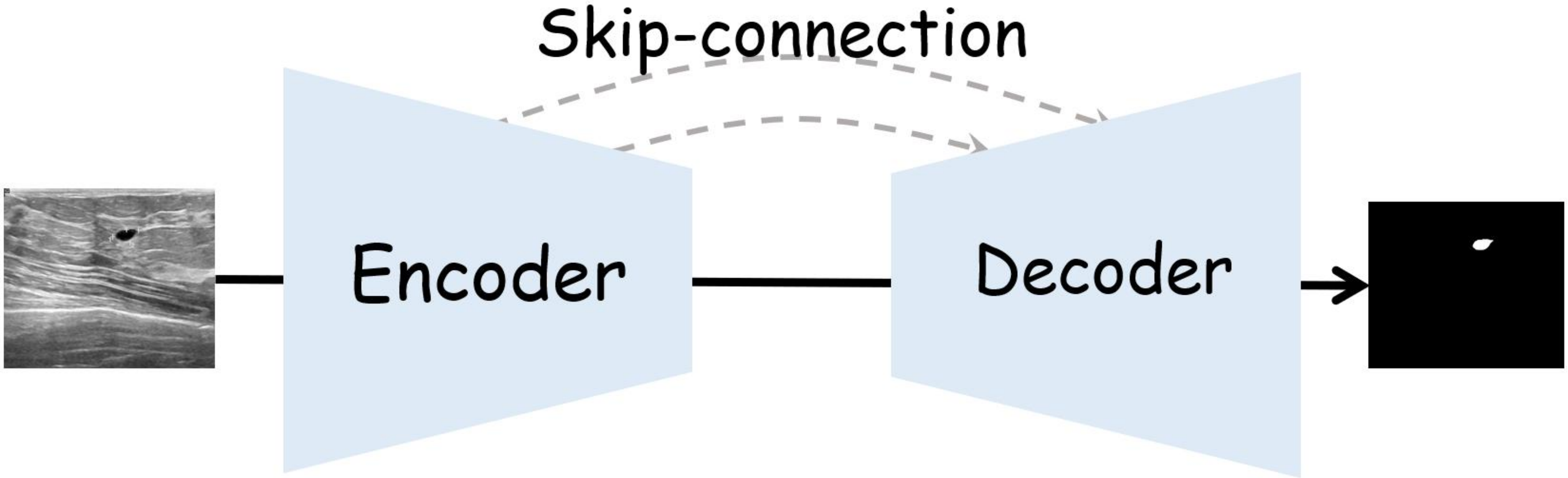




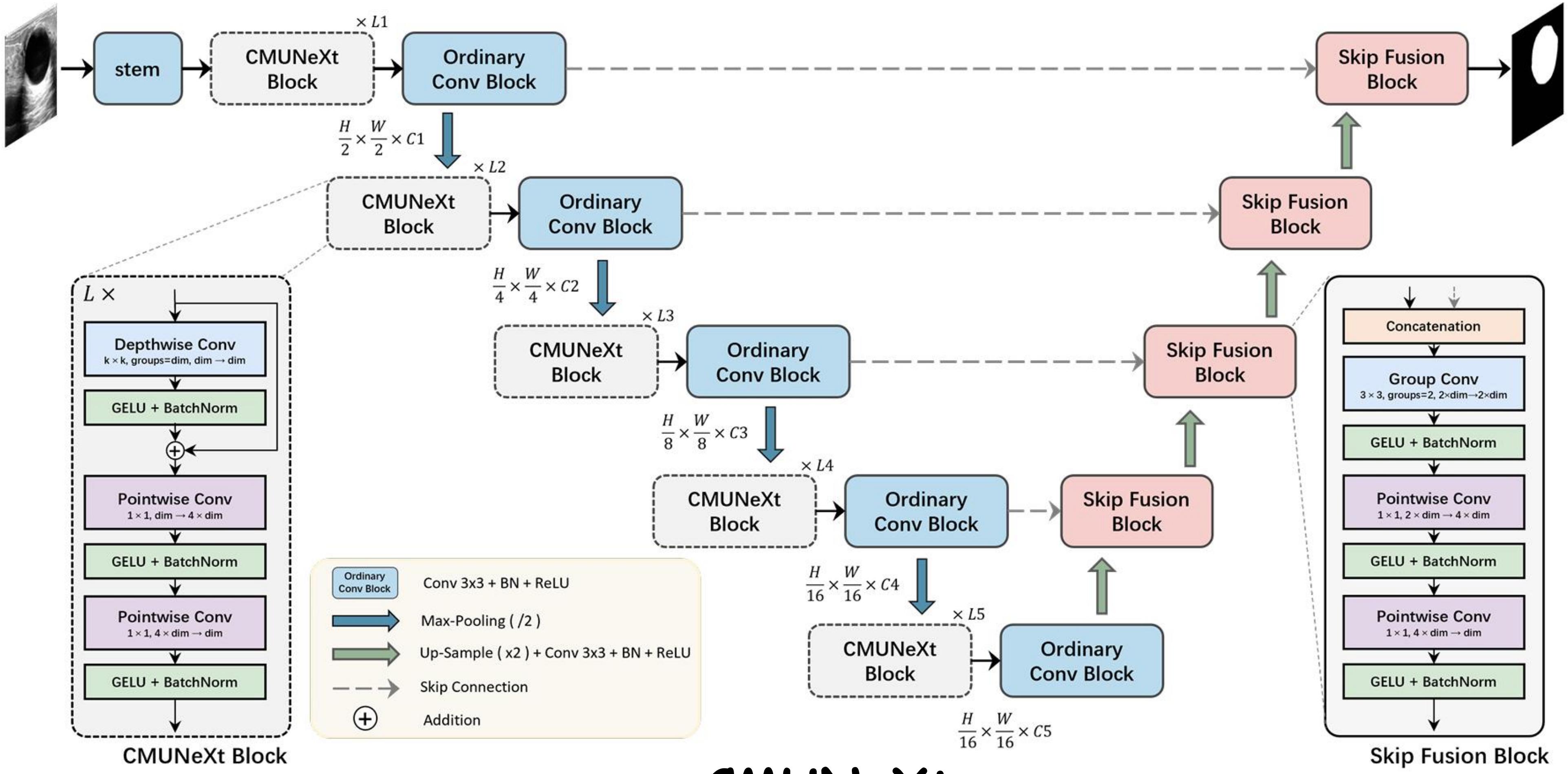
2D vs. 3D

How 2D segmentation work ?

2D Resolution: H x W x 3



Representative 2D U-variants

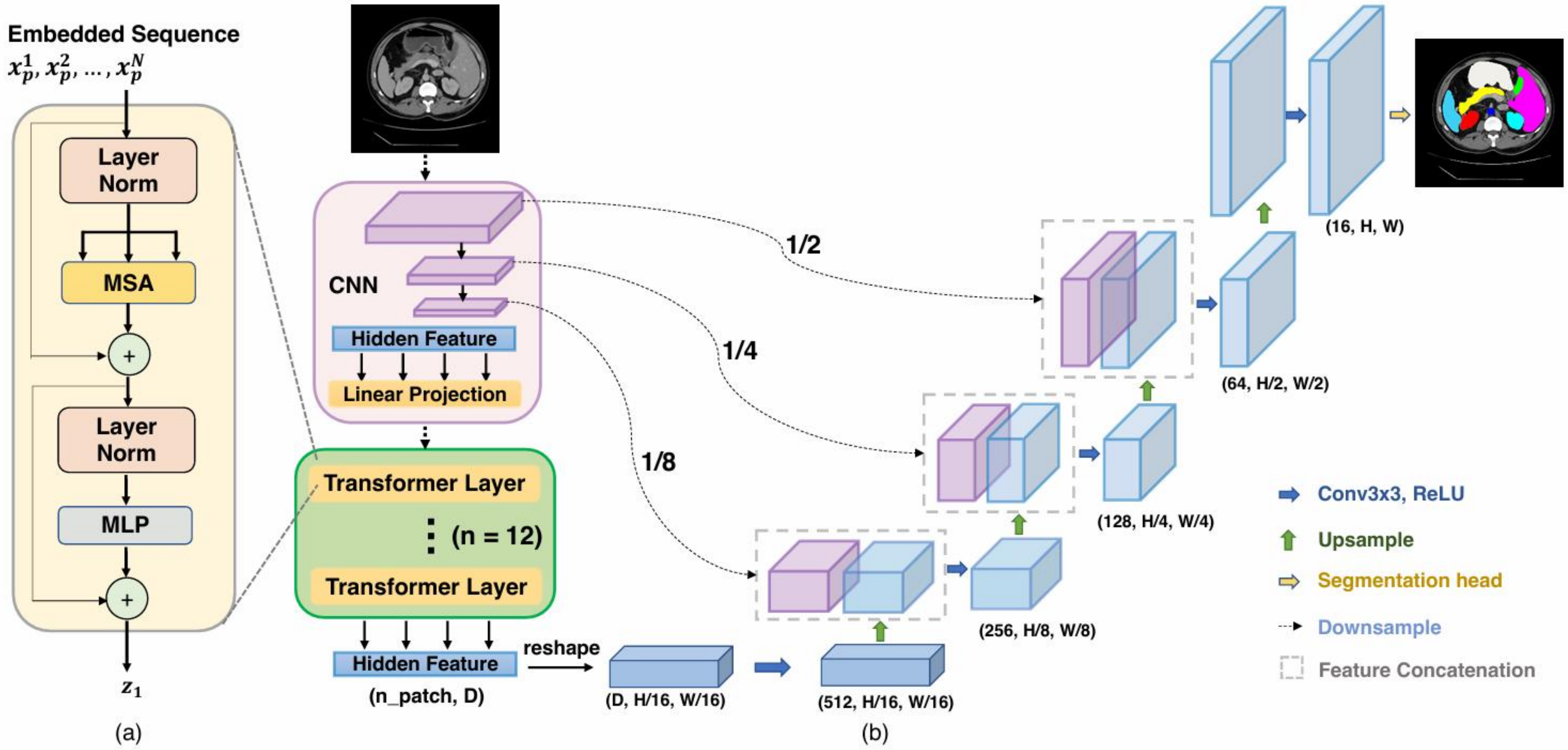


CMUNeXt

Encoder receptive field: Large kernel CNN
Skip-connection: Skip-fusion

Tang F, Ding J, Quan Q, et al. Cmunext: An efficient medical image segmentation network based on large kernel and skip fusion[C]//2024 IEEE International Symposium on Biomedical Imaging (ISBI). IEEE, 2024: 1-5.

Representative 2D U-variants

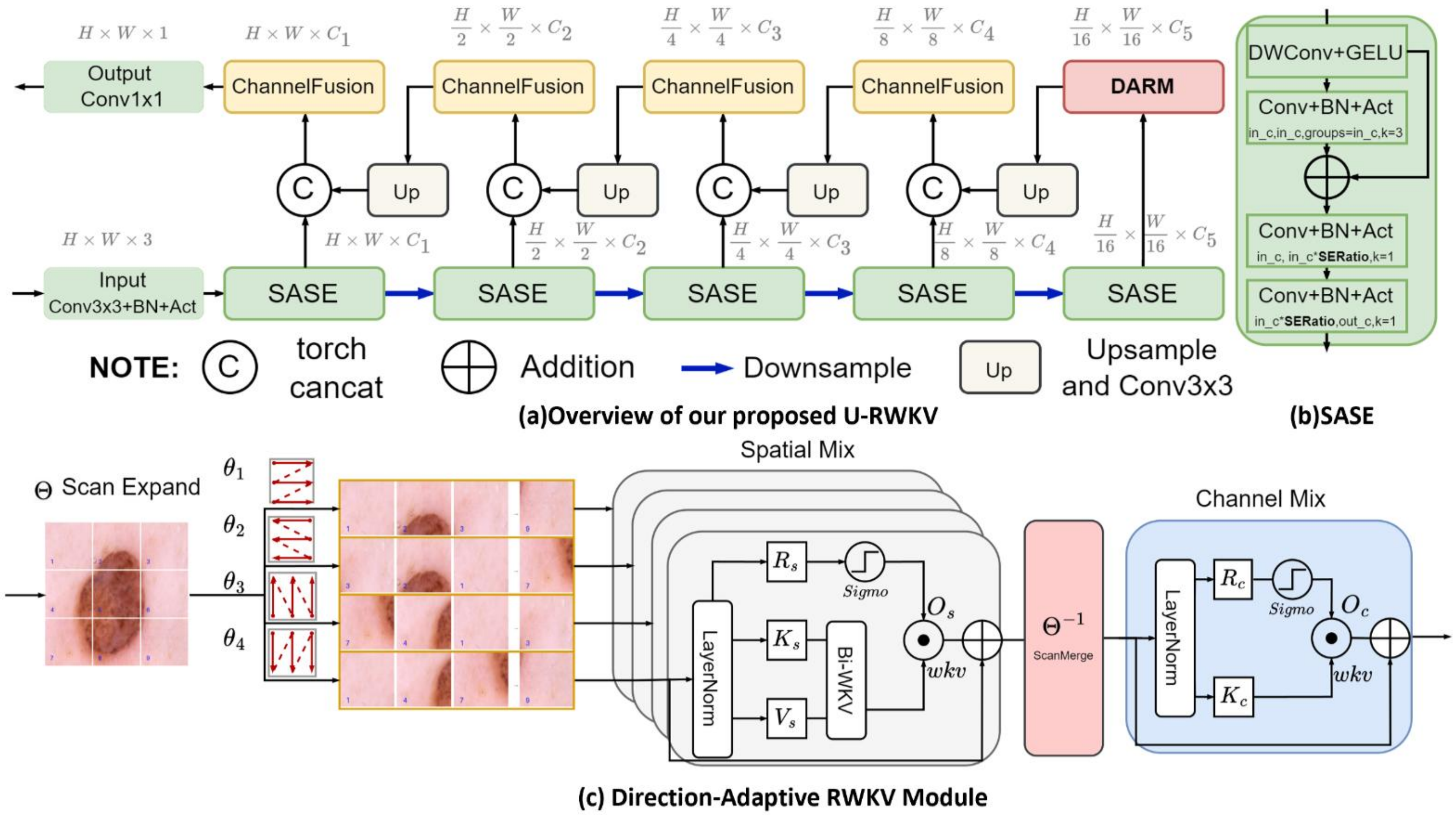


TransUnet

Encoder receptive field: Transformer
Skip-connection: CNN skip

Chen J, Lu Y, Yu Q, et al. Transunet: Transformers make strong encoders for medical image segmentation[J]. arXiv preprint arXiv:2102.04306, 2021.

Representative 2D U-variants

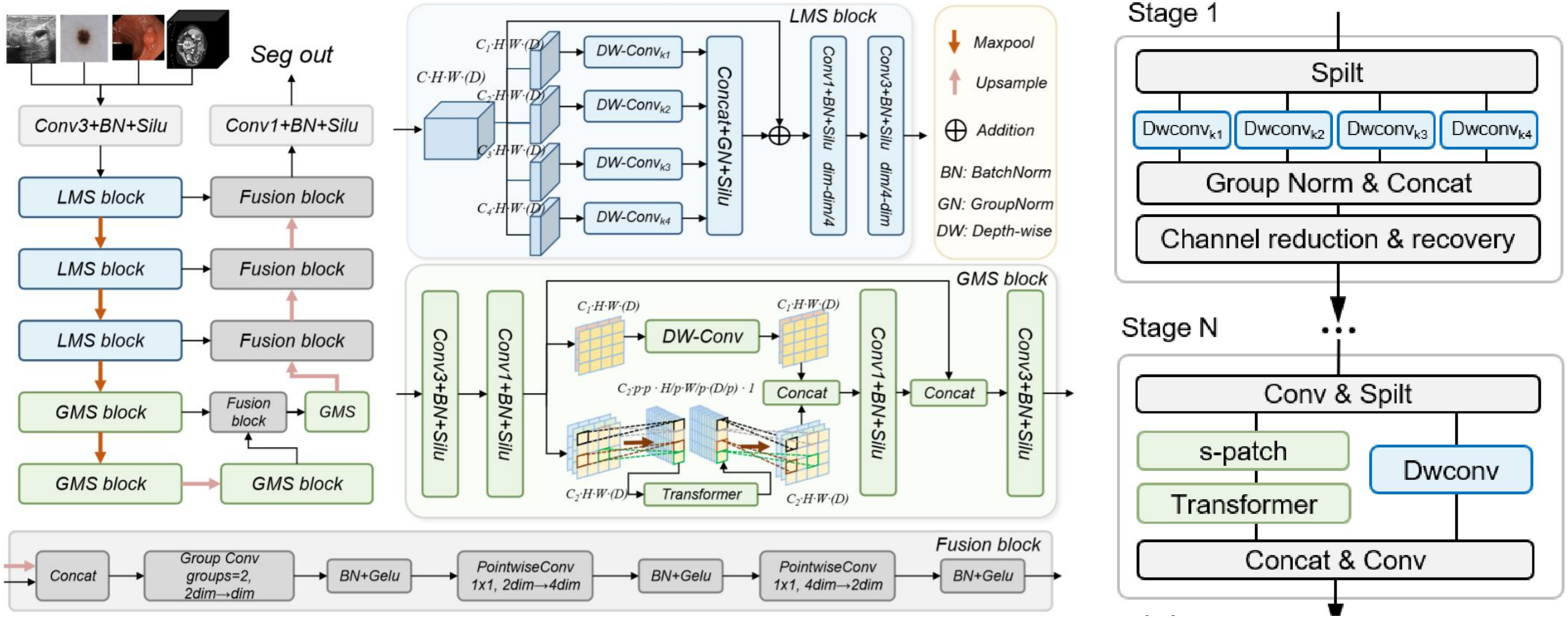


U-RWKV

Encoder receptive field: Large kernel CNN, RWKV
Skip-connection: Channel Fusion

Ye H, Tang F, Zhao P, et al. U-rwkv: Lightweight medical image segmentation with direction-adaptive rwkv[C]//International Conference on Medical Image Computing and Computer-Assisted Intervention. Cham: Springer Nature Switzerland, 2025: 613-623.

Representative 2D U-variants



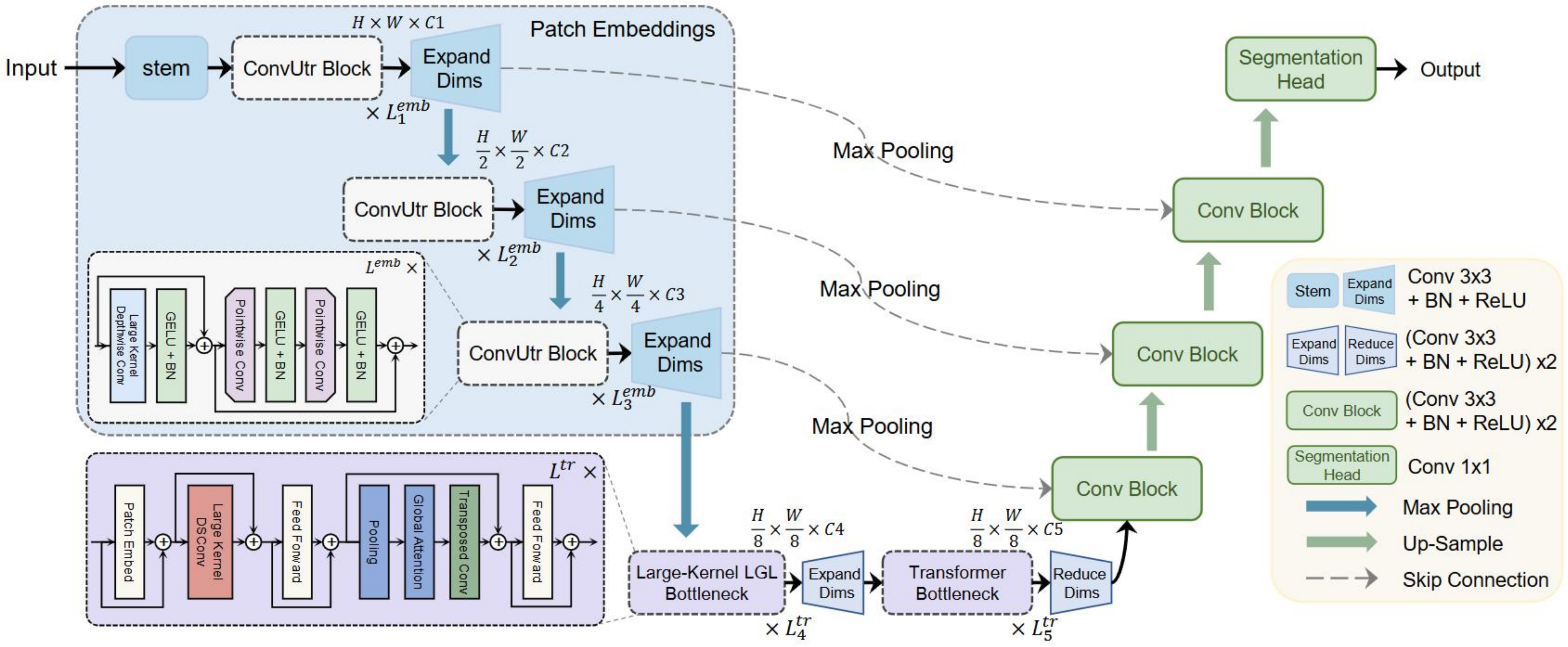
LGMSNet

Encoder receptive field: Multi-scale Large kernel CNN, Transformer

Skip-connection: Fusion Block

Dong C, Tang F, Mao R, et al. LGMSNet: Thinning a medical image segmentation model via dual-level multiscale fusion[C]. ECAI 2025.

Representative 2D U-variants

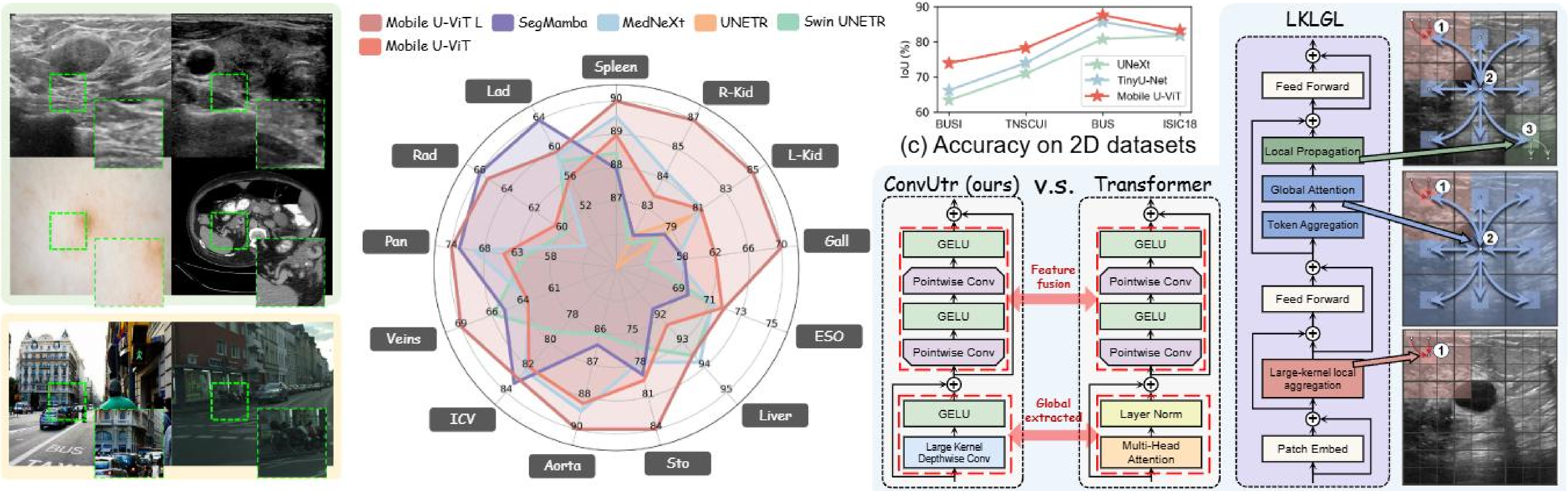


Mobile U-ViT

Encoder receptive field: Large kernel CNN, LKLGL, Transformer
Skip-connection: Downsample skip-connection

Tang F, Nian B, Ding J, et al. Mobile U-ViT: Revisiting large kernel and U-shaped ViT for efficient medical image segmentation[C]//Proceedings of the 33rd ACM International Conference on Multimedia. 2025: 3408-3417.

Representative 2D U-variants



(a) Difference between medical and natural images

(b) Accuracy comparison on 3D multi-organ segmentation dataset

(c) Accuracy on 2D datasets

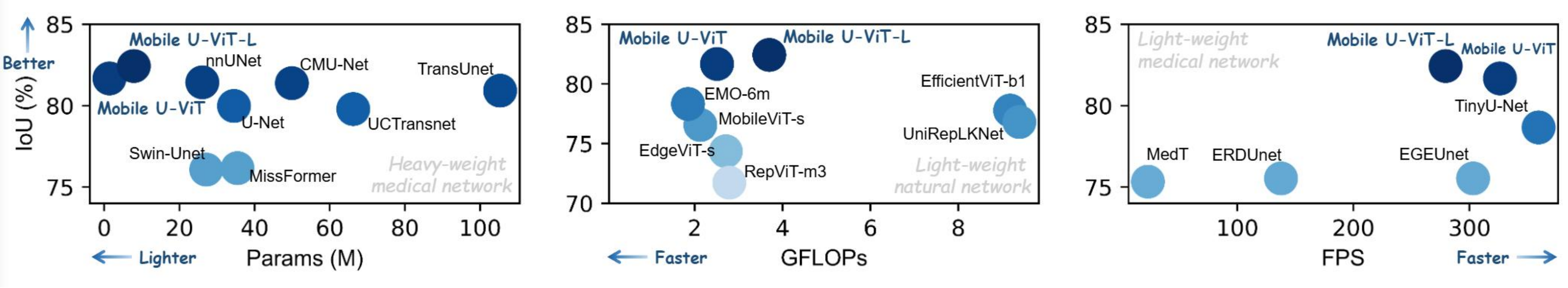
(d) Core components of Mobile U-ViT encoder

Mobile U-ViT

Encoder receptive field: Large kernel CNN, LKLGL, Transformer
Skip-connection: Downsample skip-connection

Tang F, Nian B, Ding J, et al. Mobile U-ViT: Revisiting large kernel and U-shaped ViT for efficient medical image segmentation[C]//Proceedings of the 33rd ACM International Conference on Multimedia. 2025: 3408-3417.

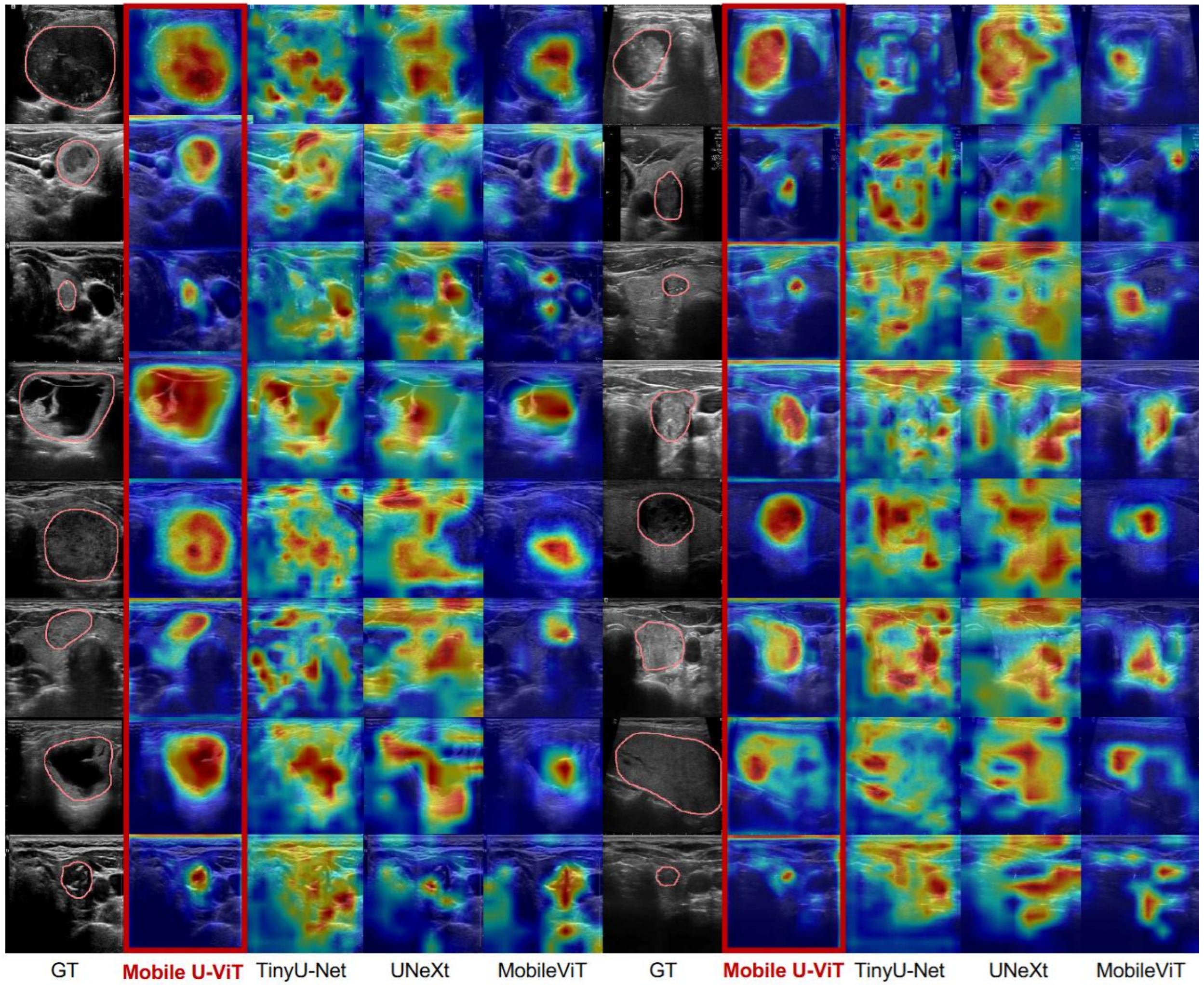
Representative 2D U-variants

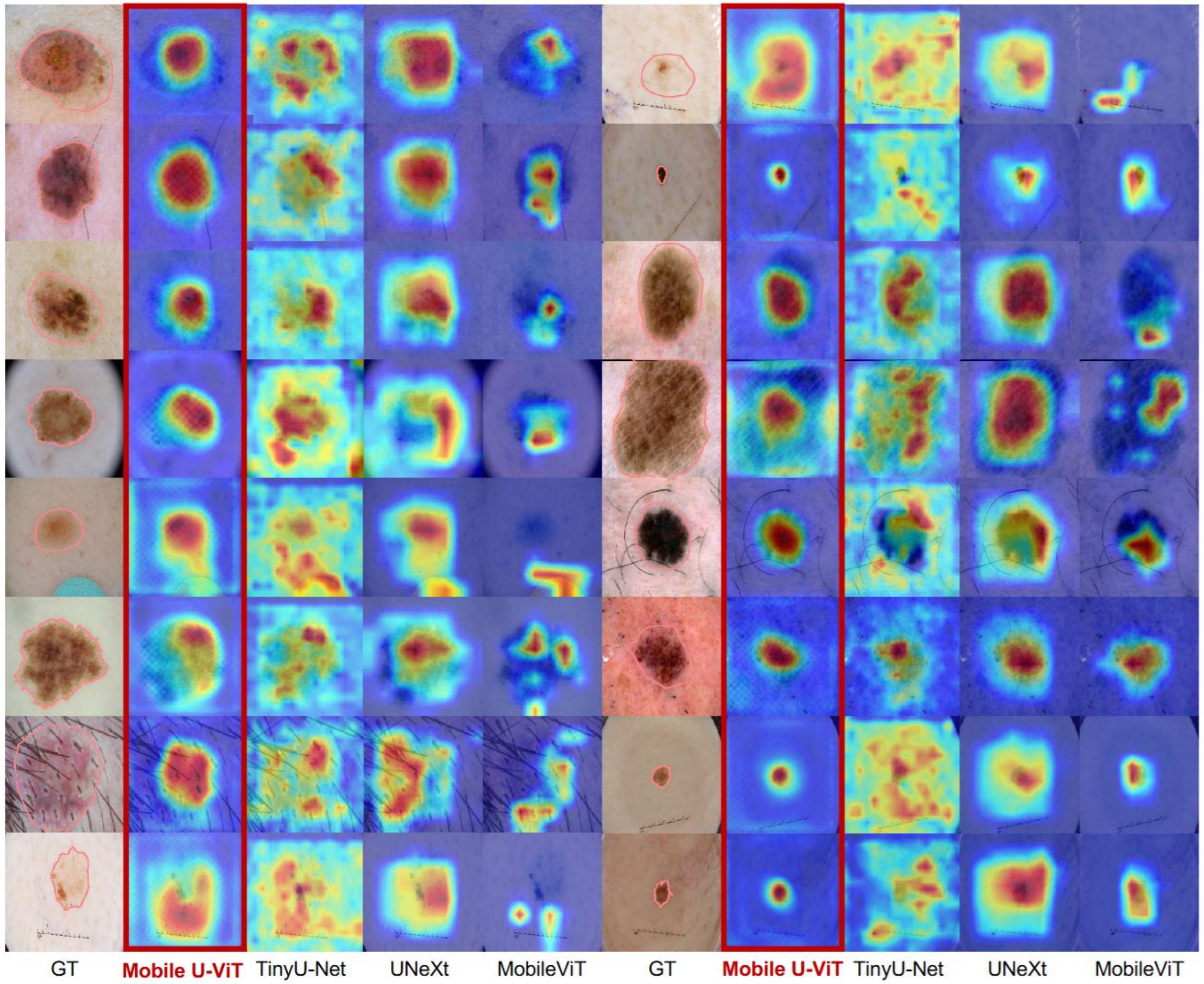


Networks	Ultrasound				Dermoscopy		Colonoscopy		Avg	
	BUSI -> BUS		TUSCUI -> TUCC		ISIC -> PH2		Kvasir -> CVC-300			
	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)
UNeXt (MICCAI'22) [38]	69.02	79.27	54.83	66.04	82.00	89.57	65.75	75.23	67.90	77.53
ERDUnet (TCSVT'23) [18]	71.75	80.72	59.66	70.06	82.90	90.06	72.23	80.62	71.64	80.37
RepViT-m3 (CVPR'24) [42]	65.07	76.38	57.28	68.88	79.95	88.27	62.96	73.68	66.32	76.80
MobileViT-s (ICLR'22) [21]	70.70	81.14	59.83	70.78	83.19	90.42	72.50	81.92	71.56	81.07
UniRepLKNet (CVPR'24) [7]	66.19	75.65	55.53	67.12	83.83	90.80	68.31	75.59	68.47	77.29
TinyU-net (MICCAI'24) [4]	64.49	73.70	57.76	69.03	81.96	89.53	71.22	78.50	68.86	77.69
Mobile U-ViT (3D) (Ours)	76.64	85.35	60.33	71.11	84.63	91.05	73.62	**81.94**	73.81	82.36
Mobile U-ViT-L (3D) (Ours)	**78.19**	**86.45**	**60.71**	**71.44**	**85.25**	**91.53**	**73.65**	81.79	**74.45**	**82.80**

Tang F, Nian B, Ding J, et al. Mobile U-ViT: Revisiting large kernel and U-shaped ViT for efficient medical image segmentation[C]//Proceedings of the 33rd ACM International Conference on Multimedia. 2025: 3408-3417.

Representative 2D U-variants



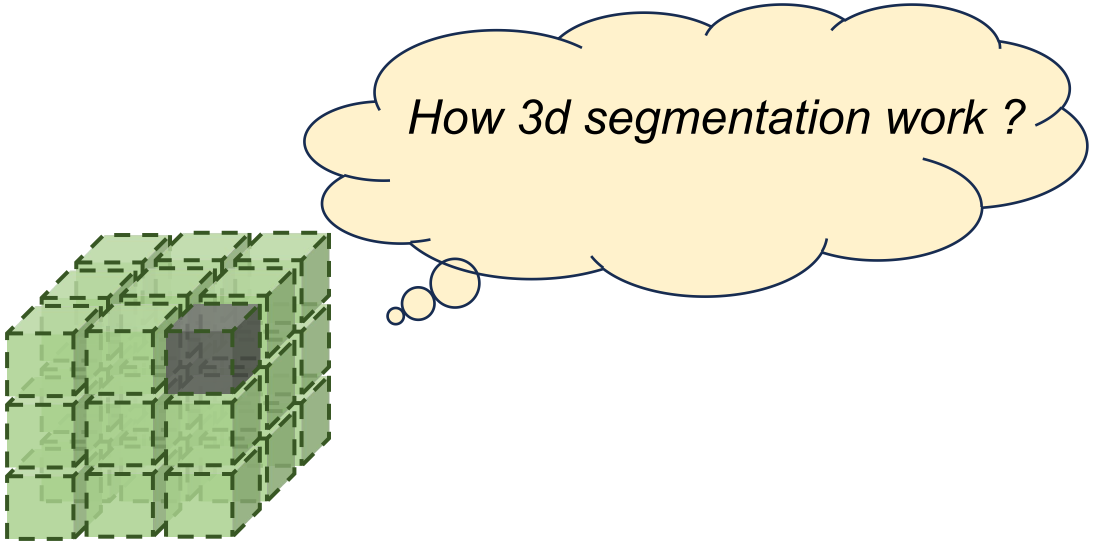


Tang F, Nian B, Ding J, et al. Mobile U-ViT: Revisiting large kernel and U-shaped ViT for efficient medical image segmentation[C]//Proceedings of the 33rd ACM International Conference on Multimedia. 2025: 3408-3417.

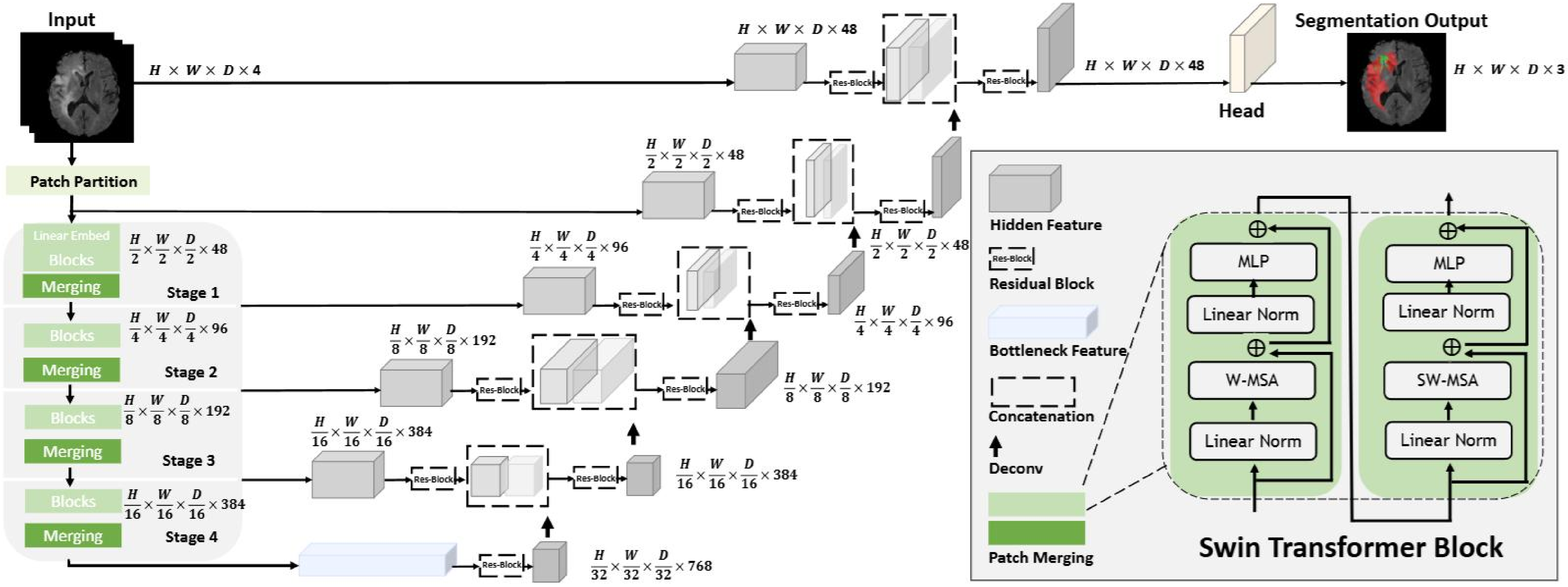
2D vs. 3D

How 3D segmentation work ?

3D Resolution: H x W x **D** x 3



Representative 3D U-variants

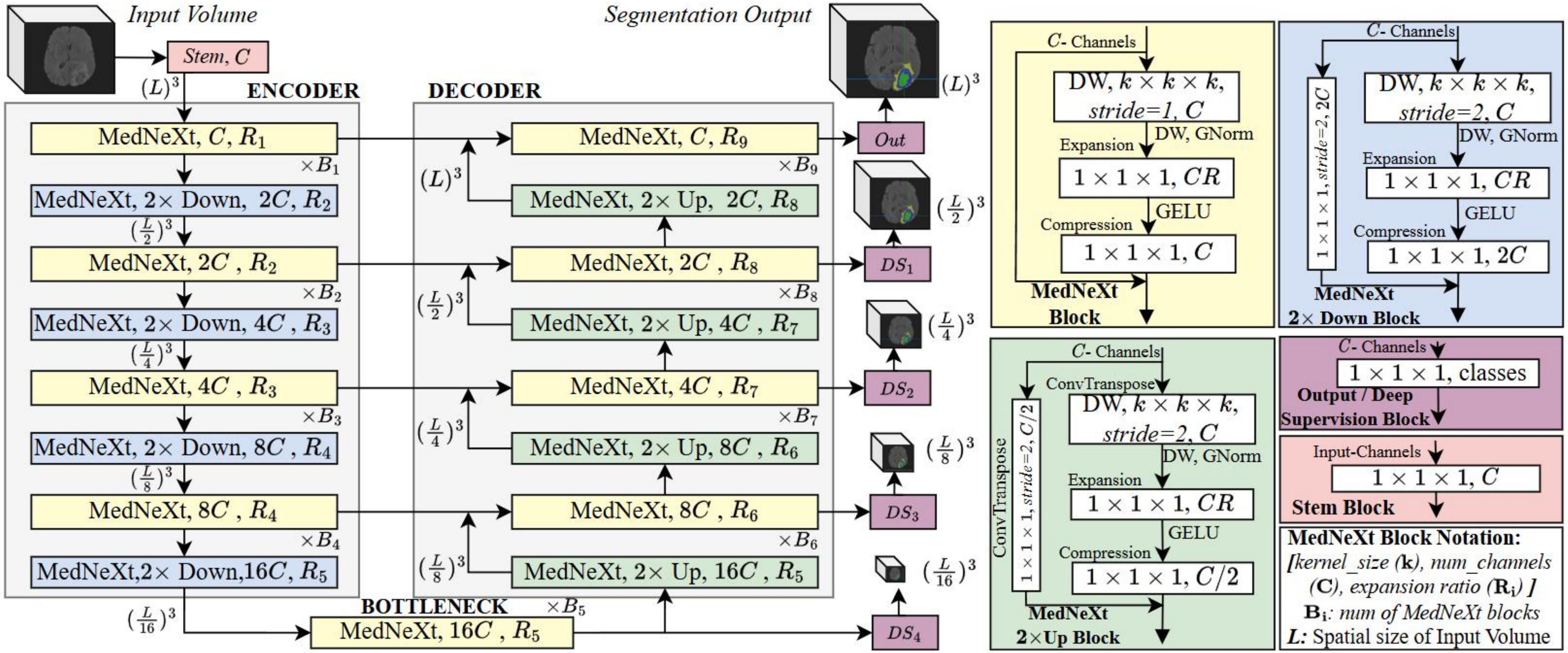


Swin UNETR

Encoder receptive field: Swin Transformer
Skip-connection: CNN Fusion skip

Tang Y, Yang D, Li W, et al. Self-supervised pre-training of swin transformers for 3d medical image analysis[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 20730-20740.

Representative 3D U-variants

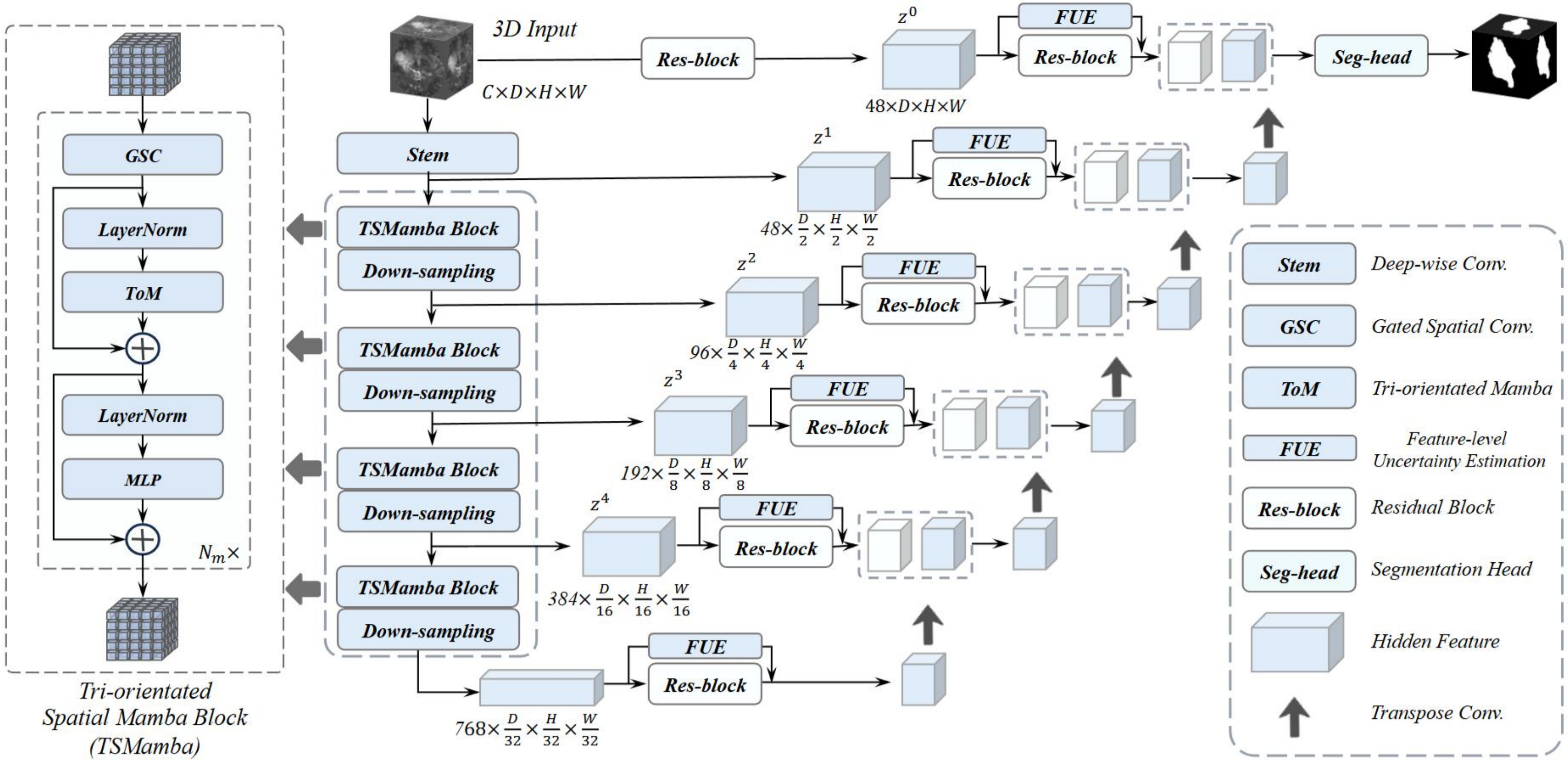


MedNeXt

Encoder receptive field: Large 3D CNN kernel
Skip-connection: CNN Fusion skip

Roy S, Koehler G, Ulrich C, et al. Mednext: transformer-driven scaling of convnets for medical image segmentation[C]//International Conference on Medical Image Computing and Computer-Assisted Intervention. Cham: Springer Nature Switzerland, 2023: 405-415.

Representative 3D U-variants

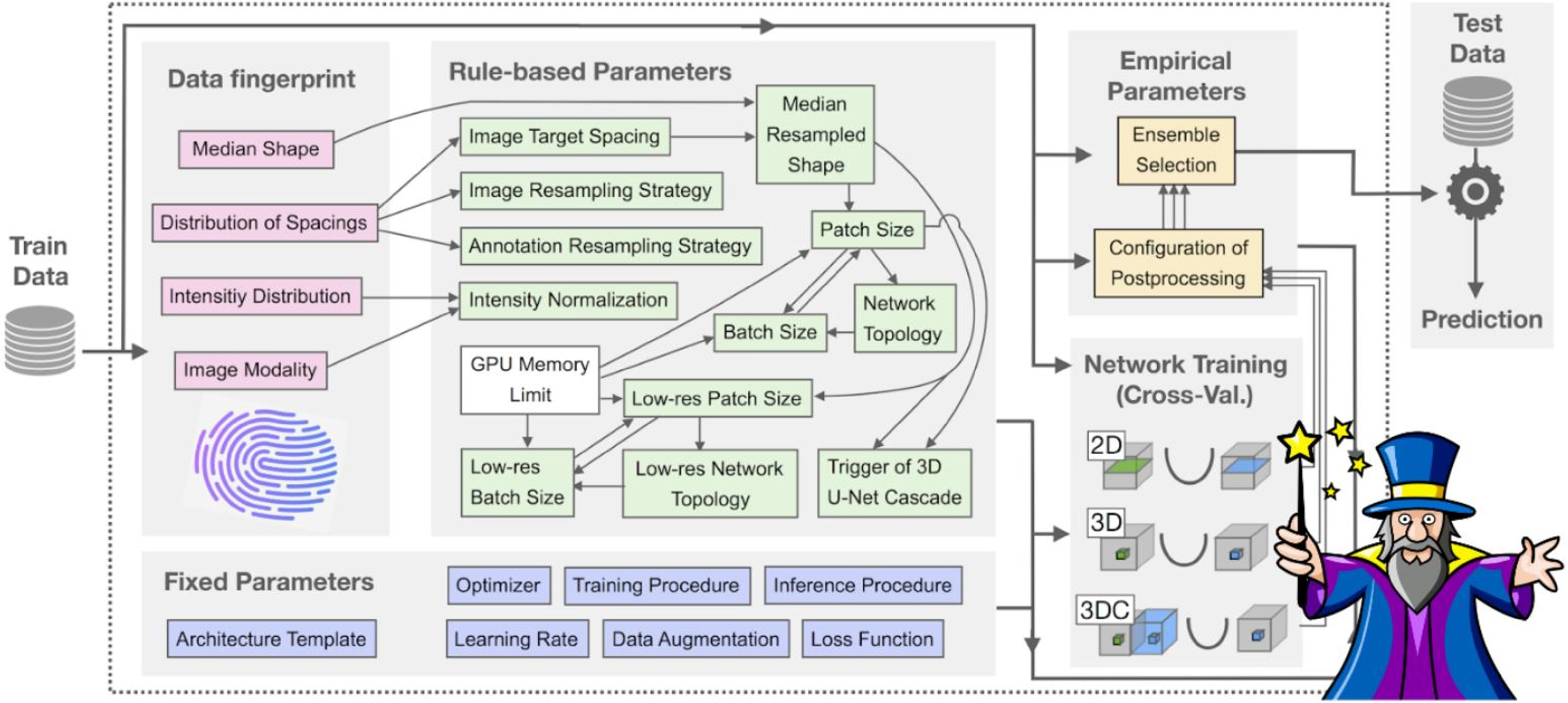


SegMamba

Encoder receptive field: Mamba
Skip-connection: CNN Fusion skip

Xing Z, Ye T, Yang Y, et al. Segmamba: Long-range sequential modeling mamba for 3d medical image segmentation[C]//International conference on medical image computing and computer-assisted intervention. Cham: Springer Nature Switzerland, 2024: 578-588.

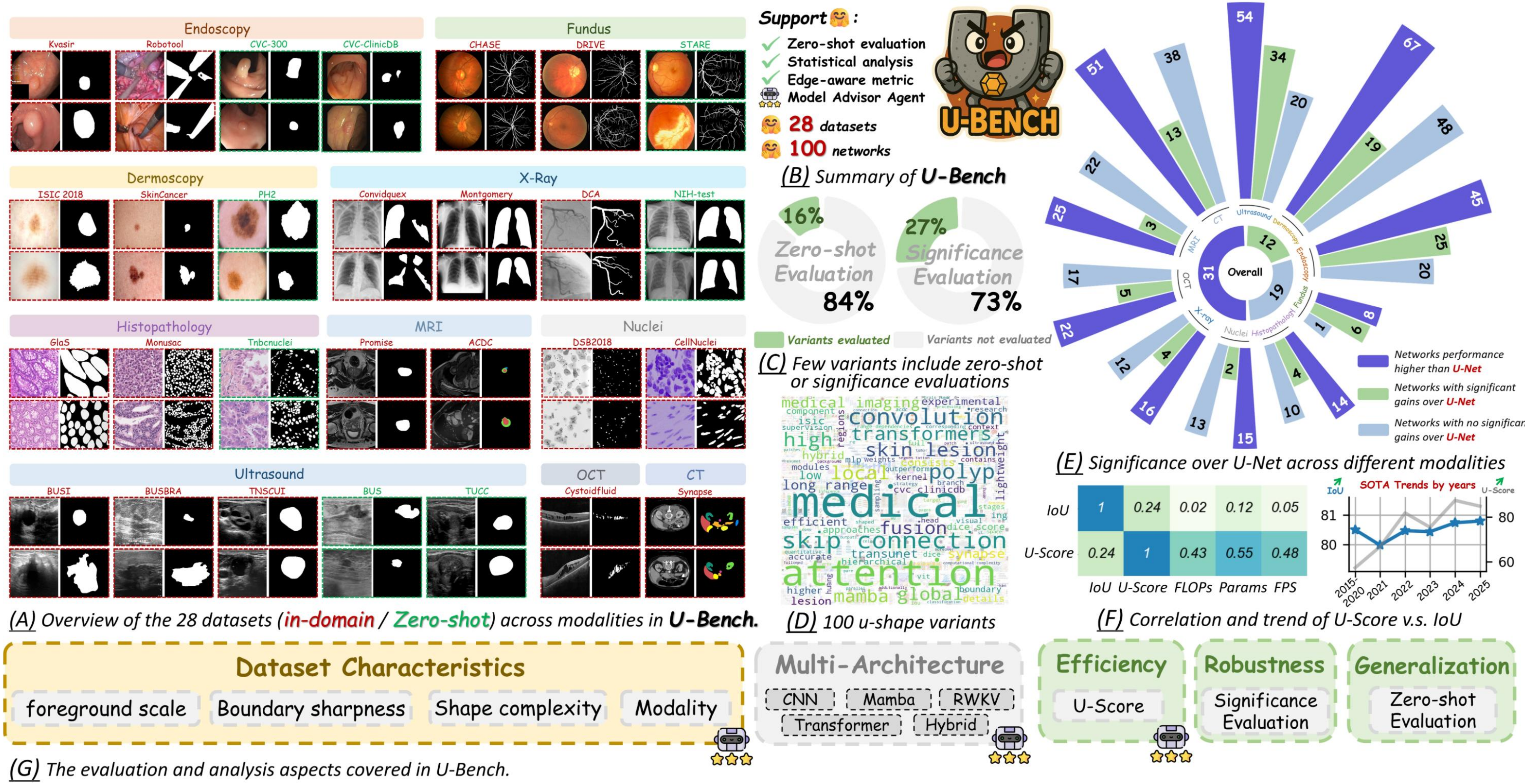
Most Representative U-Net variants



nnUNet pipeline

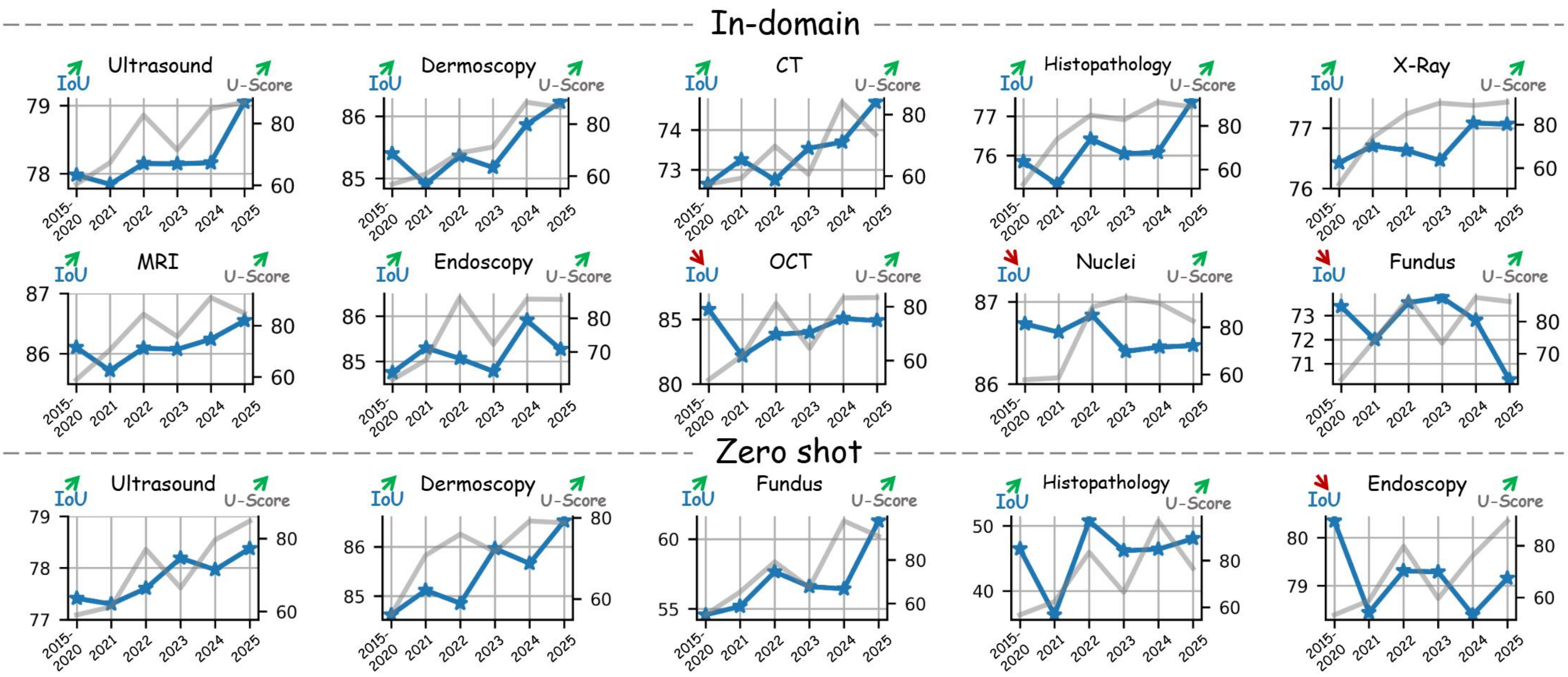
Isensee F, Jaeger P F, Kohl S A A, et al. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation[J]. Nature methods

U-Bench



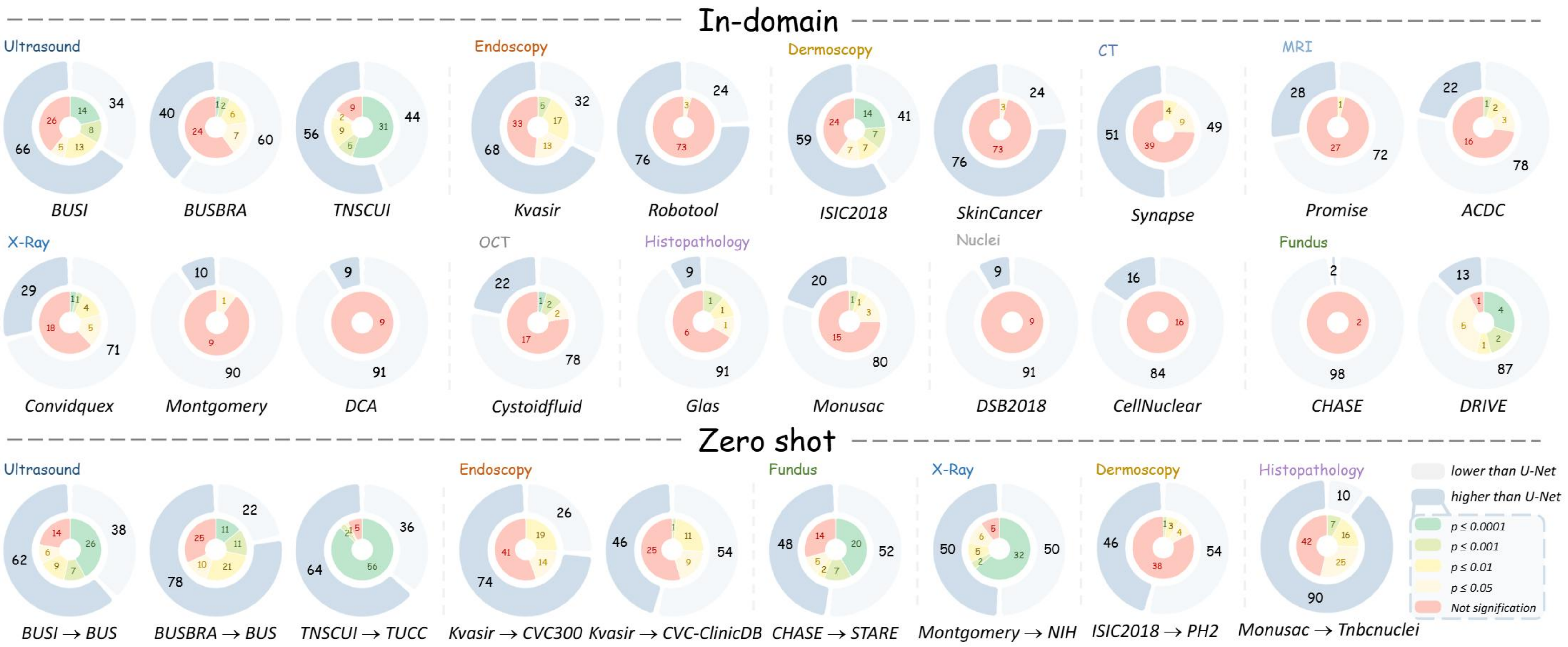
Tang F, Dong C, Ma W, et al. U-Bench: A Comprehensive Understanding of U-Net through 100-Variant Benchmarking[J]. arXiv preprint arXiv:2510.07041

U-Bench



Tang F, Dong C, Ma W, et al. U-Bench: A Comprehensive Understanding of U-Net through 100-Variant Benchmarking[J]. arXiv preprint arXiv:2510.07041

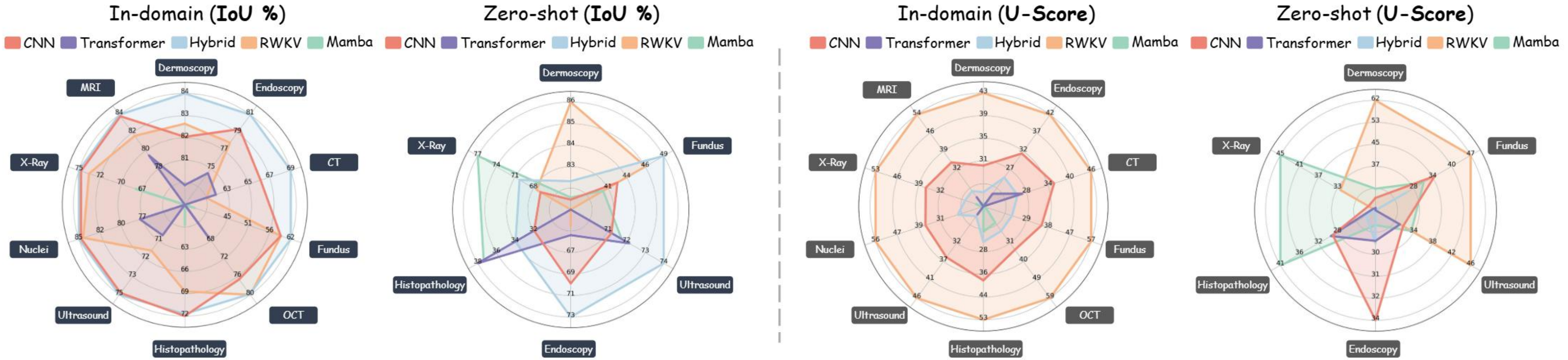
U-Bench



Tang F, Dong C, Ma W, et al. U-Bench: A Comprehensive Understanding of U-Net through 100-Variant Benchmarking[J]. arXiv preprint arXiv:2510.07041

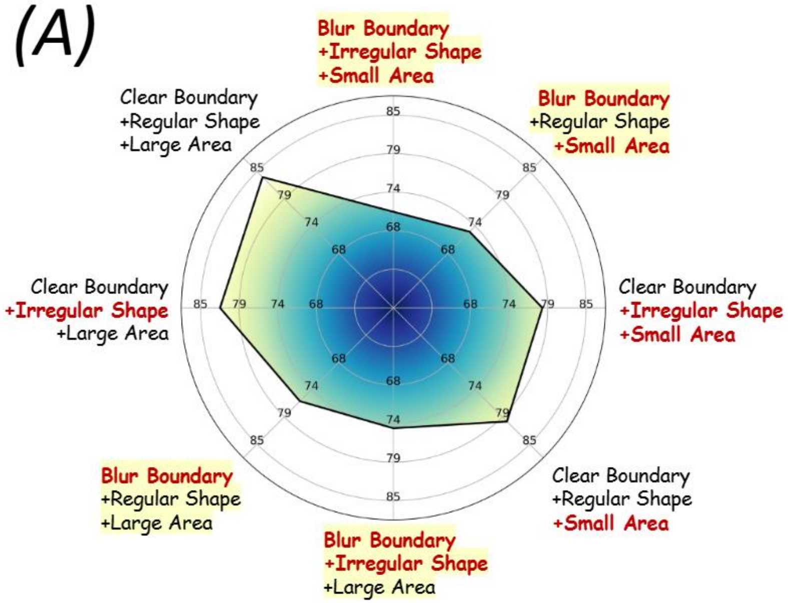
U-Bench

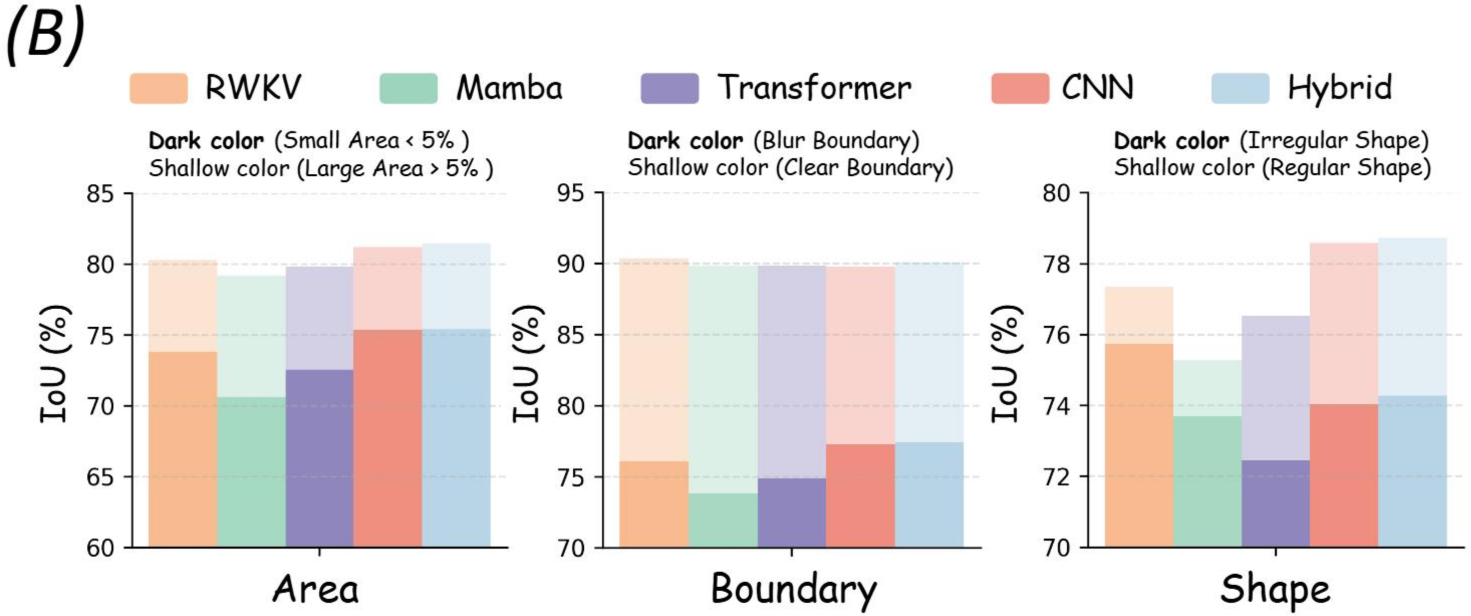
Source (IoU)			Target (IoU)			Source (U-Score)			Target (U-Score)		
Rank	Variants	# Volume (year)	Rank	Variants	# Volume (year)	Rank	Variants	# Volume (year)	Rank	Variants	# Volume (year)
🥇	RWKV-UNet	Arxiv (2025)	🥇	RWKV-UNet	Arxiv (2025)	🥇	LGMSNet	ECAI (2025)	🥇	LGMSNet	ECAI (2025)
🥈	AURA-Net	ISBI (2021)	🥈	G-CASCADE	WACV (2024)	🥈	MBSNet	MSSP (2021)	🥈	LV-UNet	BIBM (2024)
🥉	UTANet	AAAI (2025)	🥉	Swin-umamba	MICCAI (2024)	🥉	CMUNeXt	ISBI (2024)	🥉	U-KAN	AAAI (2025)
#4	MEGANet	WACV (2024)	#4	MEGANet	WACV (2024)	#4	LV-UNet	BIBM (2024)	#4	Mobile U-ViT	ACM MM (2025)
#5	Swin-umamba	MICCAI (2024)	#5	CASCADE	WACV (2023)	#5	Mobile U-ViT	ACM MM (2025)	#5	RWKV-UNet	Arxiv (2025)
#6	MFMSNet	CIBM (2024)	#6	MFMSNet	CIBM (2024)	#6	Tinyunet	MICCAI (2024)	#6	MBSNet	MSSP (2021)
#7	TransResUNet	Arxiv (2022)	#7	TransResUNet	Arxiv (2022)	#7	U-RWKV	MICCAI (2025)	#7	SwinUNETR	CVPR (2022)
#8	EViT-UNet	ISBI (2025)	#8	DS-TransUNet	TIM (2022)	#8	U-KAN	AAAI (2025)	#8	CMUNeXt	ISBI (2024)
#9	FCBFormer	MIUA (2022)	#9	CENet	MICCAI (2025)	#9	DCSAU-Net	CIBM (2023)	#9	G-CASCADE	WACV (2024)
#10	DA-TransUNet	FBB (2024)	#10	PraNet	MICCAI (2020)	#10	RWKV-UNet	Arxiv (2025)	#10	TA-Net	CIBM (2021)
#25	U-Net	MICCAI (2025)	#72	U-Net	MICCAI (2025)	#64	U-Net	MICCAI (2025)	#69	U-Net	MICCAI (2025)

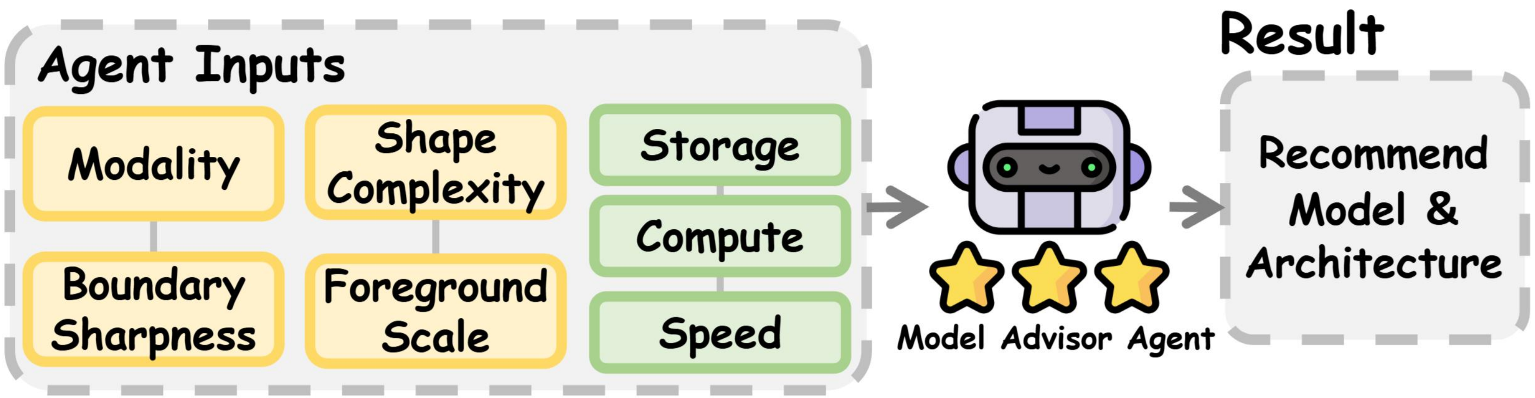


Tang F, Dong C, Ma W, et al. U-Bench: A Comprehensive Understanding of U-Net through 100-Variant Benchmarking[J]. arXiv preprint arXiv:2510.07041

U-Bench







Tang F, Dong C, Ma W, et al. U-Bench: A Comprehensive Understanding of U-Net through 100-Variant Benchmarking[J]. arXiv preprint arXiv:2510.07041

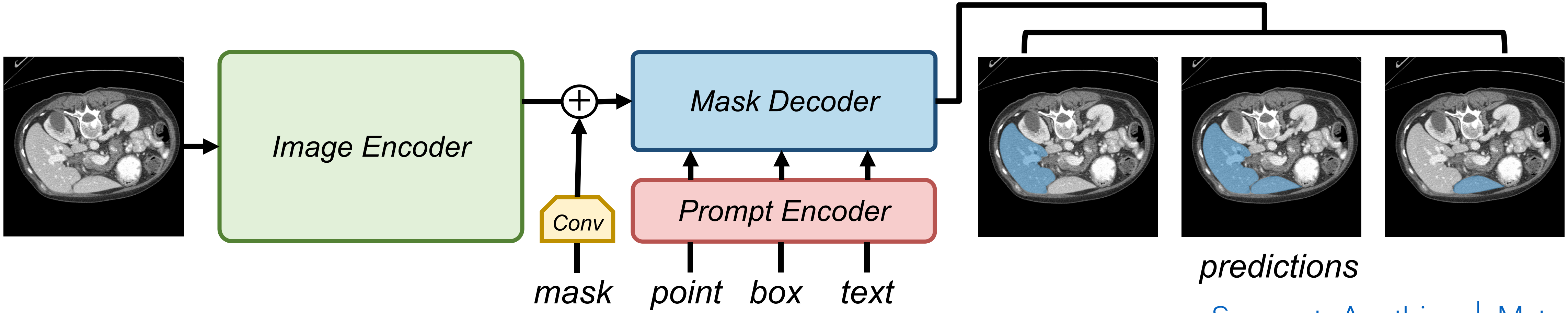
Foundation Model for medical image segmentation

Foundation Model for medical image segmentation

Inference of SAM

Pipeline of SAM

Segment Anything

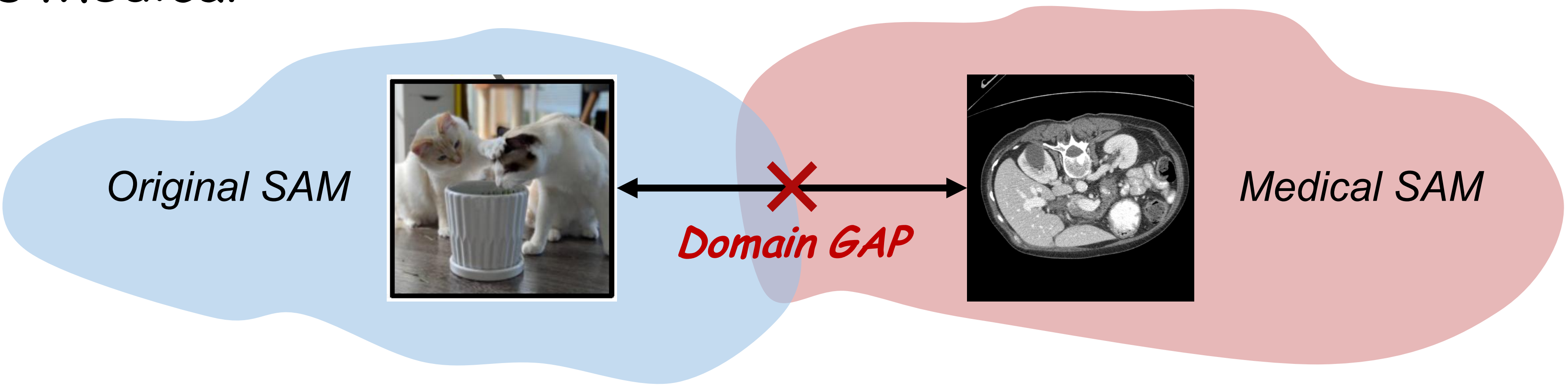


[Segment Anything | Meta AI]

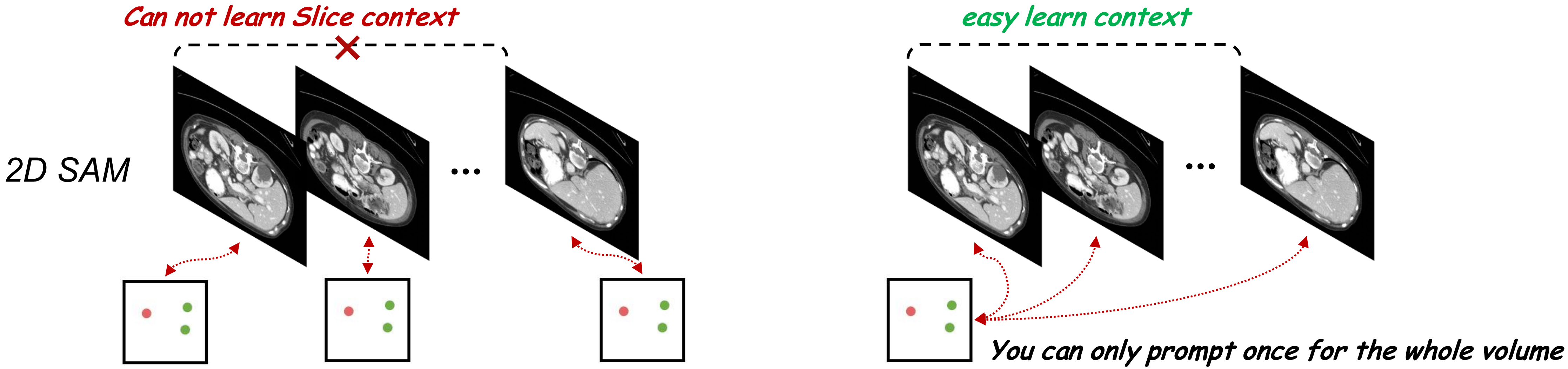
Kirillov A, Mintun E, Ravi N, et al. Segment anything[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2023: 4015-4026.

Foundation Model for medical image segmentation

Natural vs Medical

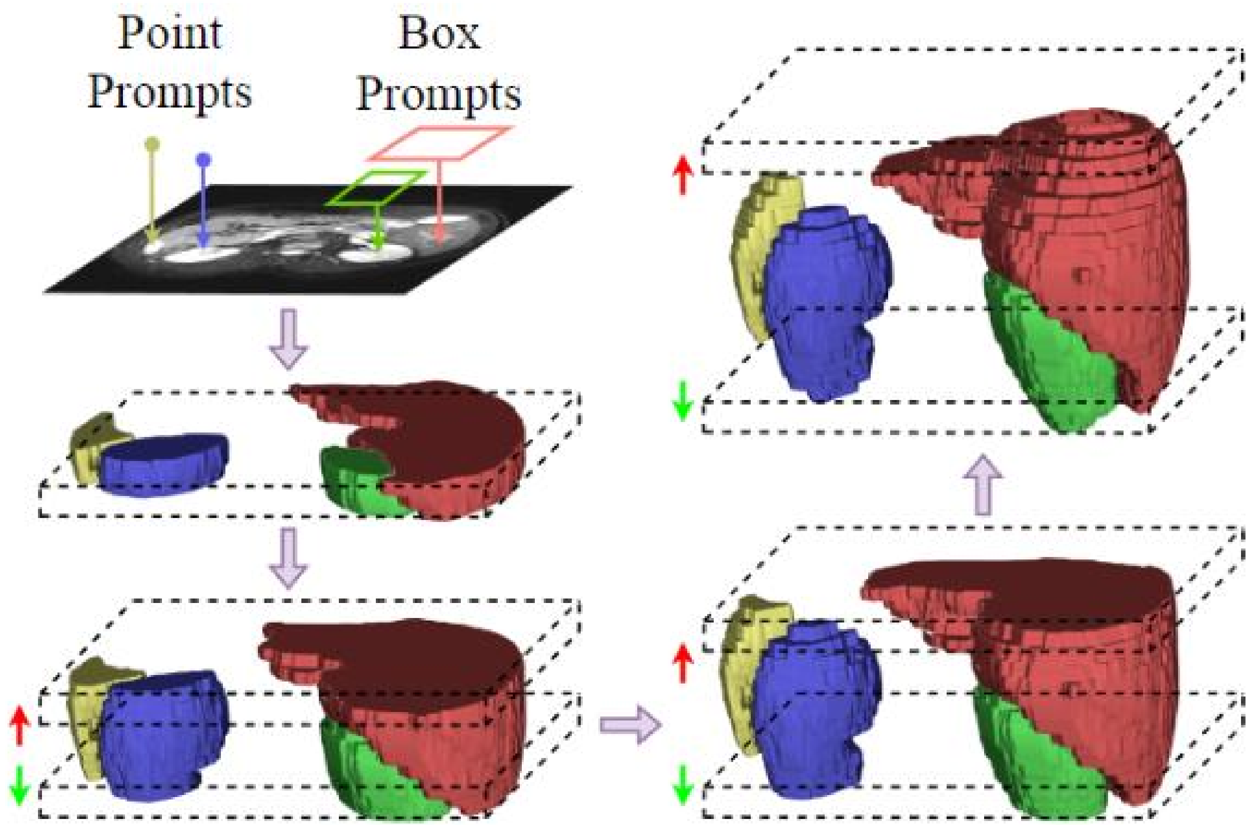


2D vs 3D



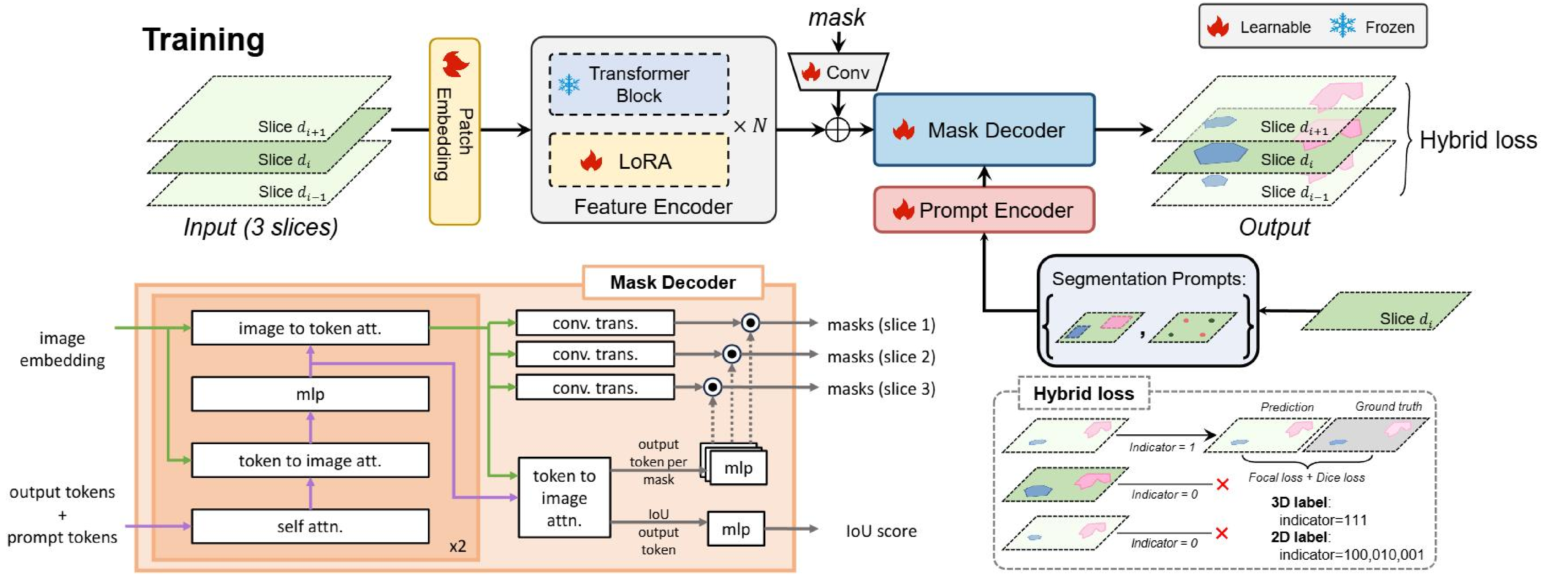
Kirillov A, Mintun E, Ravi N, et al. Segment anything[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2023: 4015-4026.

Foundation Model for medical image segmentation



Quan Q, Tang F, Xu Z, et al. Slide-SAM: Medical SAM Meets Sliding Window[C]//Medical Imaging with Deep Learning.

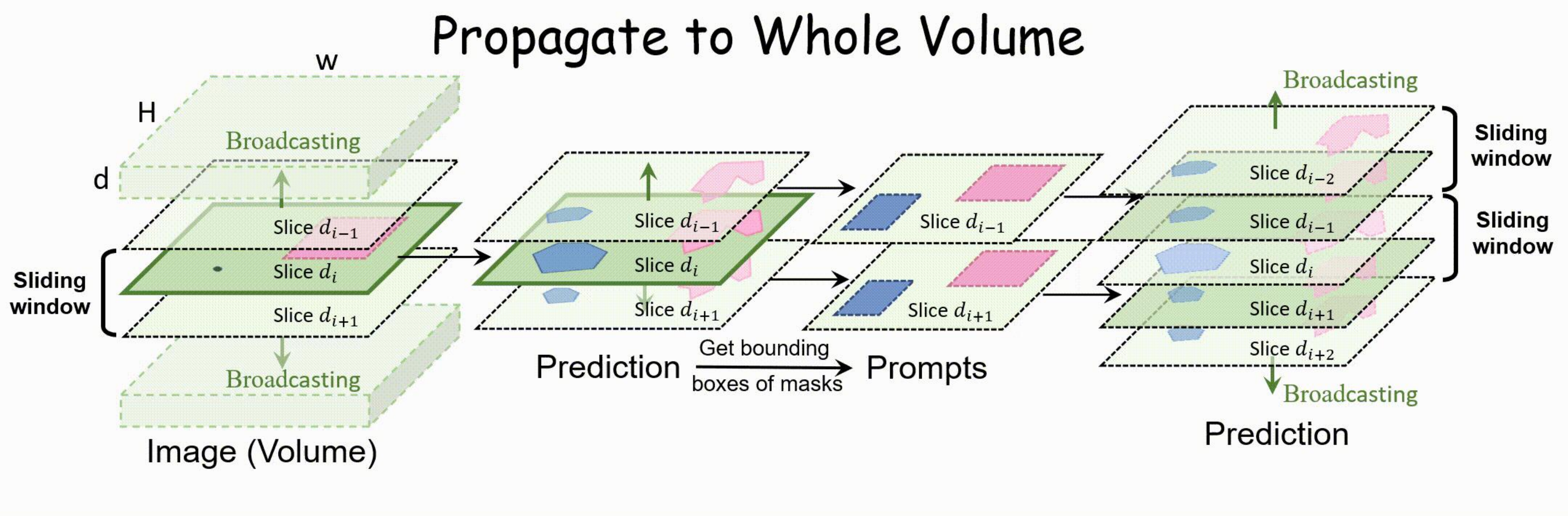
Foundation Model for medical image segmentation



Slide-SAM

Quan Q, Tang F, Xu Z, et al. Slide-SAM: Medical SAM Meets Sliding Window[C]//Medical Imaging with Deep Learning.

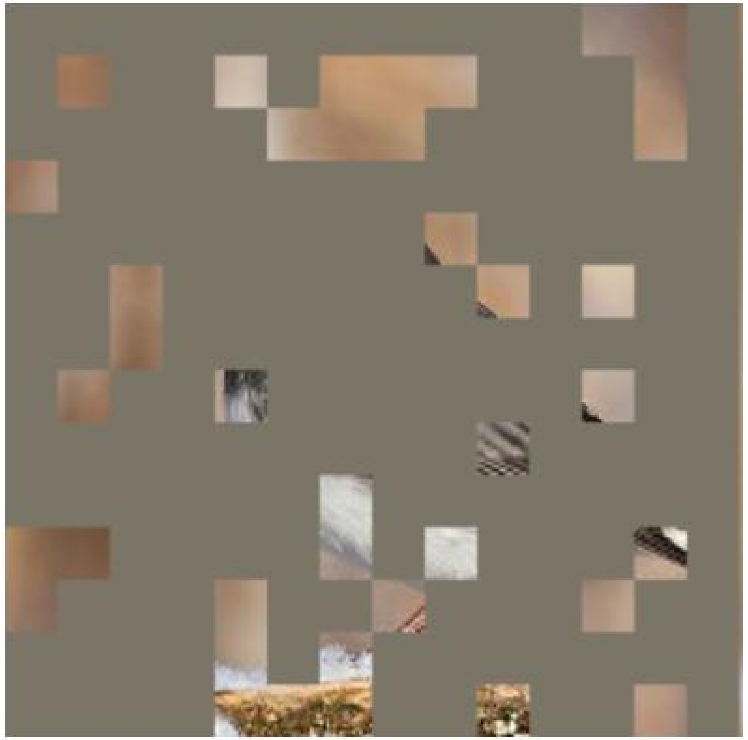
Foundation Model for medical image segmentation



Slide-SAM

Quan Q, Tang F, Xu Z, et al. Slide-SAM: Medical SAM Meets Sliding Window[C]//Medical Imaging with Deep Learning.

Foundation Model for medical image segmentation



He K, Chen X, Xie S, et al. Masked autoencoders are scalable vision learners[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 16000-16009.

Foundation Model for medical image segmentation

He K, Chen X, Xie S, et al. Masked autoencoders are scalable vision learners[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 16000-16009.

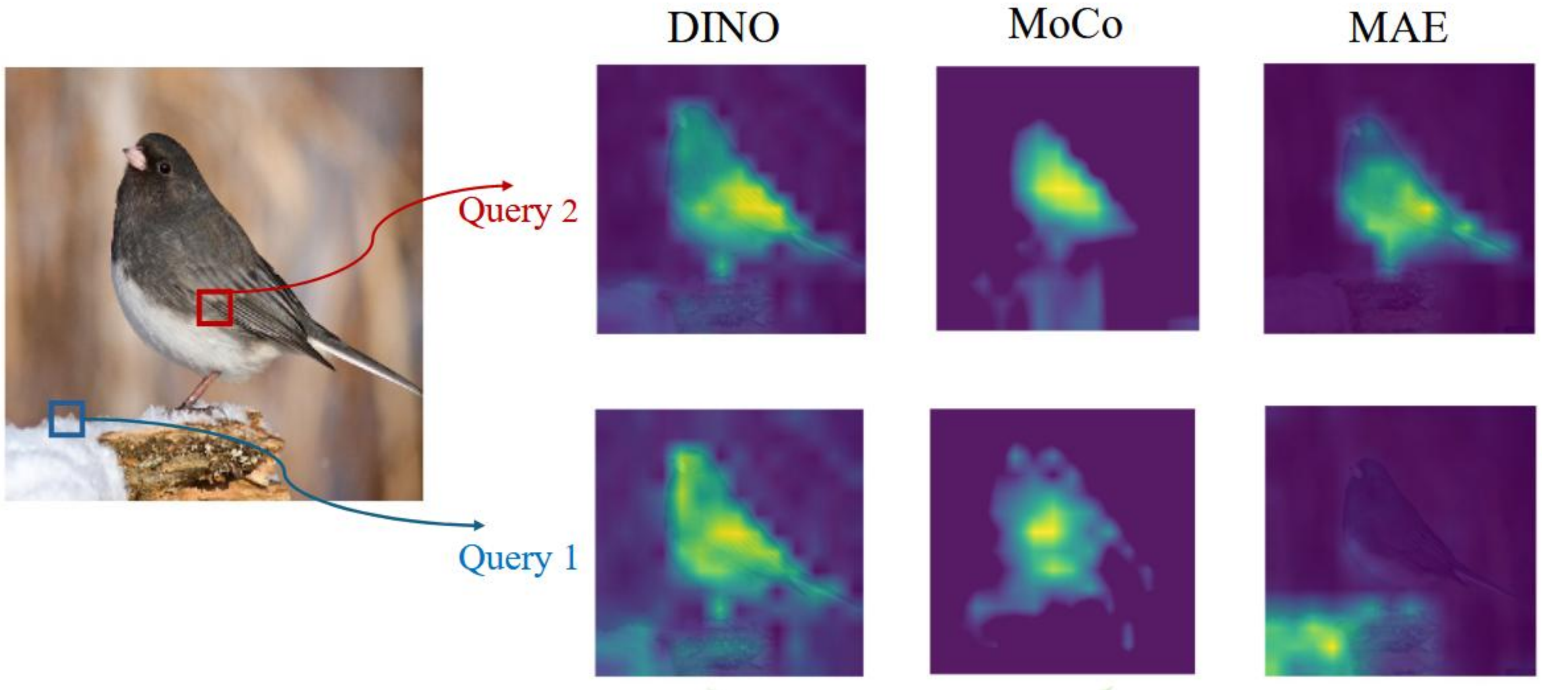
Foundation Model for medical image segmentation

Masked Image Modeling (MIM): learns diverse local attention patterns [1,2]

Huang Y, Wen Z, Chi Y, et al. How Transformers Learn Diverse Attention Correlations in Masked Vision Pretraining[J].
Zhang Q. How mask matters: Towards theoretical understandings of masked autoencoders[J]. Advances in Neural Information Processing Systems, 2022, 35: 27127-27139.

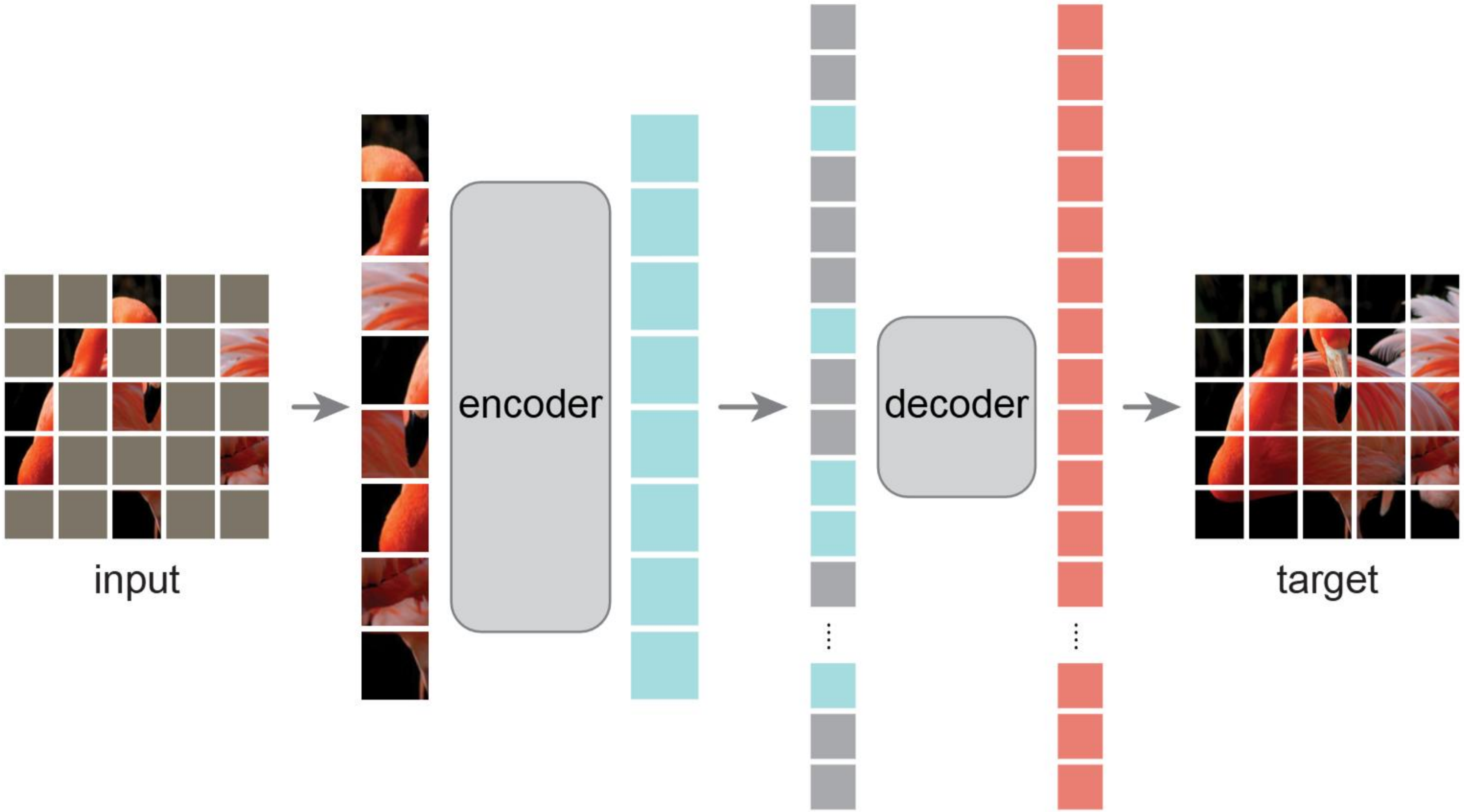
Foundation Model for medical image segmentation

Masked Image Modeling (MIM): learns diverse local attention patterns [1,2]



Huang Y, Wen Z, Chi Y, et al. How Transformers Learn Diverse Attention Correlations in Masked Vision Pretraining[J].
Zhang Q. How mask matters: Towards theoretical understandings of masked autoencoders[J]. Advances in Neural Information Processing Systems, 2022, 35: 27127-27139.

Foundation Model for medical image segmentation



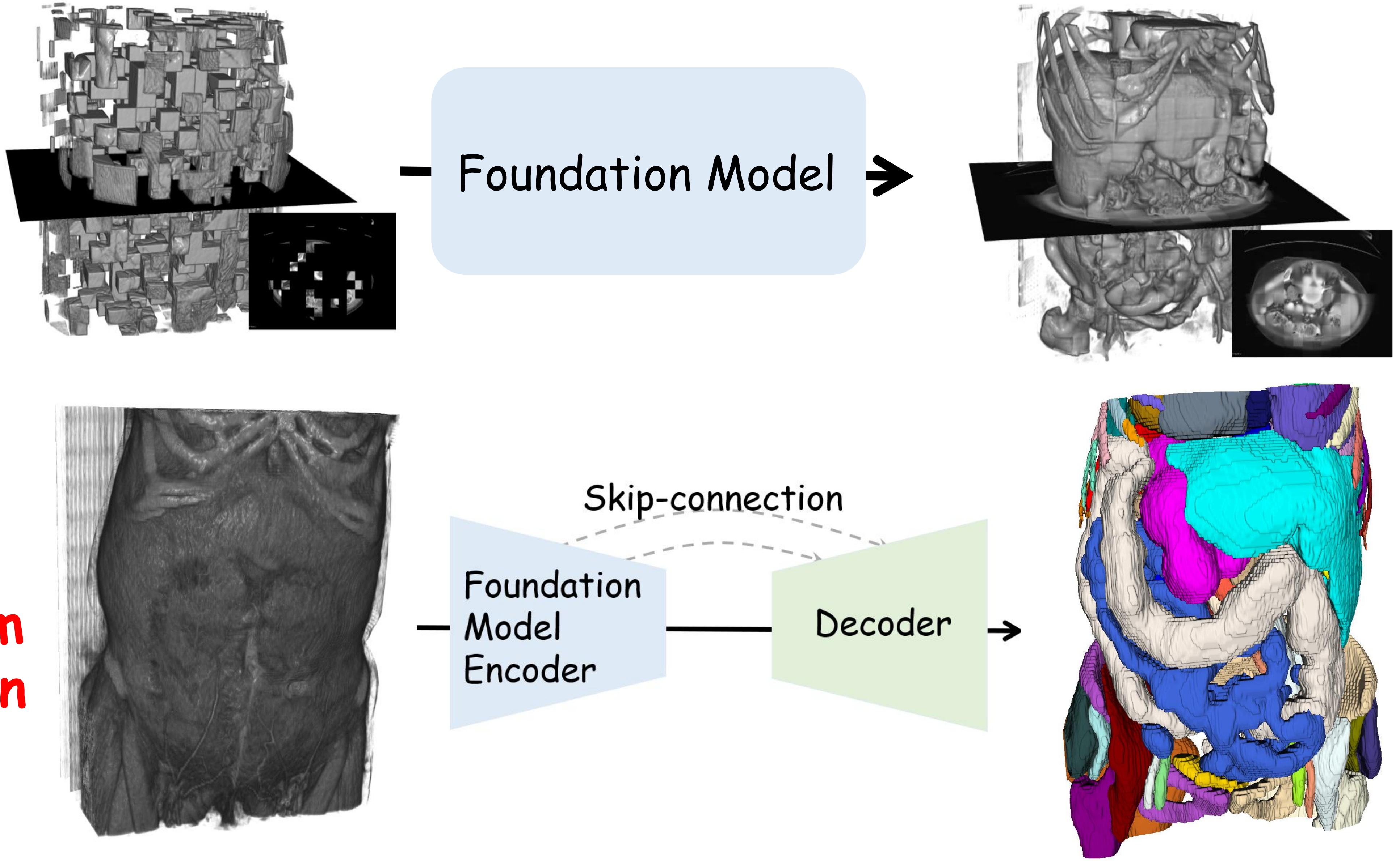
Masked Autoencoder

He K, Chen X, Xie S, et al. Masked autoencoders are scalable vision learners[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 16000-16009.

Foundation Model for medical image segmentation

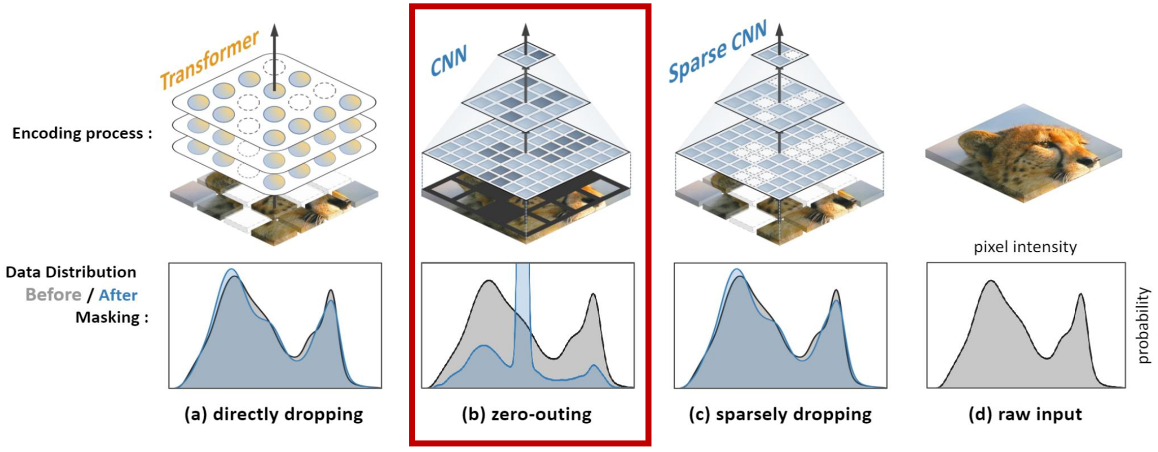
Up Stream
MIM, Dino, CLIP etc.

Down Stream
Segmentation



Foundation Model for medical image segmentation

pixel distribution shift !



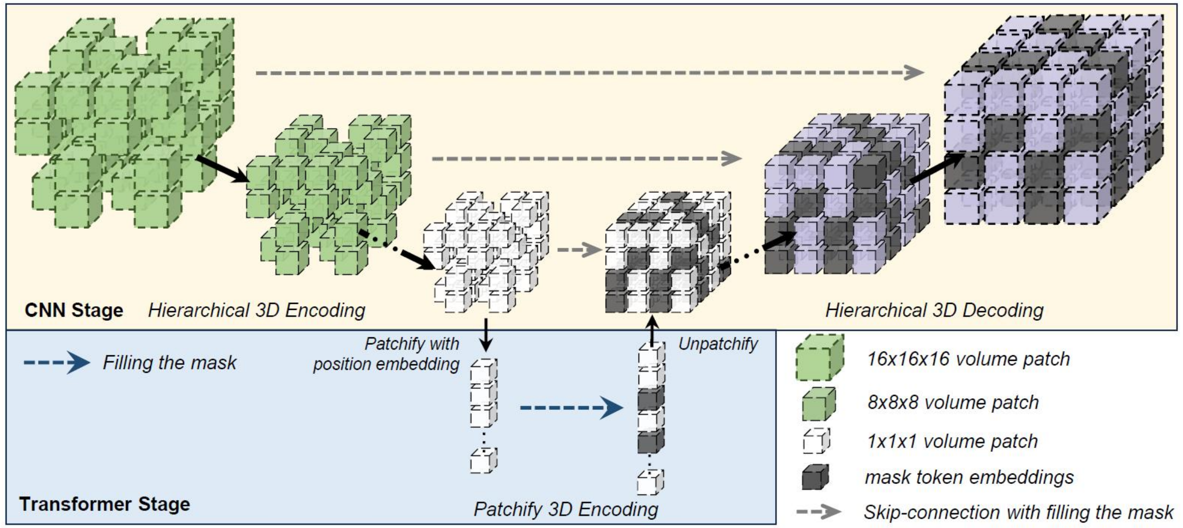


Fig. 1. Hybrid sparse masking (HySparK). The hybrid architecture comprises a CNN at the top (yellow) and a Transformer at the bottom (blue). We initiate the masking strategy at the junction between the CNN and Transformer and execute bottom-up mask modeling. The initialization unmasking patch is white, the bottom-up mapping unmasking patch is green and the masking position is **black**.

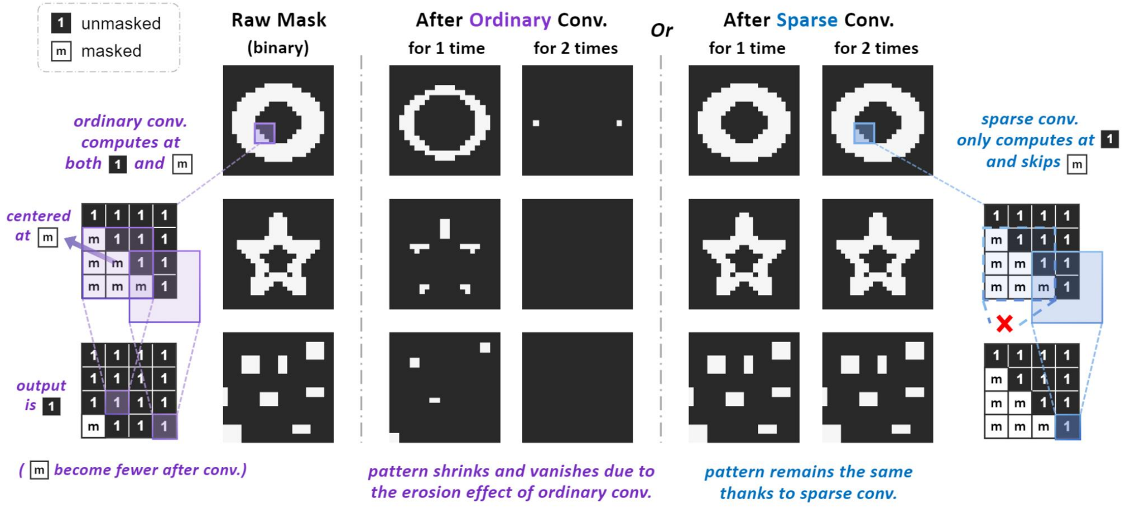


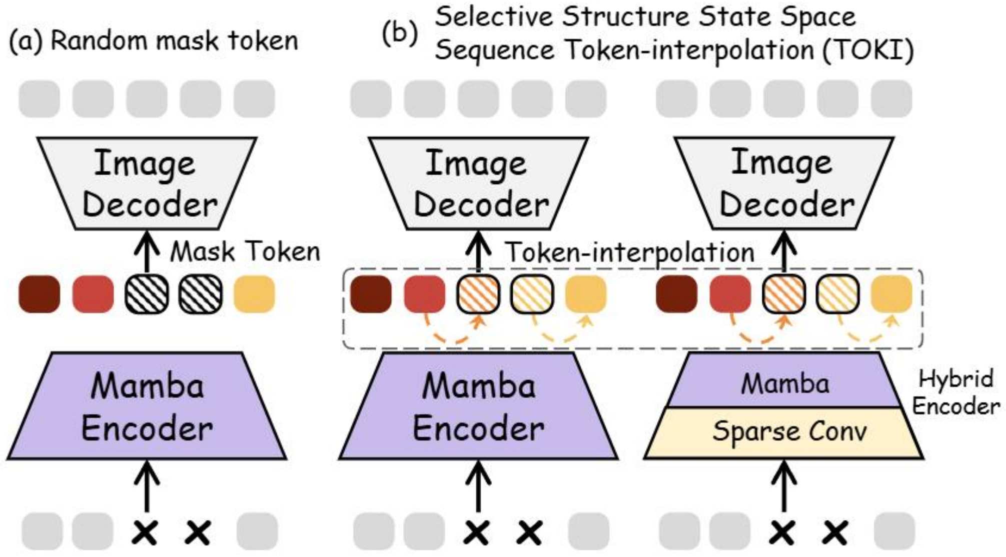
Table 3. Result on MSD. **val** (bold) / val (underline) : top method / second method.

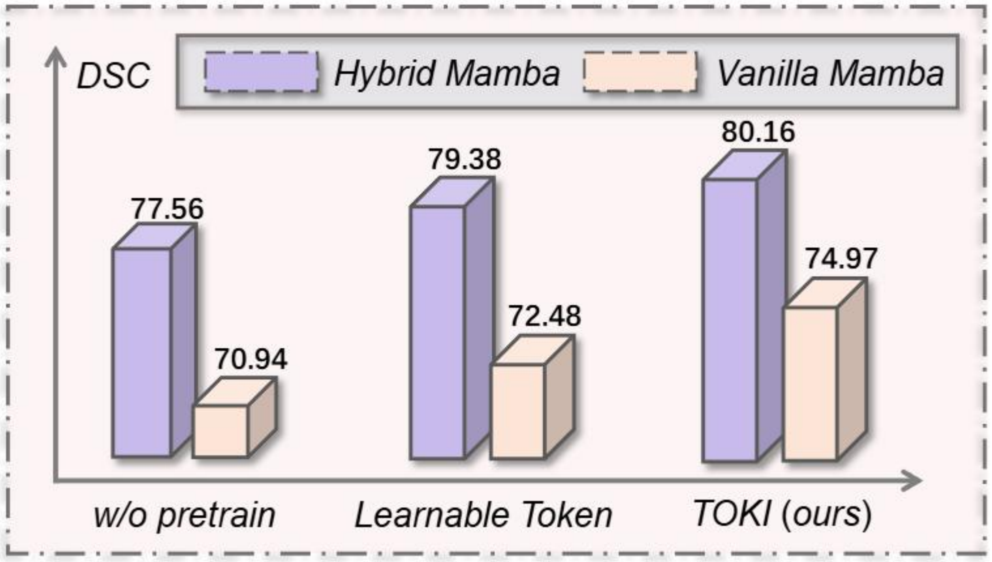
Pre-training method		Liver			Lung	Pancreas			Hepatic Vessel			Spleen	Colon	Avg
Method	Network	Dice1	Dice2	Avg	Dice	Dice1	Dice2	Avg	Dice1	Dice2	Avg	Dice	Dice	
vox2vec [21]	3D UNet(FPN) [27]	95.60	51.00	73.70	56.60	77.00	31.80	54.40	59.50	62.40	60.95	96.10	30.10	61.97
SUP [32]	Swin UNETR [32]	95.00	49.30	72.15	55.20	75.20	35.90	55.55	60.90	57.50	59.20	95.50	29.20	61.13
MAE [13]	UNETR [32]	95.49	56.47	75.98	56.42	77.76	39.29	58.52	59.99	62.22	61.10	95.28	34.53	63.63
SimMIM [13]	Swin UNETR [32]	95.32	55.25	75.28	60.31	76.16	44.96	60.56	60.67	61.79	61.23	95.64	41.11	65.68
SparK [17]	MedNeXt [29]	95.87	**62.95**	**79.41**	65.58	78.88	47.86	63.37	61.08	67.76	64.42	96.18	49.85	69.80
HySparK	MedNeXt+ViT	**96.02**	60.92	78.47	**65.96**	**79.69**	**49.67**	**64.68**	**61.58**	**69.36**	**65.47**	**96.39**	**50.78**	**70.29**

Tang F, Xu R, Yao Q, et al. Hyspark: Hybrid sparse masking for large scale medical image pre-training[C]//In MICCAI 2024. (Early Accept)

Foundation Model for medical image segmentation

(1) Limited causal token learning ability in state space modeling





(2) Masking consistency across different architectures

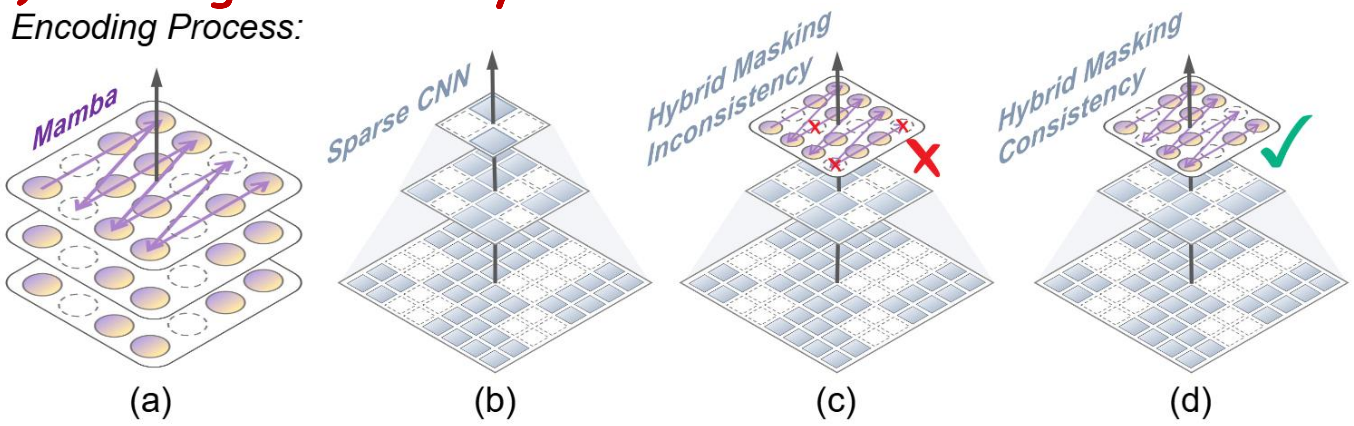
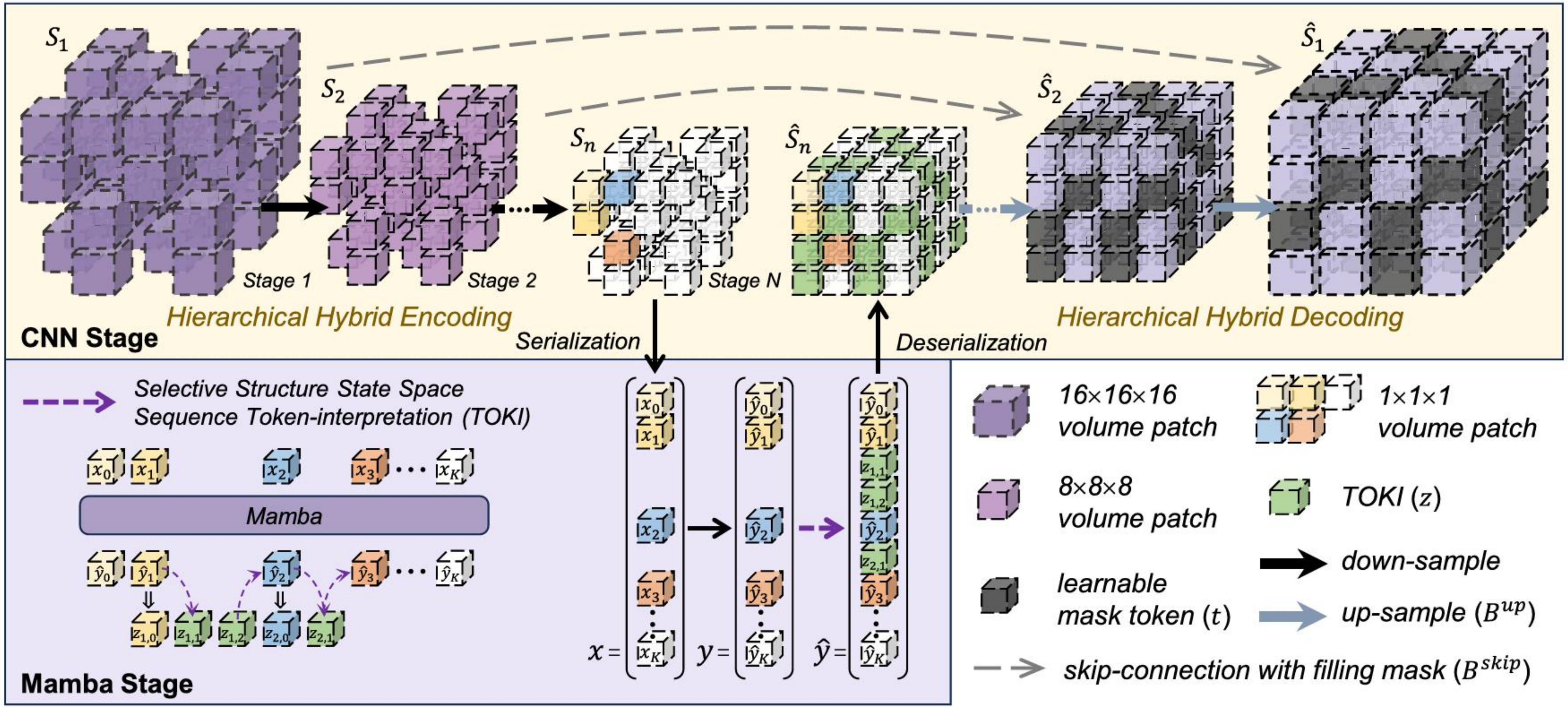


Table 9: Ablation on each component in MambaMIM. The experiment is performed on BTCV for the 3D segmentation task.

MambaMIM components	Hybrid MIM	Skip -connection	TOKI	Avg (%)
w/o pretrained	—	—	—	77.56
w/o bottom-up masking	✗	✗	✗	79.20
w/o skip	✓	✗	✗	79.25
w/o TOKI	✓	✓	✗	79.38
MambaMIM	✓	✓	✓	**80.16**

Tang F, Nian B, Li Y, et al. MambaMIM: Pre-training Mamba with state space token interpolation and its application to medical image segmentation[J]. Medical Image Analysis, 2025: 103606.

Foundation Model for medical image segmentation



MambaMIM

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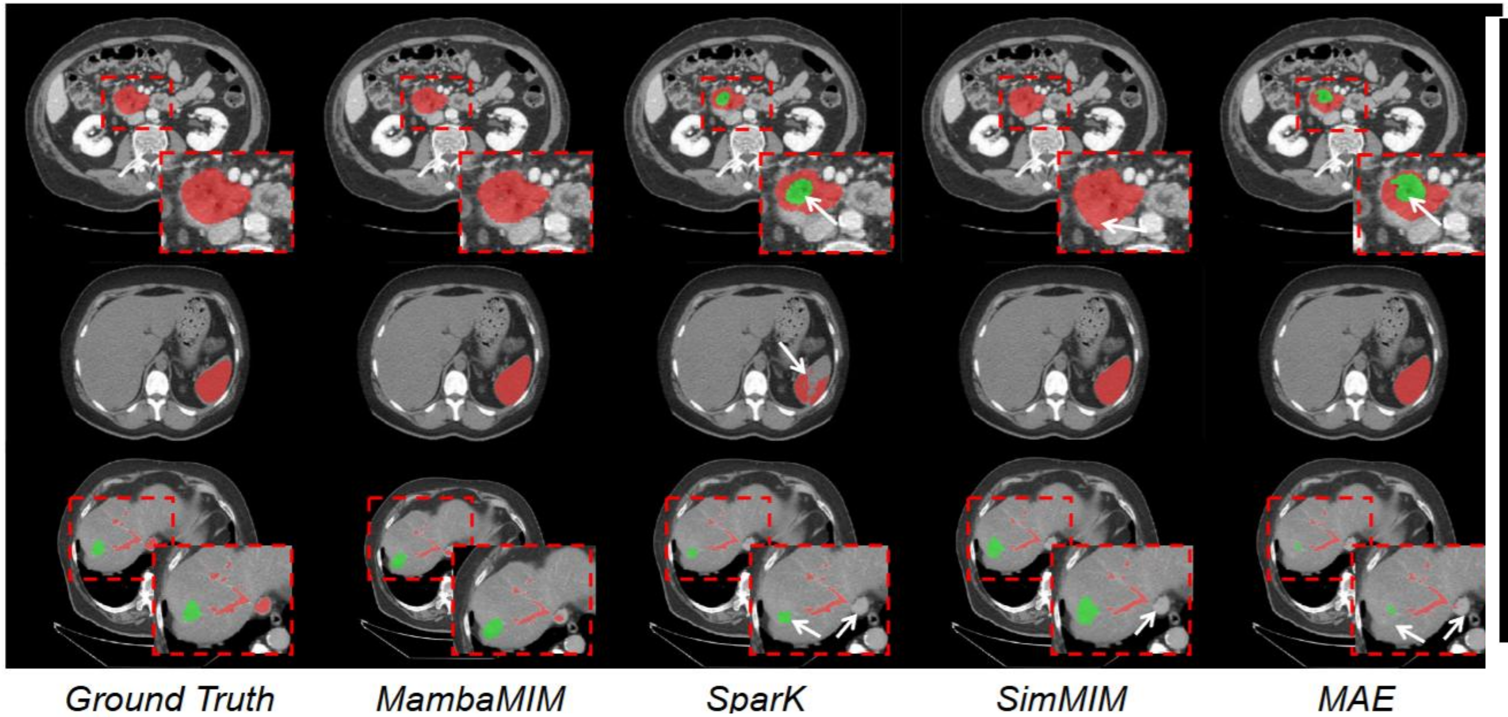
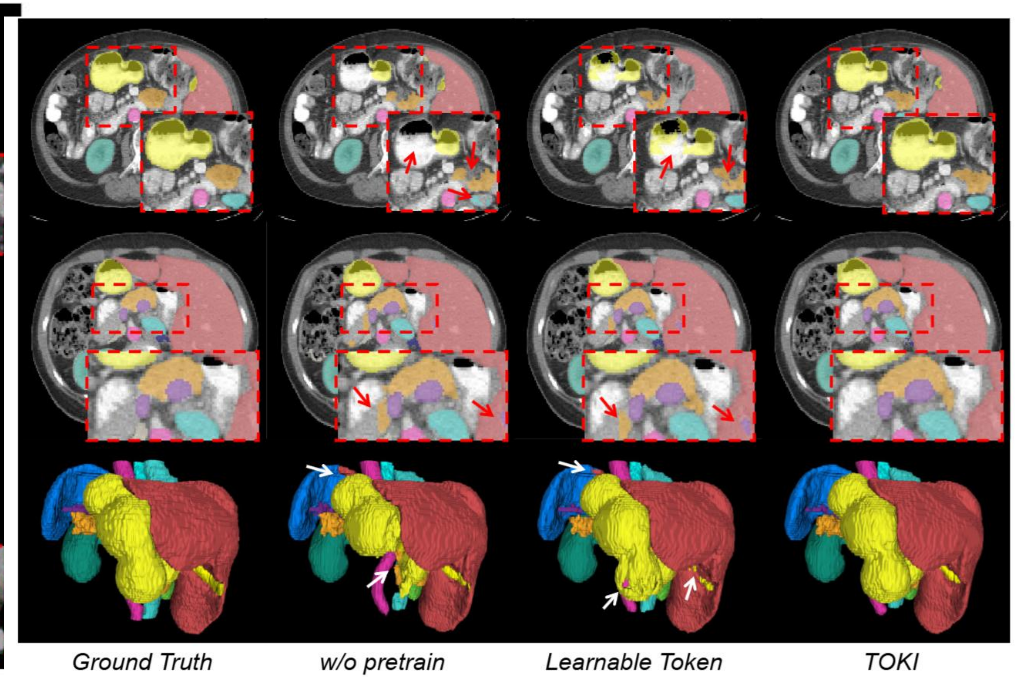
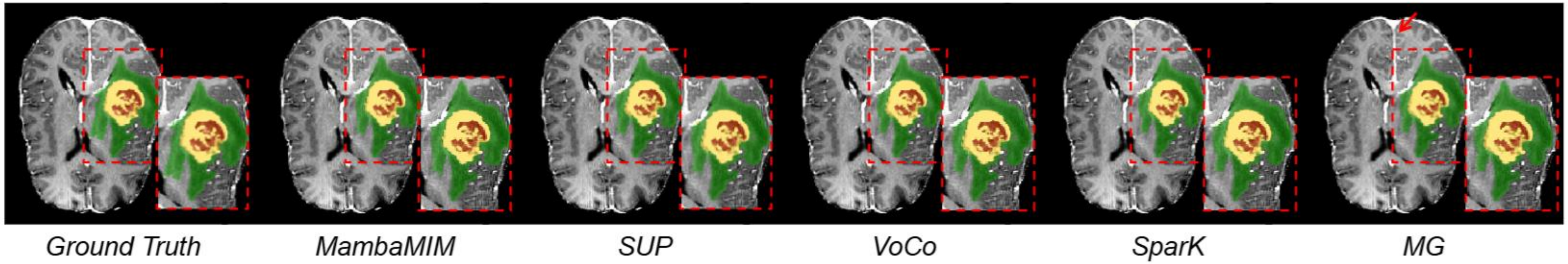
Foundation Model for medical image segmentation







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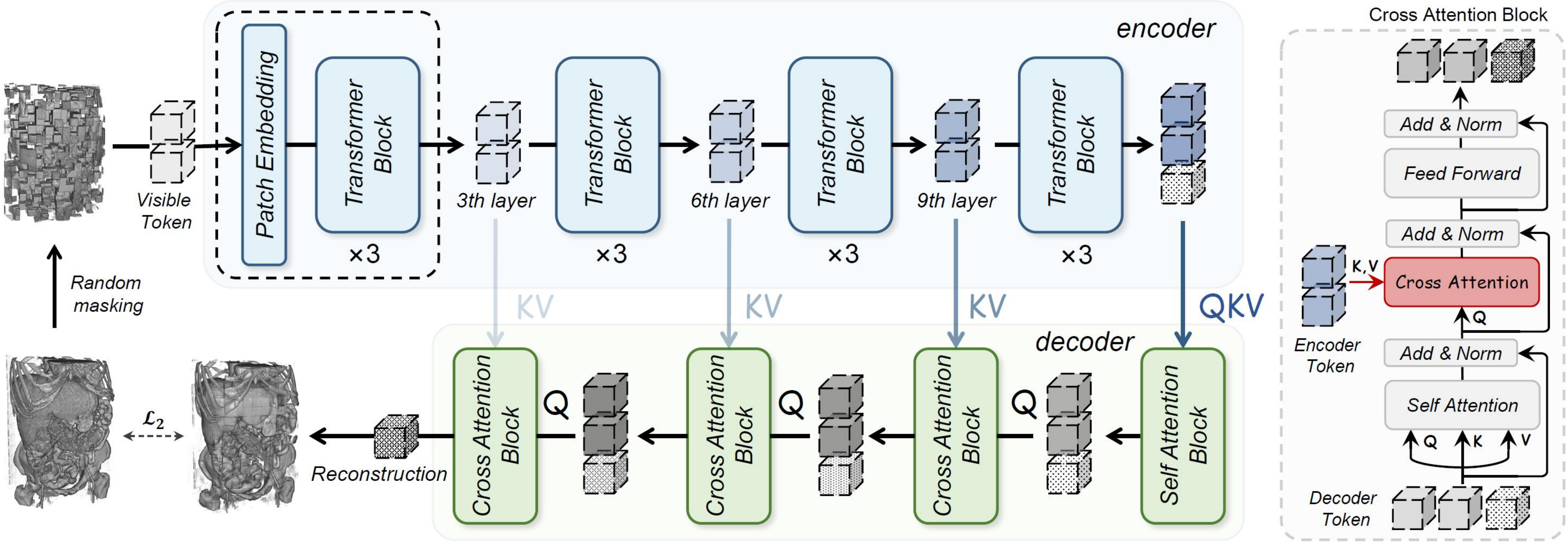
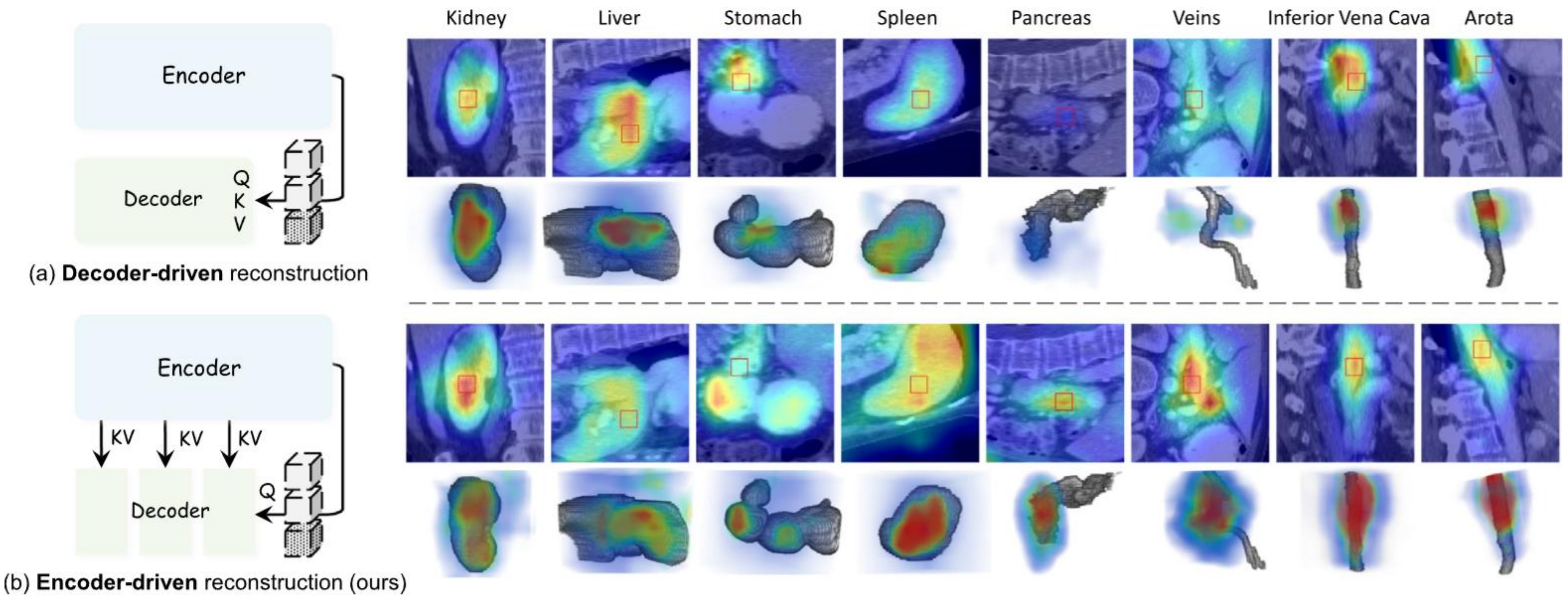
Foundation Model for medical image segmentation

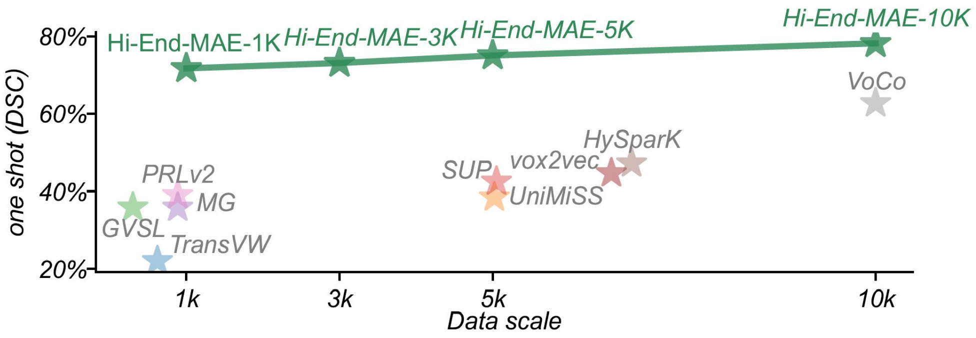


Hi-End-MAE

Tang F, Yao Q, Ma W, et al. Hi-End-MAE: Hierarchical encoder-driven masked autoencoders are stronger vision learners for medical image segmentation[J]. Medical Image Analysis, 2026: 103770.

Foundation Model for medical image segmentation



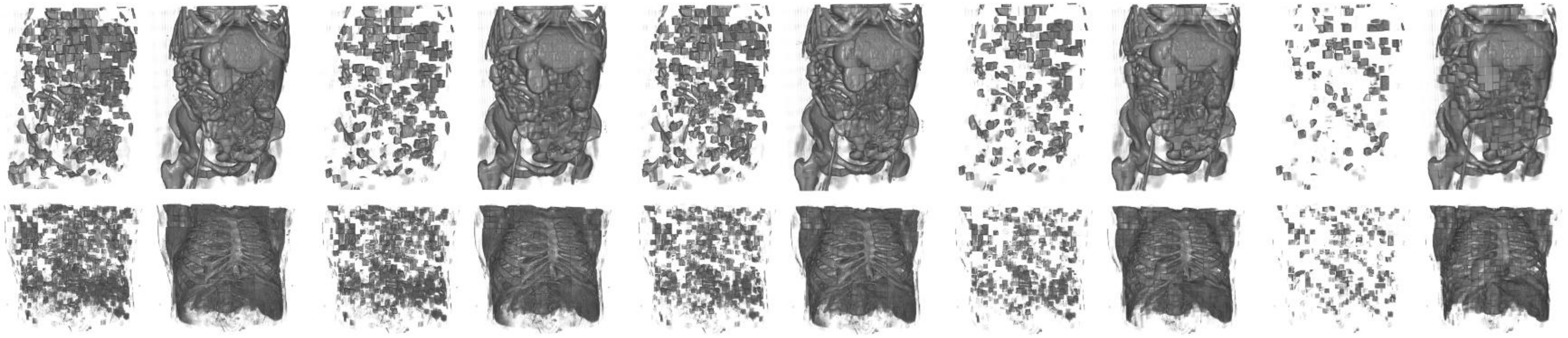


	Lun	Liv	Kid	Vein	Aor	Int	Spl	IVC	Sto	Duo	Pan	Pro/ute	ADR	Rec	Gall	Eso	Avg
MG	68.32	71.99	46.95	33.68	53.03	32.39	28.64	31.53	14.76	0.00	14.47	0.00	7.56	0.00	9.85	11.00	26.51
UniMiSS	89.14	76.96	34.38	37.77	53.13	22.64	28.80	25.77	20.16	0.00	8.42	9.55	3.40	3.91	8.64	14.28	27.31
SUP	81.92	76.85	42.67	35.69	56.58	38.69	29.26	27.85	36.04	38.41	14.91	21.31	10.14	9.98	20.22	6.99	34.22
HySparK	92.66	80.41	47.00	40.88	64.17	41.27	35.96	32.82	40.06	30.30	20.09	29.05	16.27	13.47	25.13	12.63	38.88
VoCo	85.67	89.64	71.17	74.95	65.90	75.64	67.81	55.75	45.10	31.37	45.65	49.55	29.84	34.83	38.84	15.92	54.85
Hi-End-MAE	95.93	89.93	78.35	78.66	73.36	76.50	71.29	67.15	54.38	60.28	56.13	57.43	41.88	36.96	40.06	20.58	62.42

Colour scale: 0, 20, 40, 60, 80

Tang F, Yao Q, Ma W, et al. Hi-End-MAE: Hierarchical encoder-driven masked autoencoders are stronger vision learners for medical image segmentation[J]. Medical Image Analysis, 2026: 103770.

Foundation Model for medical image segmentation



	Lun	Liv	Kid	Vein	Aor	Int	Spl	IVC	Sto	Duo	Pan	Pro/ute	ADR	Rec	Gall	Eso	Avg
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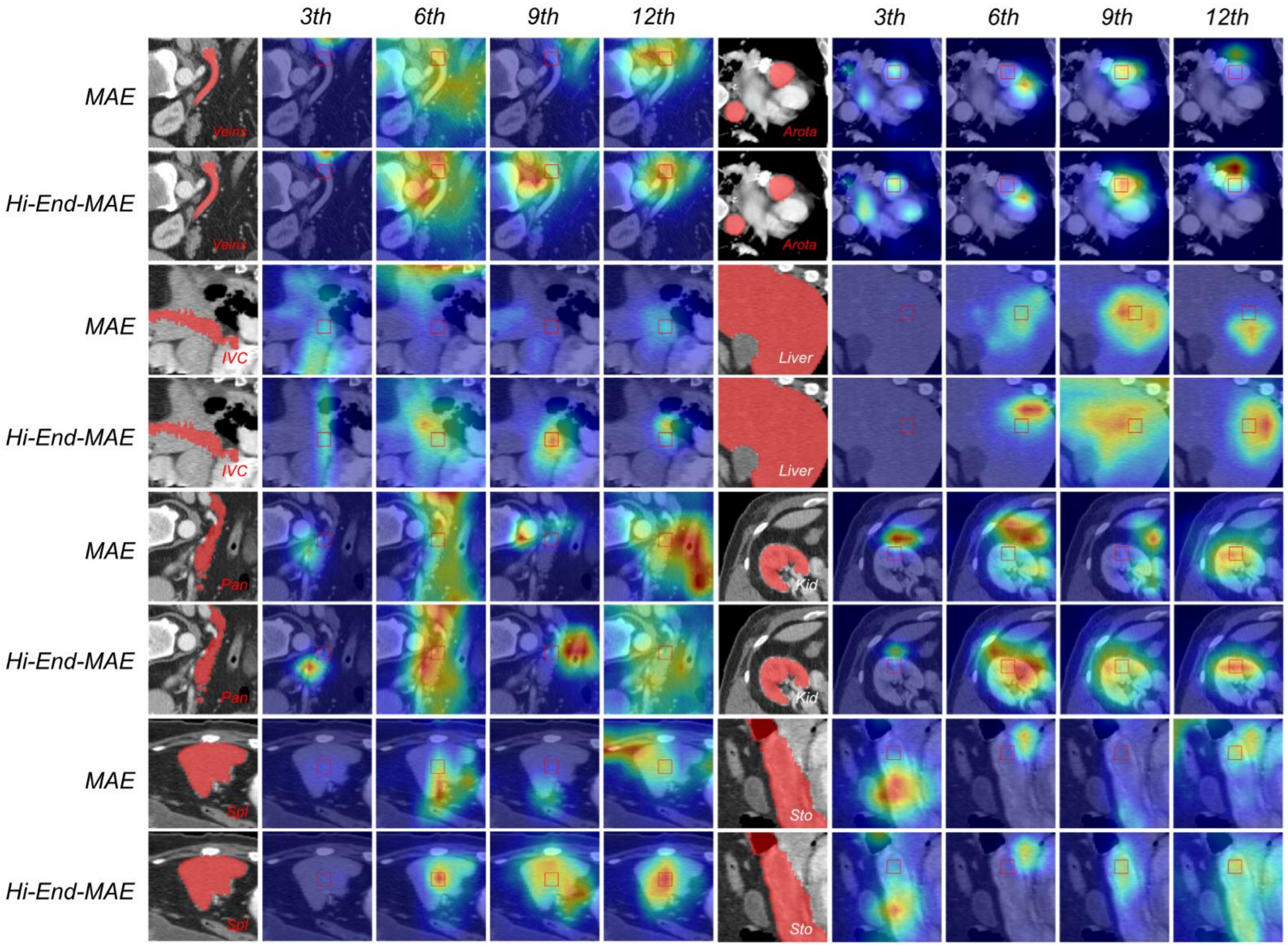
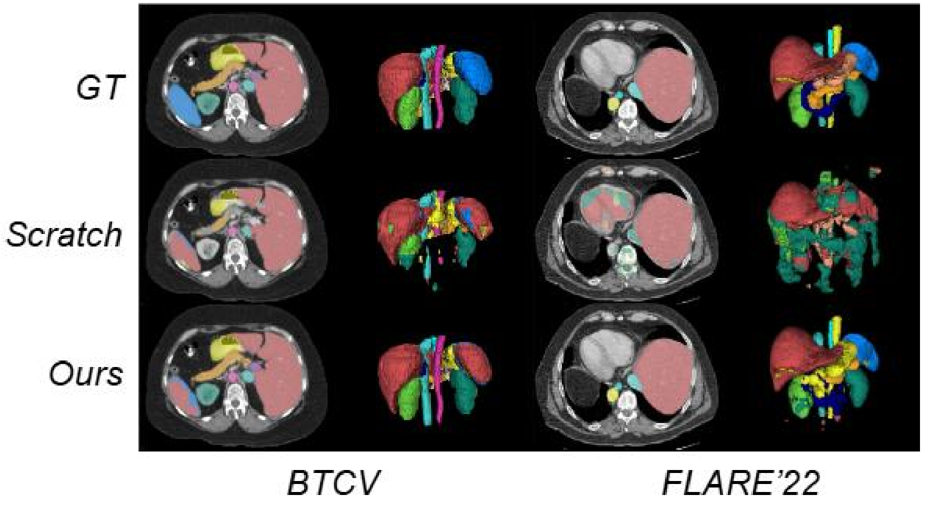
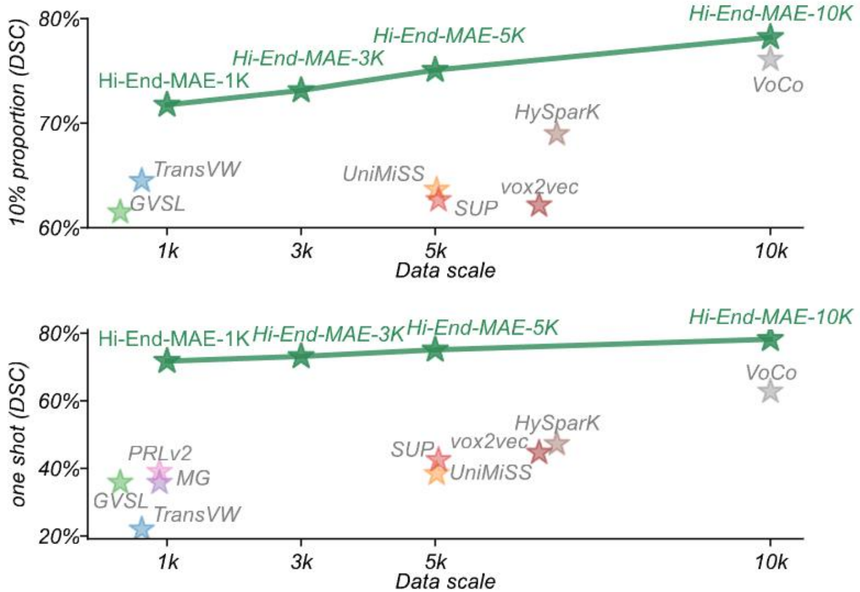
Foundation Model for medical image segmentation







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Project 1

- 下载ACDC dataset from link：ACDC Challenge（包含原始图像及标签）
- 下载 数据集对应的Scibble 标签 from https://github.com/vios-s/multiscale-adversarial-attention-gates/tree/main/DATA
- 1）基于Scibble 标签使用合适的传统方法获取分割结果（伪标签）
- 2）再利用伪标签实现基于深度学习的左/右心室及心肌分割；
- 报告：至少两种传统方法实现的分割结果及性能（可视化对比及定量指标：DSC/95HD）[2pt]
- 深度学习算法使用伪标签学习的分割结果及性能（可视化对比及定量指标：DSC/95HD）[1pt]

Project 2

Homework Assignment: U-Bench Expansion and Model Training: https://github.com/FengheTan9/U-Bench

- Train on a new medical image dataset (not included in U-Bench) [2pt]

  Construct a new dataset outside the existing U-Bench collection. Ensure proper preprocessing, split strategy.

- Train a new model architecture (not included in U-Bench) [2pt]

  Implement or adapt a novel architecture not already benchmarked in U-Bench.
  Bonus (+2 pt) for models based on **Mamba**, **RWKV**, or related sequence-state variants.
  Bonus (+5 pt) for proposing **novel architectures or new network modules** with clear innovation and justification.

- Perform zero-shot inference on a dataset within the same domain [1pt]

- Write a detailed technical report [3pt]

  The report should clearly describe: (1) Model design and training; (2) Dataset details and preprocessing pipeline; (3) Experimental setup, results, and analysis (**IoU** and **U-Score**) (4) Discussion of findings and insights

Submit both the **training logs** and **final model weights**. Include **technical report** summarizing the work.
Any **identical model weights or copied reports** found across submissions will result in a **score of 0**.